

5.2 Organization of this International Standard

This International Standard groups software system life cycle activities into seven process groups, each with a defined purpose, desired outcomes, and related activities. These process groups collectively form a Process Reference Model (PRM) that can be applied to any system or software life cycle model, providing a structured yet adaptable framework.

The seven process groups are:

- a) Agreement Processes (subclauses 5.2.2.1.1 and 6.1)
- b) Organizational Project-Enabling Processes (subclauses 5.2.2.1.2 and 6.2)
- c) Project Processes (subclauses 5.2.2.1.3 and 6.3)
- d) Technical Processes (subclauses 5.2.2.1.4 and 6.4)
- e) Software Implementation Processes (subclauses 5.2.2.2.1 and 7.1)
- f) Software Support Processes (subclauses 5.2.2.2.2 and 7.2)
- g) Software Reuse Processes (subclauses 5.2.2.2.3 and 7.3)

These process groups are introduced below and depicted in Figure 1.

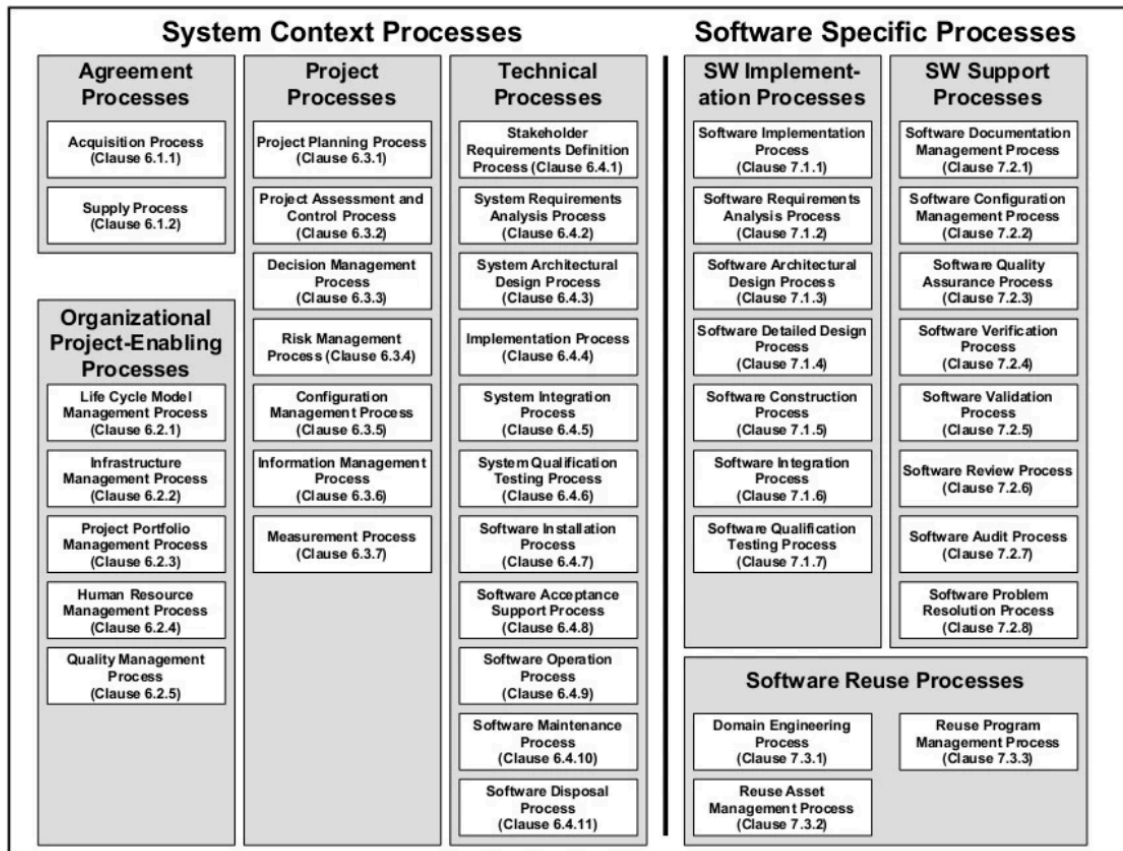


Figure 1 — Life Cycle Process groups

Figure 1 — Relationship among Process Groups

Figure 1 illustrates how these process groups interact and support each other throughout the life cycle. It shows the organizational processes enabling projects, project processes managing execution, and technical and software processes realizing and supporting system elements.

Clauses in this International Standard follow a structured numbering system:

- **6.a** and **7.a** denote a process group.
- **6.a.b** and **7.a.b** denote a specific process within that group.
- **6.a.b.1** and **7.a.b.1** describe the process's purpose.
- **6.a.b.2** and **7.a.b.2** describe the process's outcomes.
- **6.a.b.3.c** and **7.a.b.3.c** list process activities.
- **6.a.b.3.c.d** and **7.a.b.3.c.d** list tasks for each activity.

This structure supports clarity, traceability, and alignment between system and software processes. The Process Reference Model does not prescribe a specific system/software life cycle model or implementation method; it is **tailored** based on business needs, application domain, project scale, and risk. Projects typically adopt the organization's defined process in the context of customer requirements.

Process outcomes serve as objective evidence for assessing process capability and identifying improvement opportunities.

5.2.2 Summary of Life Cycle Processes

This International Standard has two major subdivisions:

- **Clause 6** provides a system context for standalone software products, services, or systems.
- **Clause 7** specifies software-specific processes for implementing software elements within larger systems.

To support concurrent application of ISO/IEC 15288 and ISO/IEC 12207, corresponding processes in Clause 6 share the same subclause number (6.x.x). The processes in this International Standard are **software-appropriate specializations or contributions** to the outcomes of ISO/IEC 15288 processes. While many are structurally similar, important distinctions remain regarding purpose, outcomes, and intended audience.

Tailoring: Organizations may tailor processes according to project characteristics, regulatory environment, and business needs, as described in Annex A.

5.2.2.1 System Context Processes

5.2.2.1.1 Agreement Processes define the activities to establish an agreement between two organizations. The Acquisition Process enables conducting business with suppliers for an operational system. The Supply Process facilitates projects resulting in products or services delivered to the acquirer. These Agreement Processes are software-appropriate specializations of those in ISO/IEC 15288.

5.2.2.1.2 Organizational Project-Enabling Processes manage the organization's capability to acquire and supply products or services through project initiation, support, and control. They provide resources and infrastructure to support projects and ensure organizational objectives are met. These processes are not a comprehensive set of business processes.

The Organizational Project-Enabling Processes include:

- Life Cycle Model Management Process
- Infrastructure Management Process
- Project Portfolio Management Process
- Human Resource Management Process
- Quality Management Process

These processes are software-appropriate specializations of the corresponding processes in ISO/IEC 15288.

5.2.2.1.3 Project Processes

This International Standard describes processes for planning, assessing, and controlling projects. These principles apply to any area of an organization's management.

There are two categories of project processes:

1. Project Management Processes: These are used to plan, execute, assess, and control project progress.

- Project Planning Process: Establishes and evolves project plans.
- Project Assessment and Control Process: Assesses actual progress.

2. Project Support Processes: These provide focused tasks for specialized management objectives. They are evident in any undertaking, from organizations to single life cycle processes.

- Decision Management Process: Manages decisions.
- Risk Management Process: Manages risks.
- Configuration Management Process: Manages configurations.
- Information Management Process: Manages information.
- Measurement Process: Measures progress.

The Project Support Processes in this International Standard are similar to those in ISO/IEC 15288, except for formatting differences. Software Supporting Processes may relate to Project Supporting Processes.

5.2.2.1.4 Technical Processes

Technical Processes define system requirements and transform them into products, enabling consistent realization, use, support, and retirement. They ensure that products and services meet requirements for timeliness, cost-effectiveness, functionality, reliability, safety, security, environmental compatibility, and societal expectations.

The Technical Process group includes:

- Stakeholder Requirements Definition
- System Requirements Analysis
- System Architectural Design
- System Implementation
- System Integration

- System Qualification Testing
- System Installation
- System Transition
- System Operation
- System Maintenance
- System Disposal

Software processes such as Software Requirements Analysis, Software Architectural Design, Software Detailed Design, Software Construction, Software Integration, Software Qualification Testing, Software Installation, Software Operation, Software Maintenance, and Software Disposal are **specializations or contributors** to these technical processes.

5.2.2.2 Software-Specific Processes

5.2.2.2.1 Software Implementation Processes

Software Implementation Processes produce a specified software item by transforming defined behavior, interfaces, and constraints into an implemented element that satisfies system requirements. This group includes:

- Software Requirements Analysis
- Software Architectural Design
- Software Detailed Design
- Software Construction
- Software Integration
- Software Qualification Testing

These processes correspond to the Implementation Process in ISO/IEC 15288, specialized for software.

5.2.2.2.2 Software Support Processes

Software Support Processes provide focused activities that ensure software quality and conformity. These processes support the implementation and maintenance of software across its life cycle:

- Software Documentation Management
- Software Configuration Management
- Software Quality Assurance
- Software Verification
- Software Validation
- Software Review
- Software Audit

- Software Problem Resolution
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5.2.2.2.3 Software Reuse Processes

Software Reuse Processes focus on systematic reuse of software components and knowledge to improve efficiency, reduce cost, and enhance quality. Operating primarily outside individual projects, this process group includes:

- a) Domain Engineering Process
 - b) Reuse Asset Management Process
 - c) Reuse Program Management Process
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5.2.3 Process Reference Model

Annex B defines the Process Reference Model (PRM) at a higher abstraction level than the detailed requirements in this International Standard. It specifies the **purpose and outcomes** of each process, enabling process capability assessment.

Organizations may evaluate their processes against the PRM, focusing on achievement of purposes and outcomes, without necessarily conforming to the detailed activity descriptions.

NOTE 1 — The term “Process Reference Model” has the same meaning as in ISO/IEC.

NOTE 2 — ISO/IEC 15504-2 defines requirements for assessing processes using the PRM.

For **mapping between system (Clause 6) and software (Clause 7) processes**, see Annex C.

6 System Life Cycle Processes

6.1 Agreement Processes

The Agreement Processes establish and maintain mutual commitments between the acquirer and the supplier. They include the Acquisition Process and the Supply Process. This clause focuses on the Acquisition Process.

6.1.1 Purpose

The purpose of the Acquisition Process is to obtain a product or service that satisfies the acquirer's needs, in accordance with organizational policies and applicable standards. This process begins with the identification of customer needs and ends with the acceptance and closure of the acquisition agreement.

6.1.2 Outcomes

As a result of successful implementation of the Acquisition Process:

- a) An acquisition strategy and plan are developed that clearly define expectations, responsibilities, liabilities, and acceptance criteria for both the acquirer and the supplier.**
 - b) One or more suppliers are selected according to defined procedures.**
 - c) A contract or agreement is established between the acquirer and the supplier.**
 - d) The acquired product or service meets the acquirer's stated requirements, constraints (cost, schedule, quality), and acceptance criteria.**
 - e) Supplier performance is monitored against contract milestones, deliverables, and quality measures.**
 - f) Acceptance is conducted according to defined strategy and criteria, with verification and validation evidence gathered.**
 - g) All contractual and technical issues are satisfactorily closed, and the acquirer assumes configuration management responsibility for the delivered product.**
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6.1.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with organizational policies and procedures.

6.1.3.1 Acquisition Preparation

This activity involves defining the acquisition need, analyzing requirements, and preparing the acquisition plan.

- 6.1.3.1.1 Describe the concept or need to acquire, develop, or enhance a system, software product, or software service.
 - 6.1.3.1.2 Define and analyze system and/or software requirements, including business, organizational, user, safety, security, and criticality requirements, referencing Stakeholder Requirements Definition (6.4.2) and System Requirements Analysis (6.4.3).
 - 6.1.3.1.3 Analyze acquisition options based on risk, cost, benefit, and feasibility, including:
 - Use of off-the-shelf software products
 - Internal development
 - Contracted development or service provision
 - Enhancement of existing software products or services
 - Combinations of the above
 - 6.1.3.1.4 Ensure that off-the-shelf acquisitions meet requirements for documentation, licensing, warranty, future support, and proprietary rights.
 - 6.1.3.1.5 Prepare, document, and execute an acquisition plan including requirements, contract type, responsibilities, support concept, risk considerations, and management methods.
 - 6.1.3.1.6 Define and document the acceptance strategy and criteria.
 - 6.1.3.1.7 Document acquisition requirements (e.g., RFP), including system/software requirements, scope, bidder instructions, software list, terms and conditions, subcontract control, and technical constraints.
 - 6.1.3.1.8 Tailor relevant International Standard processes as needed and define scope of tasks referencing the contract.
 - 6.1.3.1.9 Define contract milestones for reviews and audits (see 7.2.6 and 7.2.7).
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6.1.3.2 Acquisition Advertisement

This activity involves announcing the acquisition and soliciting responses.

- 6.1.3.2.1 Request products or services from identified suppliers.

- 6.1.3.2.2 Engage in supply chain partnership to align technical and commercial approaches.
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6.1.3.3 Supplier Selection

This activity involves evaluating proposals and selecting a supplier.

- 6.1.3.3.1 Establish a supplier selection procedure with evaluation criteria and compliance weighting.
 - 6.1.3.3.2 Evaluate proposals based on capability, technical compliance, acceptance strategy, and conditions.
 - 6.1.3.3.3 Select supplier(s) accordingly.
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6.1.3.4 Contract Agreement

This activity establishes formal agreements with the supplier.

- 6.1.3.4.1 Involve relevant parties (e.g., suppliers, regulators) to tailor standard requirements for the project.
- 6.1.3.4.2 Negotiate and prepare a contract that includes acquisition requirements, cost, schedule, proprietary and licensing rights, and quality expectations.
- 6.1.3.4.3 Establish a change control mechanism for contract modifications, analyzing their impact on plans, costs, benefits, quality, and schedule.
- 6.1.3.4.4 Determine whether to apply the term “contract” or “agreement” per organizational policy.
- 6.1.3.4.5 Express expectations and responsibilities clearly, including the level of formality for change requests.

Informative Annex F provides a sample contract change management process.

6.1.3.5 Agreement Monitoring

This activity monitors supplier performance and contract fulfillment.

- 6.1.3.5.1 Monitor supplier activities using review (7.2.6), audit (7.2.7), verification (7.2.4), and validation (7.2.5) processes as appropriate.
- 6.1.3.5.2 Cooperate with the supplier to provide necessary

information and resolve pending issues in a timely manner.

6.1.3.6 Acquirer Acceptance

This activity ensures that the delivered product or service meets acceptance criteria.

- **6.1.3.6.1 Prepare for acceptance based on the defined strategy and criteria, including test cases, data, procedures, and environment.**
 - **6.1.3.6.2 Conduct acceptance reviews and testing, referring to System Qualification Testing (6.4.6) and/or Software Qualification Testing (7.1.7).**
 - **6.1.3.6.3 Accept the deliverable when all acceptance conditions are met.**
 - **6.1.3.6.4 Assume configuration management responsibility post-acceptance (see 7.2.2).**
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6.1.3.7 Closure

This activity concludes the acquisition.

- **6.1.3.7.1 Ensure all contractual obligations are met and issues resolved.**
- **6.1.3.7.2 Process payments or other agreed considerations for products or services rendered.**
- **6.1.3.7.3 Handle incremental deliveries and payments as applicable.**

6.1.2 Supply Process

6.1.2.1 Purpose

The purpose of the Supply Process is to provide a product or service to the acquirer that meets the agreed requirements. This includes responding to acquisition requests, establishing agreements, executing the contract, delivering and supporting the product or service, and closing the contractual relationship.

6.1.2.2 Outcomes

As a result of successful implementation of the Supply Process:

- a) The acquirer and their requirements are identified.
 - b) A response is produced to the acquirer's request (e.g., proposal, bid).
 - c) An agreement is established with the acquirer for developing, maintaining, operating, packaging, delivering, and installing the product or service.
 - d) The product or service is developed and/or provided in accordance with the agreed requirements.
 - e) The product or service is delivered and, if required, installed in accordance with the agreed requirements.
 - f) The contractual relationship is successfully concluded, with responsibilities transferred to the acquirer.
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6.1.2.3 Activities and Tasks

The supplier shall perform the following activities and tasks in accordance with organizational policies and applicable standards.

6.1.2.3.1 Opportunity Identification

This activity involves identifying potential acquirers and understanding their needs.

- 6.1.2.3.1.1 Identify the existence and identity of an acquirer who has a need for a product or service.
- 6.1.2.3.1.2 Collect relevant information regarding the acquirer's needs, environment, and constraints.

6.1.2.3.2 Supplier Tendering

This activity involves analyzing the acquirer's request and preparing a formal response.

- 6.1.2.3.2.1 Review the requirements in the request for proposal, taking into account organizational policies, capabilities, and applicable regulations.
- 6.1.2.3.2.2 Decide whether to bid or accept the contract.
- 6.1.2.3.2.3 Prepare and submit a proposal or tender in response to the acquirer's request.

6.1.2.3.3 Contract Agreement

This activity establishes the formal agreement between the supplier and acquirer.

- 6.1.2.3.3.1 Negotiate and establish an agreement covering development, maintenance, operation, packaging, delivery, and installation of the product or service.
- 6.1.2.3.3.2 Ensure that the contract reflects applicable requirements for quality, performance, schedule, proprietary rights, and regulatory compliance.

6.1.2.3.4 Contract Execution

This activity involves executing the contract, which includes planning, development, operation, and maintenance.

6.1.2.3.4.1 Project Planning

The supplier shall review the acquisition requirements to define a framework for managing and assuring the project and the quality of the deliverable. This should use an organizationally defined life cycle model. The supplier shall:

- Establish requirements for project management and quality assurance plans, including resource needs and acquirer involvement.
- Evaluate development options and associated risks, including:
 - a) Internal development
 - b) Subcontracting
 - c) Procuring off-the-shelf software
 - d) Combinations of the above
- Develop and document project management plans covering:

- Organizational structure, roles, responsibilities
- Engineering and test environments
- Resources, size, schedule
- Quality characteristics and quality plans (7.2.3)
- Safety and security plans
- Subcontractor management
- Verification (7.2.4) and Validation (7.2.5) approach
- Acquirer and user involvement (7.2.6, 7.2.7)
- Risk management
- Security policy (need-to-know, access control)
- Approval and certification requirements
- Scheduling, tracking, and reporting methods
- Personnel training (6.2.4)

6.1.2.3.4.2 Development

The supplier shall develop the software product in accordance with the Technical Processes (6.4), including:

- Implementation (6.4.5)
- System/Software Integration (6.4.6 / 7.1.6)
- Qualification Testing (6.4.7 / 7.1.7)

6.1.2.3.4.3 Operation and Maintenance

The supplier shall:

- Operate the software product in accordance with the Software Operation Process (6.4.9).
- Maintain the software product in accordance with the Software Maintenance Process (6.4.10).

6.1.2.3.4.4 Subcontractor Management

The supplier shall manage and control subcontractors in accordance with the Acquisition Process (6.1.1), passing down all contractual requirements to ensure compliance with prime contract obligations.

6.1.2.3.4.5 Verification, Validation, and Quality Assurance

The supplier shall:

- Perform quality assurance in accordance with 7.2.3.
- Perform verification and validation according to 7.2.4 and 7.2.5 to demonstrate that products and services meet their requirements.
- Provide evaluation, review, audit, and test reports to the acquirer as specified in the contract.

6.1.2.3.4.6 Acquirer and Third-Party Interfaces

The supplier shall interface with the acquirer, independent verification/validation agents, and other parties as specified in the contract and project plans. This includes:

- Coordinating contract review activities, communication, and interfaces.
- Conducting or supporting reviews (7.2.6), audits (7.2.7), acceptance testing, and joint reviews with the acquirer.

6.1.2.3.4.7 Monitoring and Control

The supplier shall monitor and control the progress and quality of the software products or services throughout the contracted life cycle. This includes:

- Monitoring technical performance, costs, schedules, and status reporting.
 - Identifying, recording, analyzing, and resolving problems.
 - Tracking subcontractor performance and ensuring compliance with contractual obligations.
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6.1.2.3.5 Product/Service Delivery and Support

This activity involves delivering and, if required, installing the product or service.

- 6.1.2.3.5.1 Deliver the software product or service in accordance with the contract.
 - 6.1.2.3.5.2 Provide the acquirer with required access to supplier and subcontractor facilities for review as specified.
 - 6.1.2.3.5.3 Install the product, if required by the agreement.
 - 6.1.2.3.5.4 Assist the acquirer with the delivered product or service as specified.
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6.1.2.3.6 Closure

This activity concludes the contract.

- 6.1.2.3.6.1 Accept and acknowledge payment or other agreed consideration.
- 6.1.2.3.6.2 Transfer responsibility for the product or service to the acquirer.
- 6.1.2.3.6.3 Address terms and authorization for initiating closure as defined in the agreement.

6.2 Organizational Project-Enabling Processes

6.2.1 Life Cycle Model Management Process

6.2.1.1 Purpose

The purpose of the Life Cycle Model Management Process is to define, maintain, and make available organizational policies, life cycle models, processes, and procedures to support the consistent and effective application of this International Standard across projects.

6.2.1.2 Outcomes

As a result of successful implementation of the Life Cycle Model Management Process:

- a) Policies and procedures for managing and deploying life cycle models and processes are defined and made available to projects.
 - b) Responsibility, accountability, and authority for life cycle model management are defined and communicated.
 - c) Life cycle processes, models, and procedures are established, maintained, and continuously improved.
 - d) Process assessments are performed to determine suitability, effectiveness, and conformance.
 - e) Prioritized process improvements are implemented based on assessment results, organizational objectives, and technological advancement.
 - f) Records and historical data are maintained to support process improvement and organizational learning.
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6.2.1.3 Activities and Tasks

The organization shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.2.1.3.1 Process Establishment

This activity establishes and maintains life cycle models, processes, and procedures for organizational use.

- **6.2.1.3.1.1** Define and document organizational responsibility, accountability, and authority for life cycle model and process management.
 - **6.2.1.3.1.2** Define and document the policies and procedures that govern the selection, tailoring, and deployment of life cycle models and processes.
 - **6.2.1.3.1.3** Ensure that established life cycle models and processes are communicated to relevant stakeholders and made available to projects.
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6.2.1.3.2 Process Assessment

This activity ensures that the organization's life cycle processes remain suitable and effective over time.

- **6.2.1.3.2.1** Develop, document, and apply a process assessment procedure, including criteria for assessing process suitability, effectiveness, and compliance.
 - **6.2.1.3.2.2** Produce and maintain process assessment records.
 - **6.2.1.3.2.3** Plan and conduct regular reviews and assessments of processes at appropriate intervals.
 - **6.2.1.3.2.4** Report assessment results to management and relevant stakeholders.
 - **6.2.1.3.2.5** Identify improvement opportunities based on assessment results and feedback from projects.
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6.2.1.3.3 Process Improvement

This activity ensures that organizational processes are improved based on assessment results and historical data.

- **6.2.1.3.3.1** Implement necessary process improvements identified through assessments and reviews.
- **6.2.1.3.3.2** Update process documentation to reflect improvements and make updated versions available to projects.
- **6.2.1.3.3.3** Collect and analyze historical, technical, and evaluation data to understand process strengths and weaknesses.
- **6.2.1.3.3.4** Use the analysis to recommend process changes, project adjustments, and technological advancements.
- **6.2.1.3.3.5** Collect, maintain, and analyze quality cost data (e.g., prevention and non-conformance costs) to inform management activities.
- **6.2.1.3.3.6** Establish the cost of preventing and resolving problems and non-conformities in software products and services to guide decision-making.

6.2.2 Infrastructure Management Process

6.2.2.1 Purpose

The purpose of the Infrastructure Management Process is to provide and maintain the enabling infrastructure and services required to support organizational and project objectives throughout their life cycle. This includes ensuring that the infrastructure supports the performance of processes effectively and efficiently.

6.2.2.2 Outcomes

As a result of successful implementation of this process:

- a) Infrastructure requirements to support organizational and project processes are defined.
- b) Infrastructure elements are identified, specified, and planned.
- c) Required infrastructure elements are acquired and implemented in accordance with plans.
- d) The infrastructure is maintained, monitored, and improved to ensure stable, reliable, and secure operation throughout the life cycle.
- e) Infrastructure performance is continuously evaluated to ensure it meets process requirements and evolving organizational needs.

Infrastructure elements may include hardware, software, methods, tools, techniques, standards, networks, and facilities supporting development, operation, maintenance, and management activities.

6.2.2.3 Activities and Tasks

The organization shall implement the following activities and tasks in accordance with applicable policies and procedures.

6.2.2.3.1 Process Implementation

This activity involves defining infrastructure requirements and planning their implementation.

- 6.2.2.3.1.1 Define and document the infrastructure required to support organizational and project processes, considering applicable procedures,

standards, tools, and techniques.

- 6.2.2.3.1.2 Plan and document the establishment of the infrastructure, including timelines, responsibilities, and integration with other organizational processes.

6.2.2.3.2 Establishment of the Infrastructure

This activity involves setting up the required infrastructure according to defined plans.

- 6.2.2.3.2.1 Plan and document the configuration of infrastructure elements, taking into account functionality, performance, safety, security, availability, space, equipment, cost, and schedule constraints.
- 6.2.2.3.2.2 Acquire, configure, and install infrastructure elements in time to support process execution and project initiation.
- 6.2.2.3.2.3 Verify the installed infrastructure to ensure it meets specified requirements before operational use.

6.2.2.3.3 Maintenance of the Infrastructure

This activity involves maintaining and improving the infrastructure over time.

- 6.2.2.3.3.1 Monitor infrastructure operation to detect failures, performance degradation, or emerging needs.
- 6.2.2.3.3.2 Maintain and modify the infrastructure as necessary to ensure continued compliance with process requirements.
- 6.2.2.3.3.3 Define and implement configuration management for relevant infrastructure elements in accordance with the Configuration Management Process (7.2.2).
- 6.2.2.3.3.4 Plan for upgrades, replacements, and decommissioning of infrastructure components to maintain effectiveness and security.

6.2.3 Project Portfolio Management Process

6.2.3.1 Purpose

The purpose of the Project Portfolio Management Process is to initiate and sustain necessary, sufficient, and suitable projects that support the organization's strategic objectives. This process ensures that projects are appropriately prioritized, resourced, authorized, and continually justified throughout their life cycle.

6.2.3.2 Outcomes

As a result of successful implementation of the Project Portfolio Management Process:

- a) Business venture opportunities, investments, or necessities are qualified, prioritized, and selected in alignment with strategic objectives.
 - b) Resources and budgets are identified and allocated to each authorized project.
 - c) Project management accountabilities and authorities are defined.
 - d) Projects are sustained that continue to meet agreement and stakeholder requirements.
 - e) Projects that no longer justify continued investment are redirected or terminated in accordance with organizational policy and agreements.
 - f) Portfolio decisions are documented and traceable to strategic criteria and investment rationale.
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6.2.3.3 Activities and Tasks

The organization shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.2.3.3.1 Project Initiation

This activity identifies, prioritizes, and establishes projects aligned with strategic objectives.

- **6.2.3.3.1.1** Identify and analyze new business opportunities, strategic

initiatives, or undertakings consistent with organizational strategy, life cycle models (6.2.1), and infrastructure capabilities (6.2.2).

- **6.2.3.3.1.2** Prioritize candidate projects using defined portfolio criteria (e.g., strategic alignment, return on investment, risk, resource capacity).
- **6.2.3.3.1.3** Select and establish projects that meet prioritization thresholds and strategic needs.
- **6.2.3.3.1.4** Define project accountabilities and authorities, including designated project managers or responsible entities.
- **6.2.3.3.1.5** Identify expected outcomes, benefits, and strategic contributions of each project.
- **6.2.3.3.1.6** Allocate required organizational resources and budgets to support approved project objectives.
- **6.2.3.3.1.7** Identify and manage multi-project interfaces, including enabling systems and common elements used across projects.
- **6.2.3.3.1.8** Specify project reporting requirements and review milestones, including technical and management reviews, consistent with Project Processes (Clause 6.3).
- **6.2.3.3.1.9** Formally authorize projects to execute approved project plans, granting the necessary authority to initiate project activities.

6.2.3.3.2 Portfolio Evaluation

This activity evaluates ongoing projects to ensure their continued strategic and economic justification.

- **6.2.3.3.2.1** Establish periodic evaluation intervals or event-driven triggers (e.g., major milestones, budget cycles, strategic realignments).
- **6.2.3.3.2.2** Evaluate project progress against defined objectives, investment criteria, system life cycle plans, and compliance with organizational directives.
- **6.2.3.3.2.3** Assess ongoing viability based on service need, implementation progress, investment benefits, and evolving strategic priorities.
- **6.2.3.3.2.4** Redirect projects that remain strategically valuable but require changes in scope, schedule, or resources.
- **6.2.3.3.2.5** Cancel or suspend projects whose disadvantages (cost, risk, strategic misalignment) outweigh continued investment benefits, where agreements permit.
- **6.2.3.3.2.6** Document evaluation decisions, including rationale for redirection, continuation, or termination, for audit and traceability.

6.2.3.3.3 Project Closure

This activity manages the orderly closure of projects.

- **6.2.3.3.3.1** Close, cancel, or suspend projects in accordance with organizational policies, agreements, and evaluation decisions.
- **6.2.3.3.3.2** Ensure that all deliverables, obligations, and documentation are completed or formally terminated.
- **6.2.3.3.3.3** Retain project records and documentation as required by organizational policy and regulatory requirements.
- **6.2.3.3.3.4** Conduct post-closure reviews to capture lessons learned and strategic insights for future portfolio decisions.
- **6.2.3.3.3.5** Formally release the project from the portfolio, updating portfolio records and investment baselines.

6.2.5 Quality Management Process

6.2.5.1 Purpose

The purpose of the Quality Management Process is to ensure that products, services, and life cycle process implementations meet organizational quality objectives and contribute to customer satisfaction.

6.2.5.2 Outcomes

As a result of successful implementation of the Quality Management Process:

- a) Quality management policies, standards, and procedures are defined, maintained, and communicated throughout the organization.
 - b) Quality objectives are defined based on business strategy and stakeholder requirements.
 - c) Responsibilities and authorities for quality management are assigned and communicated.
 - d) Quality is planned, monitored, and reviewed at organizational and project levels.
 - e) Customer satisfaction is assessed, monitored, and reported.
 - f) Corrective and preventive actions are implemented when quality objectives are not achieved, and their effectiveness is verified.
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6.2.5.3 Activities and Tasks

The organization shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.2.5.3.1 Establish Quality Management Framework

This activity establishes the foundational policies, standards, and responsibilities for quality.

- 6.2.5.3.1.1 Establish and maintain organizational quality management policies, standards, and procedures aligned with business strategy and regulatory obligations.
- 6.2.5.3.1.2 Define quality management objectives that support business

goals and stakeholder expectations.

- 6.2.5.3.1.3 Assign and communicate responsibilities and authorities for implementing quality management at both organizational and project levels.

6.2.5.3.2 Quality Planning

This activity focuses on defining quality objectives and integrating them into project execution.

- 6.2.5.3.2.1 Establish quality objectives for each project based on stakeholder requirements, risk assessments, and business goals.
- 6.2.5.3.2.2 Develop or review project quality plans to ensure alignment with organizational quality policies.
- 6.2.5.3.2.3 Ensure that project quality plans address applicable verification (7.2.4), validation (7.2.5), review (7.2.6), and audit (7.2.7) processes.

6.2.5.3.3 Monitoring and Review

This activity focuses on assessing quality performance and customer satisfaction.

- 6.2.5.3.3.1 Monitor quality performance of products, services, and processes using defined quality objectives and measurement criteria (see 6.3.2 Measurement Process).
- 6.2.5.3.3.2 Assess and report customer satisfaction through surveys, feedback mechanisms, or service-level monitoring.
- 6.2.5.3.3.3 Conduct periodic reviews of project quality plans and their implementation to ensure continued alignment with organizational goals and to detect deviations.

6.2.5.3.4 Corrective and Preventive Actions

This activity ensures that deviations from quality objectives are addressed effectively.

- 6.2.5.3.4.1 Identify and analyze root causes when quality objectives are not achieved.
- 6.2.5.3.4.2 Plan and implement corrective and/or preventive actions to address identified issues.
- 6.2.5.3.4.3 Communicate actions and results to relevant stakeholders.
- 6.2.5.3.4.4 Monitor and verify the effectiveness of corrective actions through follow-up reviews or audits.

Cross-References

This process supports and is supported by the following processes:

- **Measurement Process (6.3.2)** — for defining and monitoring quality metrics.
- **Verification Process (7.2.4)** and **Validation Process (7.2.5)** — for demonstrating achievement of quality objectives.
- **Review (7.2.6)** and **Audit (7.2.7)** Processes — for assessing compliance with quality plans and policies.

6.3 Project Processes

The **Project Processes** are concerned with establishing and managing the project. They include **Project Planning** and **Project Assessment and Control**. Together, these processes ensure that projects are properly planned, executed, monitored, and controlled to meet objectives and satisfy stakeholder requirements.

6.3.1 Project Planning Process

6.3.1.1 Purpose

The purpose of the Project Planning Process is to produce and communicate effective and workable project plans and to identify criteria and resources necessary to accomplish project tasks.

6.3.1.2 Outcomes

As a result of successful implementation of the Project Planning Process:

- a) Project scope, objectives, and constraints are defined and agreed upon.
 - b) The feasibility of achieving project goals with available resources and constraints is evaluated.
 - c) Tasks and resources are sized and estimated.
 - d) Project interfaces and interactions with other projects and organizational units are identified.
 - e) Project execution plans are developed, reviewed, and approved.
 - f) Required resources, infrastructure, and management support are committed.
 - g) The project is authorized for execution.
-

6.3.1.3 Activities and Tasks

The project manager shall implement the following activities in accordance with applicable organizational policies and procedures.

6.3.1.3.1 Project Initiation

- 6.3.1.3.1.1 Establish project requirements, including objectives, motivations, and boundaries.
 - 6.3.1.3.1.2 Ensure the availability and adequacy of required resources, management structures, and timescales.
 - 6.3.1.3.1.3 Modify project requirements as necessary, with agreement by all parties, to ensure that completion criteria are achievable.
 - 6.3.1.3.1.4 Identify relevant stakeholders and establish initial communication channels.
-

6.3.1.3.2 Project Planning

- 6.3.1.3.2.1 Develop project execution plans that describe activities, tasks, deliverables, schedules, effort estimates, resource allocations, task and responsibility assignments, risk management, quality assurance measures, cost estimates, and infrastructure provisions.
 - 6.3.1.3.2.2 Define the life cycle model to be used, referencing the **Life Cycle Model Management Process (6.2.1)**.
 - 6.3.1.3.2.3 Integrate risk management planning (see 6.3.3 Risk Management Process) into the project plan.
 - 6.3.1.3.2.4 Define project measurement objectives and select indicators (see 6.3.4 Measurement Process).
 - 6.3.1.3.2.5 Establish a configuration management plan if applicable (see 7.2.2 Software Configuration Management Process).
 - 6.3.1.3.2.6 Plan for periodic reviews and quality assurance activities (see 6.2.5 Quality Management Process and 7.2.6/7.2.7 for Reviews and Audits).
 - 6.3.1.3.2.7 Ensure alignment between project planning outputs and organizational strategy, policies, and infrastructure.
-

6.3.1.3.3 Project Activation

- 6.3.1.3.3.1 Obtain formal authorization to initiate the project.
 - 6.3.1.3.3.2 Submit resource requests and secure commitments from supporting organizational units.
 - 6.3.1.3.3.3 Implement the project execution plan and establish mechanisms for project control and reporting.
-

6.3.2 Project Assessment and Control Process

6.3.2.1 Purpose

The purpose of the Project Assessment and Control Process is to determine the status of the project, ensure that project performance is in accordance with plans, and implement corrective actions when necessary.

6.3.2.2 Outcomes

As a result of successful implementation of the Project Assessment and Control Process:

- a) Project performance is regularly assessed against plans.
 - b) Deviations from plans are identified and analyzed.
 - c) Corrective actions are determined, implemented, and communicated.
 - d) Project plans are updated as necessary to reflect changes.
 - e) Decision-makers receive accurate and timely information to enable informed management actions.
-

6.3.2.3 Activities and Tasks

The project manager shall implement the following activities.

6.3.2.3.1 Project Assessment

- 6.3.2.3.1.1 Monitor project status against plans, including cost, schedule, resources, risks, and quality.
 - 6.3.2.3.1.2 Assess progress toward achieving project objectives using defined indicators (see Measurement Process 6.3.4).
 - 6.3.2.3.1.3 Identify and document deviations from the project plan.
 - 6.3.2.3.1.4 Report status and variances to relevant stakeholders at defined intervals.
-

6.3.2.3.2 Project Control

- 6.3.2.3.2.1 Analyze the impact of identified deviations on cost, schedule, quality, and risk.
- 6.3.2.3.2.2 Define and implement corrective actions to bring performance back in line with plans.
- 6.3.2.3.2.3 Communicate corrective actions and resulting plan changes

to stakeholders.

- 6.3.2.3.2.4 Re-plan as necessary to ensure continued project feasibility and alignment with objectives.
- 6.3.2.3.2.5 Track and verify the effectiveness of corrective actions.

Cross-References

This process supports and is supported by:

- **Risk Management Process (6.3.3)** — for identifying, analyzing, and mitigating project risks.
- **Measurement Process (6.3.4)** — for defining and tracking performance indicators.
- **Quality Management Process (6.2.5)** — for ensuring alignment with quality goals.
- **Configuration Management (7.2.2)** — for controlling baselines and changes.
- **Review (7.2.6)** and **Audit (7.2.7)** — for assessing project artifacts and performance.

6.3.2 Project Assessment and Control Process

6.3.2.1 Purpose

The purpose of the Project Assessment and Control Process is to determine the status of the project and ensure that it performs according to approved plans, schedules, budgets, and technical objectives. This process includes initiating corrective and preventive actions, including replanning, when deviations occur in project or technical processes.

6.3.2.2 Outcomes

As a result of successful implementation of the Project Assessment and Control Process:

- a) Project progress is monitored and reported against established plans.
 - b) Project interfaces and interactions with other organizational units and projects are monitored and controlled.
 - c) Corrective actions are taken to address deviations from plans, and preventive measures are applied to avoid recurrence.
 - d) Project objectives are met, and objective evidence of their achievement is recorded in accordance with organizational or contractual requirements.
-

6.3.2.3 Activities and Tasks

The project manager shall implement the following activities and tasks in accordance with applicable organizational policies, procedures, and contractual requirements.

6.3.2.3.1 Project Monitoring

This activity involves continuous oversight of project execution to ensure adherence to plans.

- 6.3.2.3.1.1 Monitor overall project execution, including technical and managerial progress, using defined indicators and measurement results (see 6.3.3 Measurement Process).

- 6.3.2.3.1.2 Provide internal progress reports at defined intervals and external reports to the acquirer as specified in the contract.
 - 6.3.2.3.1.3 Monitor internal project element interfaces as well as interfaces with other relevant projects and organizational units to ensure consistency and prevent integration issues.
-

6.3.2.3.2 Project Control

This activity focuses on analyzing deviations and taking appropriate action to maintain control over project execution.

- 6.3.2.3.2.1 Investigate, analyze, and resolve problems discovered during project execution, including deviations in cost, schedule, quality, or technical performance.
 - 6.3.2.3.2.2 Document the analysis, decisions, and actions taken to resolve problems.
 - 6.3.2.3.2.3 Take corrective actions to bring the project back on plan, including adjustments to schedules, budgets, resources, or technical approaches.
 - 6.3.2.3.2.4 When corrective actions are insufficient, initiate formal replanning to update baselines, with appropriate approvals.
 - 6.3.2.3.2.5 Report project progress and control actions at agreed milestones, both internally and externally, in accordance with organizational procedures and contractual agreements.
-

6.3.2.3.3 Project Assessment

This activity focuses on evaluating project products and activities against defined objectives and plans.

- 6.3.2.3.3.1 Evaluate software products, project activities, and tasks using defined acceptance and evaluation criteria to determine compliance with project plans, quality requirements, and technical objectives.
- 6.3.2.3.3.2 Analyze the results of evaluations to detect problems, trends, or nonconformities.
- 6.3.2.3.3.3 Use assessment results to identify root causes and implement preventive measures to avoid future recurrence of identified issues.
- 6.3.2.3.3.4 Feed assessment findings into quality improvement activities (see 6.2.5 Quality Management Process).

Cross-reference: Verification (7.2.4), Validation (7.2.5), Review (7.2.6), and Audit (7.2.7) processes provide mechanisms for product and process evaluations.

6.3.2.3.4 Project Closure

This activity focuses on formal closure of the project.

- 6.3.2.3.4.1 Determine whether the project is complete by verifying that all planned software products, activities, and tasks have been completed in accordance with acceptance criteria, contractual provisions, and organizational procedures.
- 6.3.2.3.4.2 Conduct final project reporting, including status, deliverables, residual risks, and lessons learned.
- 6.3.2.3.4.3 Archive all project records, results, and objective evidence in a secure and retrievable environment, as specified by organizational policy or contract.
- 6.3.2.3.4.4 Obtain formal approval or acceptance of project closure from the acquirer or relevant authority, as required.

Cross-References

This process supports and is supported by:

- **Measurement Process (6.3.3)** — for monitoring and analysis of project performance indicators.
- **Quality Management Process (6.2.5)** — for alignment with quality objectives and corrective actions.
- **Configuration Management (7.2.2)** — for control of baselines during corrective actions and replanning.
- **Verification (7.2.4), Validation (7.2.5), Review (7.2.6), and Audit (7.2.7)** — for structured evaluation and assessment of products and processes.

6.3.3 Decision Management Process

6.3.3.1 Purpose

The purpose of the Decision Management Process is to select the most beneficial course of action when alternative solutions exist. It provides a structured, transparent approach to decision-making during the system life cycle, regardless of the decision's nature or source, to achieve specified, desirable, or optimized outcomes.

6.3.3.2 Outcomes

As a result of successful implementation of the Decision Management Process:

- a) A decision-making strategy is defined, including decision categories, prioritization, and responsible parties.
 - b) Alternative courses of action are identified and analyzed against established criteria.
 - c) A preferred course of action is selected and approved.
 - d) Decisions, supporting rationale, assumptions, and outcomes are captured, tracked, and reported to relevant stakeholders.
-

6.3.3.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.3.3.3.1 Decision Planning

This activity establishes the decision-making strategy, identifies decision needs, and determines responsible parties.

- 6.3.3.3.1.1 Define a decision-making strategy, including:
 - Decision categories (e.g., technical, managerial, contractual)
 - Prioritization schemes
 - Responsible authorities for each category.
- 6.3.3.3.1.2 Identify relevant parties and stakeholders to draw on

appropriate experience and domain knowledge.

- 6.3.3.3.1.3 Identify the circumstances and triggers that require a decision, ensuring that issues, opportunities, or changes are captured in a timely manner.
- 6.3.3.3.1.4 Record, categorize, and promptly report problems or opportunities and possible alternative actions for resolution.

NOTE: Proper categorization and logging of decision requests ensure timely routing to the appropriate decision authorities.

6.3.3.3.2 Decision Analysis

This activity evaluates alternative courses of action against defined criteria to select the most beneficial outcome.

- 6.3.3.3.2.1 Select and declare a decision-making strategy for each decision situation, including analysis methods (e.g., cost-benefit, trade-off, risk assessment).
 - 6.3.3.3.2.2 Identify desired outcomes and define measurable success criteria, including cost, schedule, performance, risk, quality, and compliance factors as appropriate.
 - 6.3.3.3.2.3 Identify and document feasible alternative courses of action.
 - 6.3.3.3.2.4 Evaluate alternatives using the selected strategy, performing trade-off analyses to determine their relative merits.
 - 6.3.3.3.2.5 Select and document the preferred course of action, including rationale, assumptions, and any constraints or dependencies.
 - 6.3.3.3.2.6 Obtain appropriate authorization or approval for the selected decision.
-

6.3.3.3.3 Decision Tracking

This activity monitors and reports decision outcomes to ensure effectiveness and support organizational learning.

- 6.3.3.3.3.1 Record and maintain a decision log, including problems, opportunities, alternatives, selected actions, rationale, and disposition, in accordance with organizational procedures or agreements.
- 6.3.3.3.3.2 Track decision implementation to verify that problems are resolved, adverse trends are reversed, and opportunities are realized.
- 6.3.3.3.3.3 Periodically evaluate the impact and effectiveness of decisions against defined success criteria.
- 6.3.3.3.3.4 Report decision outcomes and any follow-up actions to

relevant stakeholders and authorities.

- 6.3.3.3.5 Retain decision records for auditing, traceability, and organizational learning.

Cross-reference:

- Configuration Management Process (6.3.4) for controlling decisions that affect baselined items.
- Measurement Process (6.3.2) for tracking effectiveness metrics.
- Review (7.2.6) and Audit (7.2.7) processes for assessing decision implementation and compliance.

6.3.4 Risk Management Process

6.3.4.1 Purpose

The purpose of the Risk Management Process is to systematically address risks throughout the life cycle of a system, software product, or service. It applies to risks related to acquisition, development, maintenance, and operation.

6.3.4.2 Outcomes

As a result of successful implementation of the Risk Management Process:

- a) The scope of risk management to be performed is determined.
- b) Appropriate risk management strategies are defined and implemented.
- c) Risks are identified as they develop and during project conduct.
- d) Risks are analyzed, prioritized, and resources allocated accordingly.
- e) Risk measures are defined, applied, and assessed to determine changes in risk status and treatment progress.
- f) Risk treatments are applied based on priority, probability, consequence, or other defined risk thresholds, to correct or avoid impacts.
- g) The effectiveness and efficiency of the Risk Management Process are periodically evaluated and improved.

Reference: ISO/IEC 16085 provides a more detailed set of activities and tasks aligned with those described below.

6.3.4.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.3.4.3.1 Risk Management Planning

This activity establishes the framework for conducting risk management throughout the project life cycle.

- 6.3.4.3.1.1 Define organizational risk management policies and

strategies.

- 6.3.4.3.1.2 Document the Risk Management Process, including its scope and applicability.
 - 6.3.4.3.1.3 Identify responsible parties and define their roles, authorities, and responsibilities for risk management.
 - 6.3.4.3.1.4 Provide adequate resources to responsible parties to execute the process effectively.
 - 6.3.4.3.1.5 Define procedures for evaluating and improving the Risk Management Process, including capturing lessons learned for continuous improvement.
-

6.3.4.3.2 Risk Profile Management

This activity defines the context and maintains a structured representation of risks and their status.

- 6.3.4.3.2.1 Define and document the risk management context, including:
 - Stakeholder perspectives
 - Risk categories (technical, managerial, operational, contractual, etc.)
 - Objectives, assumptions, and constraints.
 - 6.3.4.3.2.2 Define risk thresholds to determine acceptable levels of risk.
 - 6.3.4.3.2.3 Establish and maintain a **risk profile**, which includes:
 - Context information
 - Identified risks with probability and consequence estimates
 - Thresholds and priorities based on stakeholder criteria
 - Treatment requests and status.
 - 6.3.4.3.2.4 Update the risk profile when the state of any risk changes.
 - 6.3.4.3.2.5 Communicate relevant parts of the risk profile to stakeholders periodically, based on their needs and roles.
-

6.3.4.3.3 Risk Analysis

This activity identifies, estimates, and evaluates risks to support prioritization and treatment planning.

- 6.3.4.3.3.1 Identify risks within the defined context, including internal and external sources, during all phases of the life cycle.
- 6.3.4.3.3.2 Estimate each risk's probability of occurrence and consequence of impact using defined methods and criteria.
- 6.3.4.3.3.3 Evaluate each risk against defined thresholds to determine whether treatment is required.

- 6.3.4.3.3.4 Define and document recommended treatment strategies for risks that exceed thresholds, along with indicators to measure their effectiveness.
 - 6.3.4.3.3.5 Categorize and prioritize risks for resource allocation based on probability, consequence, and stakeholder-defined criteria.
-

6.3.4.3.4 Risk Treatment

This activity defines and implements strategies for addressing identified risks.

- 6.3.4.3.4.1 Provide stakeholders with recommended alternative treatments through risk action requests.
 - 6.3.4.3.4.2 Implement selected risk treatment alternatives.
 - 6.3.4.3.4.3 Typical treatment strategies include:
 - Eliminating the risk
 - Reducing its probability of occurrence
 - Mitigating its consequences
 - Accepting the risk under defined conditions.
 - 6.3.4.3.4.4 High-priority risks shall be monitored continuously to determine if further treatment is required.
 - 6.3.4.3.4.5 Once a treatment is selected, manage it as a problem in accordance with the assessment and control activities in 6.3.2 Measurement Process, and track through Configuration Management (6.3.5, if present).
-

6.3.4.3.5 Risk Monitoring

This activity ensures that risk states and the risk context are continuously observed and evaluated.

- 6.3.4.3.5.1 Continuously monitor all identified risks and the risk context for changes.
 - 6.3.4.3.5.2 Re-evaluate risks whose states have changed to determine if further analysis or treatment is required.
 - 6.3.4.3.5.3 Implement and monitor measures to evaluate the effectiveness of risk treatments.
 - 6.3.4.3.5.4 Continuously monitor for new risks and emerging sources throughout the life cycle.
 - 6.3.4.3.5.5 Report risk status and trends to stakeholders and decision-makers (see 6.3.3 Decision Management).
-

6.3.4.3.6 Risk Management Process Evaluation

This activity evaluates the effectiveness and efficiency of the Risk Management Process.

- 6.3.4.3.6.1 Collect risk management data throughout the project for evaluation and lessons learned.
 - 6.3.4.3.6.2 Periodically review the process for effectiveness and efficiency, identifying areas for improvement.
 - 6.3.4.3.6.3 Update risk management procedures and policies based on lessons learned and evaluation results.
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Cross-References

This process interacts with and supports:

- **Measurement Process (6.3.2)** — for monitoring risk measures and treatment effectiveness.
- **Decision Management Process (6.3.3)** — for selecting among risk treatment alternatives.
- **Configuration Management Process (6.3.5)** — for controlling risk-related changes to baselined items.
- **Review (7.2.6)** and **Audit (7.2.7)** — for assessing risk management practices and outcomes.

6.3.5 Configuration Management Process

6.3.5.1 Purpose

The purpose of the Configuration Management Process is to establish and maintain the integrity of all identified project or process outputs (configuration items) and make them available to concerned parties throughout the system life cycle.

6.3.5.2 Outcomes

As a result of successful implementation of the Configuration Management Process:

- a) A configuration management strategy is defined.
- b) Items requiring configuration management are identified.
- c) Configuration baselines are established and maintained.
- d) Changes to configuration items are identified, evaluated, approved, implemented, and verified.
- e) The configuration of released items is controlled throughout their life cycle.
- f) The status and history of configuration items are recorded and made available for traceability, control, and audit.

NOTE: The **Software Configuration Management Process** is a specialization of this process and is included in the Software Support Process Group (see 7.2.2).

6.3.5.3 Activities and Tasks

The organization shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.3.5.3.1 Configuration Management Planning

This activity establishes the configuration management framework for the project or organizational scope.

- 6.3.5.3.1.1 Define a configuration management strategy, including

applicable standards, tools, methods, roles, responsibilities, and reporting mechanisms.

- 6.3.5.3.1.2 Identify items subject to configuration control, including system elements, software components, documentation, data, and supporting artifacts.
 - 6.3.5.3.1.3 Define configuration control procedures for baselines, change requests, audits, and status accounting.
 - 6.3.5.3.1.4 Establish configuration item naming and identification conventions, ensuring uniqueness, durability, and compliance with product sector standards.
-

6.3.5.3.2 Configuration Identification

This activity identifies and records configuration items and establishes baselines.

- 6.3.5.3.2.1 Assign unique, durable identifiers or markings to configuration items in accordance with relevant standards and conventions.
 - 6.3.5.3.2.2 Describe configuration items using standardized formats and ensure traceability to design specifications, requirements, or equivalent documented descriptions.
 - 6.3.5.3.2.3 Consolidate evolving configuration states into documented **baselines** at designated times or under defined conditions (e.g., release milestones, reviews).
 - 6.3.5.3.2.4 Record baseline data including configuration states, rationale for baseline creation, and authorization details.
 - 6.3.5.3.2.5 Maintain configuration records throughout the system life cycle and archive them according to agreements, legislation, or best practices.
-

6.3.5.3.3 Configuration Control

This activity manages changes to baselined configuration items.

- 6.3.5.3.3.1 Ensure proper identification, recording, evaluation, approval, incorporation, and verification of changes to configuration baselines.
- 6.3.5.3.3.2 Use Decision Management Process (6.3.3) to support change evaluation and selection of the most beneficial course of action.
- 6.3.5.3.3.3 Control incorporation of approved changes into the configuration, including appropriate verification and validation.
- 6.3.5.3.3.4 Maintain full traceability between change requests, approvals, and implemented baselines.
- 6.3.5.3.3.5 Protect configuration information against unauthorized changes through appropriate integrity and security controls.

NOTE: Configuration control involves managing the recording, retrieval, and consolidation of current and preceding configuration states to confirm information correctness, timeliness, integrity, and security.

6.3.5.3.4 Configuration Status Accounting and Audit

This activity provides visibility into the configuration state and verifies baseline integrity.

- 6.3.5.3.4.1 Record, maintain, and report configuration status information for each configuration item throughout its life cycle.
 - 6.3.5.3.4.2 Provide configuration status reports to concerned parties, including decision makers, auditors, and implementers.
 - 6.3.5.3.4.3 Perform configuration audits to verify that the as-built configuration conforms to the approved baseline (e.g., drawings, specifications, interfaces).
 - 6.3.5.3.4.4 Use Review (7.2.6) and Audit (7.2.7) processes to assess configuration integrity and compliance.
 - 6.3.5.3.4.5 Retain configuration audit records for traceability and continuous improvement.
-

Cross-References

- **Decision Management (6.3.3)** — for evaluating and approving configuration changes.
- **Measurement (6.3.2)** — for monitoring CM performance indicators.
- **Verification (7.2.4), Review (7.2.6), Audit (7.2.7)** — for configuration validation and control.
- **Software Configuration Management (7.2.2)** — specialization for software items.

6.3.6 Information Management Process

6.3.6.1 Purpose

The purpose of the Information Management Process is to provide relevant, timely, complete, valid, and, if required, confidential information to designated parties during and after the system life cycle.

This process involves the generation, collection, transformation, retention, retrieval, dissemination, and disposal of information in accordance with organizational policy, agreements, and legal or regulatory requirements.

6.3.6.2 Outcomes

As a result of successful implementation of the Information Management Process:

- a) Information to be managed is identified and classified.
- b) Information representations (content, semantics, format, and medium) are defined and controlled.
- c) Information is generated, collected, transformed, and maintained in accordance with organizational, legal, security, and project requirements.
- d) Information status is recorded, ensuring that it remains current, complete, and valid.
- e) Designated parties have timely access to relevant and authorized information.
- f) Information is properly archived for audit, knowledge retention, and project closure, and disposed of when no longer needed in accordance with policy.

NOTE: The Software Documentation Management Process (7.2.1) is a specialization of this process and is described in the Software Support Process Group.

6.3.6.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.3.6.3.1 Information Management Planning

This activity establishes the framework for identifying, classifying, and controlling information throughout its life cycle.

- 6.3.6.3.1.1 Identify the information to be managed, including but not limited to: technical, project, organizational, contractual, regulatory, and user information.
 - 6.3.6.3.1.2 Designate authorities and responsibilities for information origination, generation, capture, archiving, transformation, transmission, retrieval, and disposal.
 - 6.3.6.3.1.3 Define rights, obligations, and commitments regarding information retention, transmission, and access, including security and privacy considerations.
 - 6.3.6.3.1.4 Define the content, semantics, formats, and media for information representation. Information may be verbal, textual, graphical, or numerical and may be stored, processed, replicated, and transmitted using electronic, printed, magnetic, or optical media.
 - 6.3.6.3.1.5 Define information maintenance actions, including:
 - Status reviews of stored information for integrity, validity, and availability.
 - Replication or transformation to alternative media when required.
 - Retaining infrastructure to access archived media or migrating archives to new technology as media change.
 - 6.3.6.3.1.6 Identify and apply relevant domain storage, transmission, and presentation standards, ensuring compliance with policy, contractual agreements, and applicable legislation.
 - 6.3.6.3.1.7 Define retention periods and archival policies in accordance with organizational policy, legal/regulatory requirements, and contractual obligations.
-

6.3.6.3.2 Information Management Execution

This activity implements the plans for capturing, maintaining, disseminating, and disposing of information.

- 6.3.6.3.2.1 Obtain identified information items, including generating or collecting information from appropriate sources.
- 6.3.6.3.2.2 Maintain information items and storage records to ensure integrity. Records should include version identifiers, distribution lists, security classifications, and traceability metadata.
- 6.3.6.3.2.3 Ensure information is legible, accessible, and stored in facilities that protect it from damage, deterioration, and loss.

- 6.3.6.3.2.4 Retrieve and distribute information to designated parties in appropriate forms and formats, ensuring timely delivery and access controls.
 - 6.3.6.3.2.5 Provide official documentation as required by agreements, regulation, or policy (e.g., certifications, accreditations, audit reports, assessment ratings).
 - 6.3.6.3.2.6 Archive designated information for audit, knowledge retention, traceability, and project closure, in accordance with defined retention policies.
 - 6.3.6.3.2.7 Dispose of unwanted, invalid, or unverifiable information securely and in compliance with organizational policy, legal requirements, security, and privacy considerations.
-

Cross-References

- **Configuration Management Process (6.3.4)** — for maintaining baselines of controlled information.
- **Quality Management Process (6.2.5)** — for ensuring information validity and completeness as part of quality objectives.
- **Software Documentation Management Process (7.2.1)** — specialization of this process for software-related information.
- **Audit (7.2.7)** — for verification of archival and retention procedures.

6.3.7 Measurement Process

6.3.7.1 Purpose

The purpose of the Measurement Process is to collect, analyze, and use measurement data to support management and technical decision-making and to objectively demonstrate product and process quality throughout the system life cycle.

6.3.7.2 Outcomes

As a result of successful implementation of the Measurement Process:

- a) Information needs of relevant technical and management processes are identified and prioritized.
 - b) Appropriate measures are identified, defined, or developed to satisfy the information needs.
 - c) Measurement activities are planned, resourced, and conducted in accordance with organizational policies and procedures.
 - d) Data is collected, stored, analyzed, and interpreted using defined procedures.
 - e) Information products are generated and used to support decision-making and provide objective communication to stakeholders.
 - f) The effectiveness of the Measurement Process and the utility of measures are evaluated.
 - g) Potential improvements to the Measurement Process and measures are identified and communicated to the process owner.
-

6.3.7.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

NOTE 1 — ISO/IEC 15939 provides a more detailed set of activities and tasks aligned with the ones below.

NOTE 2 — Clause 8 of ISO 9001:2000 specifies Quality Management System requirements for measurement and monitoring of processes and products.

6.3.7.3.1 Measurement Planning

This activity defines how measurement will support organizational and project objectives.

- 6.3.7.3.1.1 Describe relevant organizational characteristics that influence measurement (e.g., structure, roles, information flows, project types).
 - 6.3.7.3.1.2 Identify and prioritize information needs for technical and management decision-making.
 - 6.3.7.3.1.3 Select and document measures that satisfy these information needs, including definitions, data sources, units, scales, and responsible parties.
 - 6.3.7.3.1.4 Define data collection, storage, analysis, and reporting procedures.
 - 6.3.7.3.1.5 Define criteria for evaluating the quality and utility of information products and measurement results.
 - 6.3.7.3.1.6 Review and approve measurement plans and allocate required resources.
 - 6.3.7.3.1.7 Acquire and deploy supporting technologies and tools for data management and analysis.
 - 6.3.7.3.1.8 Integrate measurement procedures into relevant organizational and project processes.
-

6.3.7.3.2 Data Collection

This activity gathers measurement data in accordance with the defined procedures.

- 6.3.7.3.2.1 Collect data from defined sources using approved methods and tools.
 - 6.3.7.3.2.2 Store data in accordance with organizational data management and security policies.
 - 6.3.7.3.2.3 Verify and validate collected data for completeness, accuracy, and consistency.
-

6.3.7.3.3 Data Analysis and Reporting

This activity transforms data into information products that support decisions.

- 6.3.7.3.3.1 Analyze collected data using defined procedures to derive meaningful information products.
- 6.3.7.3.3.2 Interpret analysis results against defined success criteria and decision thresholds.

- 6.3.7.3.3 Document and communicate results to measurement users, decision authorities, and relevant stakeholders in a clear and timely manner.
 - 6.3.7.3.3.4 Use information products to support technical and management decisions (see 6.3.3 Decision Management Process).
 - 6.3.7.3.3.5 Ensure traceability of measurement data and results for audit and review purposes.
-

6.3.7.3.4 Evaluation and Improvement

This activity assesses the effectiveness of the Measurement Process and identifies improvements.

- 6.3.7.3.4.1 Evaluate the quality, timeliness, and usefulness of information products and measurement results.
 - 6.3.7.3.4.2 Evaluate the performance and adequacy of the Measurement Process itself against defined criteria.
 - 6.3.7.3.4.3 Identify and communicate potential improvements to the process owner or designated authority.
 - 6.3.7.3.4.4 Implement approved improvements in accordance with organizational policies.
-

Cross-References

This process supports and is supported by:

- **Quality Management Process (6.2.5)** — for defining quality objectives and using measurement for quality monitoring.
- **Decision Management Process (6.3.3)** — for providing objective data to support trade-offs and selections.
- **Configuration Management (6.3.4)** — for measuring baseline integrity and change impacts.
- **Verification (7.2.4)** and **Validation (7.2.5)** — for collecting objective evidence of compliance.
- **Review (7.2.6)** and **Audit (7.2.7)** — for assessing measurement implementation and results.

6.4 Technical Processes

6.4.1 Stakeholder Requirements Definition Process

This International Standard specializes the Stakeholder Requirements Definition Process of ISO/IEC 15288. Users may claim conformance to the 15288 process where appropriate. The process can be tailored according to organizational policies, application domains, and project contexts.

6.4.1.1 Purpose

The purpose of this process is to define the requirements for a system that provides services to users and other stakeholders in a defined environment. It identifies stakeholders, their needs, and stakeholder classes throughout the system life cycle. It analyzes and transforms these needs into a common set of stakeholder requirements that:

- Express the system's intended interaction with its environment;
 - Serve as a reference for validating each resulting operational service;
- and
- Provide a basis for defining system requirements in 6.4.2.
-

6.4.1.2 Outcomes

As a result of successful implementation of the Stakeholder Requirements Definition Process:

- a) The required characteristics and context of use of services are specified.
 - b) Constraints on a system solution are defined.
 - c) Traceability of stakeholder requirements to stakeholders and their needs is achieved.
 - d) The basis for defining system requirements is described.
 - e) The basis for validating the conformance of services is defined.
 - f) A basis for negotiating and agreeing to supply a service or product is provided.
-

6.4.1.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.4.1.3.1 Stakeholder Identification

This activity identifies all relevant stakeholders and stakeholder classes.

- 6.4.1.3.1.1 Identify individual stakeholders or stakeholder classes with legitimate interests in the system throughout its life cycle.
 - 6.4.1.3.1.2 Include users, operators, supporters, developers, producers, trainers, maintainers, disposers, acquirers and suppliers, parties responsible for external interfacing entities, regulatory bodies, and members of society.
 - 6.4.1.3.1.3 Where direct communication is not practicable, select representatives or designated proxy stakeholders.
-

6.4.1.3.2 Requirements Identification

This activity elicits, analyzes, and documents stakeholder requirements and constraints.

- 6.4.1.3.2.1 Elicit stakeholder requirements using appropriate methods (e.g., interviews, workshops, surveys, models).
- 6.4.1.3.2.2 Use product quality models and quality requirements (e.g., ISO/IEC 9126-1, ISO/IEC 25030) to guide identification of quality characteristics.
- 6.4.1.3.2.3 Include societal needs, acquiring organization constraints, and user/operator capabilities and operational characteristics.
- 6.4.1.3.2.4 Cite requirement sources, including stakeholder assumptions and the importance placed on satisfying each requirement.
- 6.4.1.3.2.5 Define measures of effectiveness for key stakeholder needs to assess operational performance.
- 6.4.1.3.2.6 Identify significant human-system interaction risks across the system life cycle. Refer to ISO PAS 18152 for guidance on identifying and treating human factors issues.
- 6.4.1.3.2.7 Define unavoidable constraints on the system solution due to existing agreements, management decisions, and technical decisions.
- 6.4.1.3.2.8 Define representative sequences of activities to identify all required services for anticipated operational and support scenarios and environments, including unstated legal, regulatory, and social requirements.
- 6.4.1.3.2.9 Identify user-system interactions, considering human capabilities and limitations (physical, mental, learned skills), workplace conditions, environments, equipment, and scenarios (normal, unusual, emergency).

- 6.4.1.3.2.10 Specify health, safety, security, environmental, and other stakeholder requirements.
 - If usability is important, plan, specify, and implement usability requirements throughout the life cycle, referencing ISO 9241-11, ISO 13407, and ISO/TR 18529.
 - Address potential adverse effects on human health and safety for critical qualities.
 - Identify safety risks and specify safety requirements and functions, referencing applicable standards (e.g., IEC 61508) and accepted professional practices.
 - Identify security risks and specify security functions across physical, procedural, communications, computing, data, and emissions domains. Include access control, information protection, and denial-of-service considerations.
-

6.4.1.3.3 Requirements Evaluation

This activity analyzes the set of elicited requirements to ensure they are suitable for further system definition.

- 6.4.1.3.3.1 Analyze the complete set of stakeholder requirements to detect conflicts, omissions, incompleteness, ambiguity, inconsistency, incongruity, or unverifiability.
 - 6.4.1.3.3.2 Use appropriate evaluation techniques, such as inspections, reviews, model checks, or requirements quality checklists.
 - 6.4.1.3.3.3 Prioritize and resolve issues to produce a consistent, complete, and feasible set of stakeholder requirements.
-

6.4.1.3.4 Requirements Agreement

This activity ensures stakeholder requirements are agreed upon and accepted.

- 6.4.1.3.4.1 Resolve problems with requirements that are conflicting, impractical, or unrealizable.
 - 6.4.1.3.4.2 Feed back analyzed requirements to stakeholders to verify that needs and expectations are adequately captured and expressed.
 - 6.4.1.3.4.3 Present and obtain agreement on proposals to resolve conflicting or impractical requirements.
 - 6.4.1.3.4.4 Confirm with stakeholders that requirements are expressed correctly and fully represent their needs.
-

6.4.1.3.5 Requirements Recording

This activity records and maintains stakeholder requirements and traceability.

- 6.4.1.3.5.1 Record stakeholder requirements in an appropriate format to establish baselines, retain changes, and provide traceability to system requirements.
- 6.4.1.3.5.2 Maintain traceability between stakeholder requirements and their sources, ensuring changes in needs are identified and reflected at key decision points.
- 6.4.1.3.5.3 Use these records as a knowledge base for subsequent system development and for communicating requirement status to stakeholders.

Cross-References

- **System Requirements Analysis Process (6.4.2)** — for transforming stakeholder requirements into system requirements.
- **Validation Process (7.2.5)** — for validating that the delivered system satisfies stakeholder requirements.
- **Decision Management (6.3.3)** and **Configuration Management (6.3.4)** — for managing requirement conflicts and baselines.

6.4.2 System Requirements Analysis Process

The System Requirements Analysis Process in this International Standard is a specialization of the Requirements Analysis Process of ISO/IEC 15288. Users may consider claiming conformance to the ISO/IEC 15288 process instead.

This process **transforms stakeholder requirements** (6.4.1) into **system technical requirements** that guide System Architectural Design (6.4.3). It provides a structured basis for traceability, prioritization, and controlled evolution of requirements throughout the system life cycle.

6.4.2.1 Purpose

The purpose of the System Requirements Analysis Process is to transform defined stakeholder requirements into desired system technical requirements that guide system design.

6.4.2.2 Outcomes

As a result of successful implementation of the System Requirements Analysis Process:

- a) A defined set of system functional and non-functional requirements describing the problem to be solved is developed.
 - b) System requirements are analyzed and trade-offs are performed to optimize the preferred solution.
 - c) System requirements are analyzed for correctness and testability.
 - d) The impact of system requirements on the operating environment is understood.
 - e) System requirements are prioritized, approved, and updated as needed.
 - f) Consistency and traceability are established between system requirements and the stakeholder/customer requirements baseline.
 - g) Changes to the requirements baseline are evaluated for cost, schedule, and technical impact.
 - h) System requirements are communicated to all affected parties and baselined.
-

6.4.2.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with

applicable organizational policies and procedures for the System Requirements Analysis Process.

6.4.2.3.1 Requirements Specification

This activity defines, analyzes, and documents the system requirements.

- 6.4.2.3.1.1 Analyze the intended use of the system to identify system interfaces, operational and maintenance requirements, design constraints, and qualification requirements.
 - 6.4.2.3.1.2 Develop and document the System Requirements Specification (SyRS) using appropriate modeling and analysis techniques (e.g., use cases, scenarios, data flow diagrams, state models) as applicable.
 - 6.4.2.3.1.3 Perform trade-off analyses to optimize the preferred solution and balance stakeholder priorities, constraints, and technical feasibility.
 - 6.4.2.3.1.4 Analyze and document the impact of the system requirements on the operating environment, including interoperability, performance, and compliance considerations.
 - 6.4.2.3.1.5 Prioritize and approve system requirements based on criticality, feasibility, and stakeholder needs.
 - 6.4.2.3.1.6 Communicate approved system requirements to all affected parties and establish them as the initial baseline.
-

6.4.2.3.2 Requirements Evaluation

This activity evaluates system requirements for quality, feasibility, and alignment with acquisition needs.

- 6.4.2.3.2.1 Evaluate system requirements using the following criteria:
 - Traceability to acquisition and stakeholder needs
 - Consistency with stakeholder baseline
 - Testability and verifiability
 - Feasibility of system architectural design (6.4.3)
 - Feasibility of operation and maintenance
 - 6.4.2.3.2.2 Document the evaluation results, including identified issues, discrepancies, or risks.
 - 6.4.2.3.2.3 Resolve identified issues through clarification, refinement, or change requests to the stakeholder requirements baseline (6.4.1), as appropriate.
-

6.4.2.3.3 Requirements Baseline and Change Evaluation

This activity maintains the integrity of the requirements baseline and evaluates proposed changes.

- 6.4.2.3.3.1 Establish the system requirements baseline once approved and evaluated.
 - 6.4.2.3.3.2 Evaluate proposed updates or changes to the baseline for their impact on cost, schedule, performance, and technical feasibility.
 - 6.4.2.3.3.3 Approve or reject proposed changes in accordance with Configuration Management (6.3.4) and Decision Management (6.3.3) processes.
 - 6.4.2.3.3.4 Communicate approved changes to affected parties and maintain traceability to ensure consistency between updated system requirements and the stakeholder baseline.
-

Cross-References

This process supports and is supported by:

- **6.4.1 Stakeholder Requirements Definition Process** — for initial input requirements.
- **6.4.3 System Architectural Design Process** — for assessing feasibility and guiding design.
- **6.3.3 Decision Management Process** — for evaluating requirement trade-offs and changes.
- **6.3.4 Configuration Management Process** — for maintaining requirement baselines.

6.4.3 System Architectural Design Process

The **System Architectural Design Process** in this International Standard is a specialization of the **Architectural Design Process** of ISO/IEC 15288. Users may consider claiming conformance to the ISO/IEC 15288 process instead of this process.

6.4.3.1 Purpose

The purpose of the System Architectural Design Process is to transform the system requirements into a structured system architecture. This architecture identifies system elements (hardware, software, and manual operations), allocates requirements to these elements, and defines their internal and external interfaces. It provides a coherent framework that supports implementation, integration, verification, and validation throughout the life cycle.

6.4.3.2 Outcomes

As a result of successful implementation of the System Architectural Design Process:

- a) A system architecture design is developed that identifies system elements and meets the defined system requirements.
- b) Functional and non-functional requirements (including performance, security, safety, and environmental considerations) are addressed in the system architecture.
- c) System requirements are allocated to system elements in a structured and traceable manner.
- d) Internal and external interfaces of each system element are defined.
- e) Verification between system requirements and the system architecture is performed.
- f) Bidirectional traceability between allocated requirements and the customer's requirements baseline is established and maintained.
- g) Consistency and traceability between system requirements and the system architecture design are maintained.
- h) The system requirements, architecture design, and their relationships are baselined and communicated to all affected parties.
- i) Human factors and ergonomic principles are incorporated in the architecture, including identification and execution of human-centred design activities.

6.4.3.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.4.3.3.1 Establishing the System Architecture

This activity develops a top-level architecture and allocates requirements to system elements.

- 6.4.3.3.1.1 Develop a top-level system architecture that identifies the system elements, including hardware, software, and manual operations.
- 6.4.3.3.1.2 Allocate all system requirements among the identified system elements, ensuring completeness and avoiding ambiguity or overlap.
- 6.4.3.3.1.3 Define internal and external interfaces of each system element in the architecture, including data, control, and human interfaces.
- 6.4.3.3.1.4 Incorporate human factors and ergonomic knowledge into system design and identify required human-centred design activities.
- 6.4.3.3.1.5 Document the system architecture and allocated system requirements.
- 6.4.3.3.1.6 Baseline and communicate the system architecture and its relationship with system requirements to all affected parties.

Cross-reference: This activity builds upon the **System Requirements Analysis Process (6.4.2)** and provides the basis for **System Implementation (6.4.4)** and **Software Architectural Design (7.1.3)**.

6.4.3.3.2 Architectural Evaluation

This activity evaluates the system architecture to ensure its adequacy, consistency, and feasibility.

- 6.4.3.3.2.1 Evaluate the system architecture and the allocated requirements using defined criteria, including:
 - Bidirectional traceability to system and stakeholder requirements.
 - Consistency with system requirements.
 - Appropriateness of design standards, architectural principles, and methods applied.
- Feasibility of software, hardware, and manual elements in meeting their

allocated requirements.

- Compliance with performance, safety, security, usability, environmental, and other non-functional requirements.
 - Feasibility of system operation, integration, verification, and maintenance.
 - 6.4.3.3.2 Document evaluation results and address identified deficiencies through architectural refinement as required.
 - 6.4.3.3.3 Maintain evaluation records for traceability and audit purposes.
-

Cross-References

- **System Requirements Analysis Process (6.4.2)** — source of system requirements for architectural design.
- **System Implementation Process (6.4.4)** — uses the architecture as the basis for realization.
- **Software Architectural Design Process (7.1.3)** — derives the software architecture from system architecture.
- **Verification (7.2.4)** — checks conformance between architecture and system requirements.

6.4.4 Implementation Process

6.4.4.1 Purpose

The purpose of the Implementation Process is to realize a specified system element in accordance with its design and requirements.

NOTE: Users of this International Standard intending to implement software products or software elements of a larger system shall apply the **Software Implementation Process** (subclause 7.1.1), which is a conforming instance of the Implementation Process.

6.4.4.2 Outcomes

As a result of successful implementation of the Implementation Process:

- a) An implementation strategy is defined that aligns with system design and applicable standards.
 - b) The system element is realized in accordance with its specified design and requirements.
 - c) The implemented element is verified against its allocated requirements.
 - d) The implemented element is packaged, stored, and made available for subsequent integration, testing, or deployment, with configuration status recorded.
-

6.4.4.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.4.4.3.1 Implementation Planning

- 6.4.4.3.1.1 Define and document the implementation strategy, including applicable standards, methods, tools, environments, and responsibilities.
- 6.4.4.3.1.2 Identify system elements to be implemented, referencing System Architectural Design (6.4.3) for interfaces and dependencies.
- 6.4.4.3.1.3 Establish verification and acceptance criteria for the implemented element.

6.4.4.3.2 Implementation Execution

- 6.4.4.3.2.1 Realize the system element according to the defined strategy, applying required standards, methods, and tools.
- 6.4.4.3.2.2 Maintain traceability between implementation artifacts, requirements, and design documentation.
- 6.4.4.3.2.3 Apply configuration management controls (6.3.4) to all implementation baselines.

6.4.4.3.3 Verification and Packaging

- 6.4.4.3.3.1 Verify the implemented element against allocated requirements using appropriate methods (inspection, analysis, demonstration, or test).
- 6.4.4.3.3.2 Record verification results and handle nonconformances per organizational procedures.
- 6.4.4.3.3.3 Package, store, and deliver the verified element for integration, ensuring documentation and configuration status accounting are complete.

6.4.5 System Integration Process

6.4.5.1 Purpose

The purpose of the System Integration Process is to integrate system elements—including software items, hardware items, manual operations, and other systems—into a complete system that satisfies the system design and customer expectations expressed in the system requirements.

6.4.5.2 Outcomes

As a result of successful implementation of the System Integration Process:

- a) A strategy to integrate the system according to the priorities of the system requirements is developed.
- b) Criteria are defined to verify compliance with system requirements allocated to system elements, including interfaces between system elements.

- c) System integration is verified using the defined criteria.
 - d) A regression strategy is developed and applied for re-testing the system when changes are made.
 - e) Consistency and traceability are established between the system design and the integrated system elements.
 - f) An integrated system is constructed that demonstrates compliance with the system design and includes a complete set of usable deliverable system elements.
-

6.4.5.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.4.5.3.1 Integration Planning

This activity establishes the strategy, criteria, and sequence for system integration.

- 6.4.5.3.1.1 Define and document an integration strategy that prioritizes system requirements and establishes the integration sequence of system elements.
 - 6.4.5.3.1.2 Define integration interfaces, responsibilities, tools, and methods.
 - 6.4.5.3.1.3 Establish criteria and procedures for integration verification.
 - 6.4.5.3.1.4 Ensure consistency with System Architectural Design (6.4.3) and Configuration Management (6.3.4).
 - 6.4.5.3.1.5 Plan for regression testing when changes occur.
-

6.4.5.3.2 System Integration

This activity integrates elements and verifies the integrated aggregates.

- 6.4.5.3.2.1 Integrate software configuration items, hardware configuration items, manual operations, and other systems as necessary into the system according to the integration strategy.
- 6.4.5.3.2.2 Test integrated aggregates against their allocated requirements as integration progresses.
- 6.4.5.3.2.3 Document integration steps and test results to ensure traceability.
- 6.4.5.3.2.4 Address consistency and traceability between the system

design and the integrated system configuration.

6.4.5.3.3 Test Readiness

This activity ensures the integrated system is ready for qualification testing.

- 6.4.5.3.3.1 Develop and document tests, test cases, criteria, and procedures for each qualification requirement of the system.
- 6.4.5.3.3.2 Evaluate the readiness of the integrated system for System Qualification Testing (6.4.6) using the following criteria:
 - Test coverage of system requirements
 - Appropriateness of test methods and standards
 - Conformance to expected results
 - Feasibility of executing system qualification testing
- 6.4.5.3.3.3 Develop and maintain a regression testing strategy to support re-testing after changes.
- 6.4.5.3.3.4 Record and report test readiness evaluations to relevant stakeholders.

6.4.6 System Qualification Testing Process

The System Qualification Testing Process contributes to demonstrating that the integrated system satisfies its defined system requirements and is ready for delivery and operation.

6.4.6.1 Purpose

The purpose of the System Qualification Testing Process is to ensure that the implementation of each system requirement is tested for compliance and that the system, as integrated, is verified to be ready for delivery in its intended operational environment.

NOTE — This process focuses on qualification of the integrated system as a whole and provides objective evidence that system requirements have been met.

6.4.6.2 Outcomes

As a result of successful implementation of the System Qualification Testing Process:

- a) Criteria are developed for evaluating compliance with all applicable system requirements.
 - b) The integrated system is tested using the defined qualification criteria.
 - c) Qualification test results are recorded and maintained as verification evidence.
 - d) The readiness of the system for delivery is assured based on qualification results and supporting evidence.
-

6.4.6.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.4.6.3.1 Qualification Testing

This activity ensures that the integrated system satisfies its defined requirements

and that qualification evidence is collected and documented.

- **6.4.6.3.1.1 Document Qualification Requirements**

Document the system requirements that must be verified during qualification testing. Ensure traceability between requirements, qualification criteria, and test cases.

- **6.4.6.3.1.2 Develop Qualification Criteria**

Define qualification criteria for evaluating compliance with each requirement.

Evaluation criteria should address at least the following:

- a) Test coverage of all system requirements
- b) Conformance to expected results and performance specifications
- c) Feasibility of system operation and maintenance under intended

conditions

Qualification criteria should also address any regulatory, safety, security, or interoperability requirements applicable to the system.

- **6.4.6.3.1.3 Plan and Prepare Qualification Tests**

Develop qualification test plans, procedures, and required resources. Establish and configure the test environment to replicate operational conditions as closely as practicable.

- **6.4.6.3.1.4 Execute Qualification Tests**

Conduct tests of the integrated system according to the qualification plan. Monitor and control test execution to ensure consistency and reliability.

- **6.4.6.3.1.5 Record and Evaluate Results**

Record the results of all qualification tests, including observed behavior, measured values, anomalies, and deviations from expected results. Evaluate the system against the qualification criteria to determine compliance.

- **6.4.6.3.1.6 Perform Audits (if applicable)**

Support and document qualification-related audits in accordance with the Software Audit Process (7.2.7). If audits are conducted, record their results and ensure any non-conformities are addressed.

NOTE — This subclause does not apply to software configuration items for which audits have already been conducted.

- **6.4.6.3.1.7 Confirm Readiness for Delivery**

Based on the results of qualification testing and any related audits, update and prepare the integrated system for Software Installation (6.4.7) and Software Acceptance Support (7.2.8), if applicable. Provide qualification evidence to relevant stakeholders for approval.

Cross-References

This process is closely related to:

- **Software Verification Process (7.2.4)** — for verifying software components against their requirements, which may be incorporated into overall

system qualification.

- **Software Audit Process (7.2.7)** — for auditing qualification evidence and configuration items.
- **Software Installation Process (6.4.7)** — for subsequent installation of the qualified system into the operational environment.

6.4.7 Software Installation Process

This International Standard's Software Installation Process contributes to the outcomes of ISO/IEC 15288's Transition Process. Users may consider claiming conformance to the ISO/IEC 15288 Transition Process instead of this process where appropriate. This process addresses the activities required to plan and execute the installation of a software product in its target environment to ensure readiness for operational use.

6.4.7.1 Purpose

The purpose of the Software Installation Process is to install the software product that meets the agreed requirements in the target environment.

6.4.7.2 Outcomes

As a result of successful implementation of the Software Installation Process:

- a) A software installation strategy is developed.
 - b) Criteria for software installation are established to demonstrate compliance with installation requirements.
 - c) The software product is installed in the target environment in accordance with the plan.
 - d) The readiness of the software product for use in its intended environment is assured.
-

6.4.7.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.4.7.3.1 Installation Planning

This activity focuses on preparing for the installation, ensuring that the environment, strategy, and resources are defined and agreed.

- 6.4.7.3.1.1 Develop a **software installation strategy** in agreement with

the customer and operating organization. The strategy shall define installation objectives, responsibilities, required resources, rollback procedures, and success criteria.

- 6.4.7.3.1.2 Develop an **installation plan** for the software product, detailing the installation sequence, environment preparation, resources, timelines, roles, and dependencies, as designated in the contract or project agreements.
- 6.4.7.3.1.3 Ensure that necessary resources (hardware, software, personnel, documentation) and information are available for installation.
- 6.4.7.3.1.4 Support any **parallel running activities** or phased installations required by the contract or operational strategy.
- 6.4.7.3.1.5 Establish **rollback and recovery procedures**, including making a complete backup of the system before installation to enable reinstallation of the last working version if required.
- 6.4.7.3.1.6 Define and document **environment criteria** for installation, including configuration requirements, network dependencies, permissions, and operational constraints.
- 6.4.7.3.1.7 Specify **system adaptation requirements** to ensure the software product can be correctly configured for its intended environment.
- 6.4.7.3.1.8 Document the installation strategy and plan, including environment criteria, adaptation procedures, rollback procedures, and acceptance criteria.

Audit Evidence Storage: All installation plans, environment specifications, and rollback procedures shall be stored under /docs/audit/installation/ with digital signatures where applicable, to support traceability and compliance audits.

6.4.7.3.2 Software Installation Execution

This activity involves carrying out the installation in accordance with the approved plan and documenting the results.

- 6.4.7.3.2.1 Install the software product in the target environment in accordance with the installation plan.
- 6.4.7.3.2.2 Verify that software code and databases initialize, execute, and terminate correctly as specified in the contract or product documentation.
- 6.4.7.3.2.3 Adapt the system as needed to meet the requirements for operation in the intended environment.
- 6.4.7.3.2.4 Record installation events, including start and end times, environment details, issues encountered, deviations from the plan, and their resolution.
- 6.4.7.3.2.5 Document installation results and store them as part of the installation audit trail.
- 6.4.7.3.2.6 Confirm that the software product is ready for use in its

intended environment, meeting the defined installation success criteria.

Cross-References

This process supports and is supported by the following processes:

- **Configuration Management (7.2.2)** — for managing environment baselines and version control.
- **Verification Process (7.2.4)** — for verifying that installation meets defined criteria.
- **Software Operation Process (7.2.8)** — for transitioning the installed product into operational use.
- **Problem Resolution Process (7.2.9)** — for managing issues during installation.

6.4.8 Software Acceptance Support Process

The Software Acceptance Support Process contributes to the outcomes of the **Transition** and **Validation** Processes of ISO/IEC 15288. Organizations may claim conformance to either the relevant ISO/IEC 15288 processes or this subclause (6.4.8).

6.4.8.1 Purpose

The purpose of the Software Acceptance Support Process is to assist the acquirer in achieving confidence that the delivered software product meets its requirements and is ready for deployment in the intended operational environment.

6.4.8.2 Outcomes

As a result of successful implementation of the Software Acceptance Support Process:

- a) The software product is completed and delivered to the acquirer in accordance with contractual agreements.
 - b) The developer supports the acquirer during acceptance reviews and testing of the software product.
 - c) The software product is deployed in the customer's environment, as contractually required.
 - d) Problems detected during acceptance are identified, documented, communicated to responsible parties, and tracked to resolution.
-

6.4.8.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.4.8.3.1 Preparation for Acceptance

This activity ensures that the software product and supporting materials are ready for acceptance.

- 6.4.8.3.1.1 Deliver the software product to the acquirer in accordance with contractual requirements, including supporting documentation and training materials where applicable.
 - 6.4.8.3.1.2 Provide initial and continuing training and support to the acquirer, as required by the contract or organizational policy.
 - 6.4.8.3.1.3 If incremental delivery is specified, provide the product in completed, testable increments and coordinate acceptance activities accordingly.
 - 6.4.8.3.1.4 Coordinate with the acquirer regarding the planned acceptance environment, schedule, and resources.
-

6.4.8.3.2 Acceptance Support

This activity involves supporting the acquirer in conducting acceptance reviews and testing.

- 6.4.8.3.2.1 Support the acquirer's acceptance review and testing of the software product, providing necessary technical information and assistance.
 - 6.4.8.3.2.2 Ensure that acceptance review and testing consider:
 - Results of the **Software Review Process** (7.2.6)
 - Results of the **Software Audit Process** (7.2.7)
 - Results of **Software Qualification Testing** (7.1.7) and, where applicable, **System Qualification Testing** (6.4.6).
 - 6.4.8.3.2.3 Document acceptance review and testing results, including test data, anomalies, and any deviations from acceptance criteria.
 - 6.4.8.3.2.4 Where required by contract, support deployment of the software product in the customer's environment following acceptance, referencing the **Software Installation Process** (6.4.7).
-

6.4.8.3.3 Problem Resolution Support

This activity ensures that problems detected during acceptance are communicated and resolved effectively.

- 6.4.8.3.3.1 Identify and document problems detected during acceptance review and testing.
 - 6.4.8.3.3.2 Communicate problems to the responsible parties for analysis and resolution, following organizational problem reporting procedures (see 7.2.8 Problem Resolution Process, if applicable).
 - 6.4.8.3.3.3 Support the acquirer in verifying that problems have been resolved or dispositioned appropriately before final acceptance.
-

Cross-References

This process interacts with the following processes:

- **System Qualification Testing Process (6.4.6)** — to verify system-level compliance.
- **Software Installation Process (6.4.7)** — for deployment into the operational environment.
- **Software Review (7.2.6)** and **Software Audit (7.2.7)** — for formal verification and compliance assessments.
- **Software Qualification Testing (7.1.7)** — to provide acceptance evidence.
- **Problem Resolution Process (7.2.8)** — for tracking and resolving acceptance issues.

6.4.9 Software Operation Process

This International Standard specializes the Software Operation Process, a specialization of the **Operation Process** in ISO/IEC 15288. Users may claim conformance to either the ISO/IEC 15288 process or this subclause (6.4.9).

6.4.9.1 Purpose

The purpose of the Software Operation Process is to operate the software product in its intended environment and provide customer support throughout its operational life.

6.4.9.2 Outcomes

As a result of successful implementation of the Software Operation Process:

- a) An operation strategy is defined.
 - b) Conditions for correct software operation in its intended environment are identified and evaluated.
 - c) Software operation in its intended environment is tested and determined.
 - d) The software is operated in its intended environment under defined conditions.
 - e) Assistance and consultation are provided to customers in accordance with agreements.
-

6.4.9.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

6.4.9.3.1 Preparation for Operation

This activity establishes the plans, procedures, and environment for software operation.

- 6.4.9.3.1.1 Develop and document an operation plan that defines operational standards, responsibilities, procedures, and resources for performing operational tasks.

- 6.4.9.3.1.2 Establish procedures for receiving, recording, resolving, tracking, and providing feedback on problems. Problems shall be recorded and entered into the **Software Problem Resolution Process** (7.2.8).
 - 6.4.9.3.1.3 Establish procedures for testing the software in its operational environment prior to release for operational use. Problem reports and modification requests shall be entered into the **Software Maintenance Process** (6.4.10).
 - 6.4.9.3.1.4 Define procedures for releasing the software product for operational use once operational testing criteria are satisfied.
-

6.4.9.3.2 Operational Use

This activity executes the software in its intended environment and manages transitions between releases or systems.

- 6.4.9.3.2.1 For each software product release, perform operational testing and release the product for operational use if the specified operational criteria are met.
 - 6.4.9.3.2.2 Initialize, execute, and operate the software code and databases according to their intended purpose.
 - 6.4.9.3.2.3 Activate the software in its intended operational situation to deliver instances of service or continuous service.
 - 6.4.9.3.2.4 Where agreed, maintain continuous service capacity and quality when replacing an existing system being retired. During any specified changeover or concurrent operation period, manage the transfer of services to ensure stakeholder needs continue to be met.
 - 6.4.9.3.2.5 Terminate the system according to the operation plan when required (e.g., during replacement, decommissioning, or upgrades).
-

6.4.9.3.3 Monitoring and Evaluation

This activity ensures that operational performance is monitored and risks are identified.

- 6.4.9.3.3.1 Operate the system in its intended environment with the required resources and under specified conditions.
 - 6.4.9.3.3.2 Identify and monitor risks to product operation on an ongoing basis.
 - 6.4.9.3.3.3 Monitor operational service regularly against defined criteria, performance objectives, and service-level agreements.
-

6.4.9.3.4 Customer Support

This activity provides assistance and consultation to customers and users.

- 6.4.9.3.4.1 Provide assistance and consultation to users as requested, including training, documentation, and other support services necessary for effective product use.
- 6.4.9.3.4.2 Record and monitor user requests and actions taken in accordance with organizational procedures.
- 6.4.9.3.4.3 Forward user requests that require corrective action to the **Software Maintenance Process** (6.4.10) for resolution. Inform originators of planned and completed actions.
- 6.4.9.3.4.4 Forward identified operational problems to the **Software Problem Resolution Process** (7.2.8) for handling.
- 6.4.9.3.4.5 When a temporary workaround exists for a reported problem, communicate this to the originator. Apply permanent corrections, previously omitted functions or features, and system improvements to the operational software product through the **Software Maintenance Process** (6.4.10).

NOTE: Assistance may include training, operational documentation, or helpdesk services.

Cross-References

This process interacts with the following:

- **Software Problem Resolution Process (7.2.8)** — for tracking and resolving operational issues.
- **Software Maintenance Process (6.4.10)** — for applying fixes, enhancements, and updates.
- **Software Installation Process (6.4.7)** — for deploying software into the operational environment prior to operation.
- **Measurement Process (6.3.2)** — for tracking operational performance and service levels.

6.4.10 Software Maintenance Process

This International Standard specializes the Software Maintenance Process (aligned with the Operation & Maintenance concepts in ISO/IEC 15288). Organizations may claim conformance to the relevant 15288 processes or this subclause (6.4.10).

6.4.10.1 Purpose

The purpose of the Software Maintenance Process is to provide cost-effective support for a delivered software product throughout its operational life. Pre-delivery activities include planning for post-delivery operations and supportability; post-delivery activities include software modification and operational support (e.g., training, logistics, and service continuity).

6.4.10.2 Outcomes

As a result of successful implementation of the Software Maintenance Process:

- a) A maintenance strategy is defined for managing product modification, migration, and release cadence.
- b) Impacts of changes on the organization, operations, security, safety, data/privacy, and interfaces are identified and evaluated.
- c) Affected system/software documentation, configuration baselines, and records are updated.
- d) Modified products are developed with associated verification/validation and regression tests to ensure requirements and non-functional constraints are not compromised.
- e) Product upgrades are migrated/deployed into the customer's environment with defined rollback and recovery strategies.
- f) Modifications, deprecations, and end-of-support notices are communicated to all affected parties.
- g) Maintenance records (requests, analyses, approvals, test evidence, release notes) are captured and retained per policy.

6.4.10.3 Activities and tasks

The maintainer shall implement the following activities in accordance with applicable organizational policies and procedures.

6.4.10.3.1 Maintenance planning and process implementation

- 6.4.10.3.1.1 Develop, document, and execute plans and procedures for conducting maintenance activities and tasks, including SLAs/SLOs, triage policy, classification, environments, and tooling.
- 6.4.10.3.1.2 Establish or integrate with the **Configuration Management Process** (6.3.4) for managing modifications to baselined items (code, data,

infrastructure, docs).

- 6.4.10.3.1.3 Define criteria for acceptance, regression scope, rollback/recovery, and release documentation (release notes, advisories).
- 6.4.10.3.1.4 Define interfaces with **Problem Resolution** (7.2.8), **Verification** (7.2.4), **Validation** (7.2.5), **Review** (7.2.6), **Audit** (7.2.7), **Installation** (6.4.7), and **Operation** (6.4.9).

6.4.10.3.2 Problem and modification analysis

- 6.4.10.3.2.1 Analyze the impact of problem reports or modification requests on the organization, existing system, and interfacing systems. Classify by:
 - a) **Type**: corrective, improvement, preventive, adaptive.
 - b) **Scope**: size/complexity, cost, schedule.
 - c) **Criticality**: performance, safety, security, compliance, availability.
 - 6.4.10.3.2.2 Reproduce or verify the reported problem.
 - 6.4.10.3.2.3 Develop options for implementing the modification (including do-nothing, workaround, phased fix, full remediation).
 - 6.4.10.3.2.4 Document the request, analysis, options, recommended option, and assumptions; link to CM items and test impact.
 - 6.4.10.3.2.5 Obtain approval for the selected option per contract or **Decision Management** (6.3.3).
 - 6.4.10.3.2.6 Register and track the item in **Problem Resolution** (7.2.8).

6.4.10.3.3 Modification implementation

- 6.4.10.3.3.1 Determine which documentation, software units, data schemas, configurations, and versions require change; update traceability.
- 6.4.10.3.3.2 Enter the **Technical Processes** (6.4) to implement the change; define test & evaluation criteria for modified and unaffected areas (regression).
- 6.4.10.3.3.3 Perform development, peer reviews, static/dynamic analysis, and security checks; maintain branch/merge discipline under CM.
- 6.4.10.3.3.4 Execute verification (7.2.4) and, where appropriate, validation (7.2.5); ensure original requirements remain satisfied.
- 6.4.10.3.3.5 Record test results and evidence; update user/admin docs, runbooks, and training materials.

6.4.10.3.4 Maintenance review and acceptance

- 6.4.10.3.4.1 Conduct reviews with the authorizing organization to assess integrity, completeness, and risks of the modified system (see 7.2.6).
- 6.4.10.3.4.2 Support audits as required (7.2.7).
- 6.4.10.3.4.3 Obtain approval/sign-off for satisfactory completion per contract/quality plan.

6.4.10.3.5 Migration

- 6.4.10.3.5.1 Ensure software and data produced/modified during migration comply with this International Standard and contractual requirements.
- 6.4.10.3.5.2 Develop, document, and execute a migration plan with user involvement, covering requirements analysis, tooling, data/model conversion, dry-runs, execution, verification/validation, back-out/rollback, and future support for the old environment.
- 6.4.10.3.5.3 Notify users of migration plans, including rationale for ending support, details of the new environment, timelines, and support options.
- 6.4.10.3.5.4 Execute changeover/cutover or concurrent operation as planned; maintain service continuity and quality; coordinate with **Operation** (6.4.9).
- 6.4.10.3.5.5 Announce commencement of the scheduled migration; upon completion, archive old-environment documentation, logs, and code per retention policy.
- 6.4.10.3.5.6 Perform a post-operation review to assess migration outcomes; report results to the appropriate authorities.
- 6.4.10.3.5.7 Ensure data from the old environment remains accessible per contract and applicable data protection/audit requirements.

6.4.10.3.6 Release to operation and support handover

- 6.4.10.3.6.1 Package the release (artifacts, configs, infrastructure changes, release notes, advisories) and coordinate **Installation** (6.4.7).
- 6.4.10.3.6.2 Handover to **Operation** (6.4.9): update runbooks, monitoring thresholds/SLOs, on-call procedures, and training.
- 6.4.10.3.6.3 Communicate deprecations and end-of-support timelines; provide workarounds when applicable.

6.4.10.3.7 Records and communication

- 6.4.10.3.7.1 Maintain maintenance records: requests, analyses, approvals, test evidence, reviews/audits, release notes, deprecation/EOS notices.
- 6.4.10.3.7.2 Communicate changes and impacts to affected stakeholders and partners; update stakeholder-facing documentation and portals.

Cross-references

- **Problem Resolution** (7.2.8) — log, track, and close issues.
- **Configuration Management** (6.3.4) — control of baselined items.
- **Verification** (7.2.4), **Validation** (7.2.5), **Review** (7.2.6), **Audit** (7.2.7) — assurance activities.
- **Installation** (6.4.7) and **Operation** (6.4.9) — deployment and service continuity.
- **Measurement** (6.3.2) — maintenance KPIs (MTTR, change fail rate, rollback rate, availability).

6.4.11 Software Disposal Process

The Software Disposal Process in this International Standard is a specialization of the **Disposal Process** in ISO/IEC 15288. Users may consider claiming conformance to either the ISO/IEC 15288 process or this subclause (6.4.11).

6.4.11.1 Purpose

The purpose of the Software Disposal Process is to end the use of a system's software entity in a controlled manner, ensuring that software, data, and supporting documentation are retired responsibly while preserving organizational integrity and stakeholder trust.

NOTE: Disposal must consider organizational operations, long-term impacts, and societal obligations such as health, safety, security, privacy, and environmental protection.

6.4.11.2 Outcomes

As a result of successful implementation of the Software Disposal Process:

- a) A software disposal strategy is defined.
 - b) Disposal constraints are identified and provided as inputs to requirements for planning and execution.
 - c) System software elements are destroyed, archived, or otherwise stored in accordance with organizational and regulatory requirements.
 - d) The target environment is restored to an agreed-upon state after software removal.
 - e) Records of disposal actions, including long-term impact analysis, are created and preserved for compliance, auditing, and organizational learning.
-

6.4.11.3 Activities and Tasks

The project shall implement the following activities in accordance with applicable organizational policies and procedures.

6.4.11.3.1 Software Disposal Planning

This activity defines the strategy and prepares the plan for disposal.

- 6.4.11.3.1.1 Define and document a software disposal strategy that aligns with business, technical, and regulatory requirements.
 - 6.4.11.3.1.2 Develop and document a disposal plan that removes active support responsibilities from the operation and maintenance organizations.
 - 6.4.11.3.1.3 Include users and stakeholders in the planning activities to ensure acceptance and readiness for change.
 - 6.4.11.3.1.4 Address in the disposal plan:
 - Scheduled cessation of full or partial support.
 - Archiving of the software product, documentation, and associated logs.
 - Responsibility assignment for residual support issues.
 - Transition approach to any replacement software product.
 - Accessibility of archived copies of data.
 - 6.4.11.3.1.5 Identify disposal constraints such as health, safety, security, privacy, and environmental considerations.
-

6.4.11.3.2 Software Disposal Execution

This activity executes the disposal plan and ensures proper retirement of the software.

- 6.4.11.3.2.1 Execute the software disposal plan according to the defined strategy.
 - 6.4.11.3.2.2 Notify users of the retirement schedule, replacement or upgrade options, reasons for discontinuation, and available support options.
 - 6.4.11.3.2.3 Where a new system is introduced, conduct parallel operations as needed to ensure continuity of service. Provide user training and guidance during the transition.
 - 6.4.11.3.2.4 Upon scheduled retirement, terminate operational use and archive associated development documentation, source code, databases, and operational logs.
 - 6.4.11.3.2.5 Ensure data used by or associated with the retired software product remains accessible in accordance with data protection, legal, and audit requirements.
 - 6.4.11.3.2.6 Verify that the environment is restored to an acceptable operational state after software removal.
 - 6.4.11.3.2.7 Create and preserve records of disposal actions and assess long-term impacts on the organization, stakeholders, and environment.
-

Cross-References

This process interacts with the following:

- **Software Maintenance Process (6.4.10)** — for managing updates and changes prior to retirement.
- **Software Problem Resolution Process (7.2.8)** — for resolving issues discovered during disposal.
- **Information Management Process (6.3.5)** — for handling archiving and accessibility of records.
- **Configuration Management Process (6.3.4)** — for controlling baselines and ensuring proper archival.

7 Software-Specific Processes

7.1 Software Implementation Processes

The Software Implementation Process is a conforming instance of the **Implementation Process of ISO/IEC 15288**, specialized for implementing software products or services. Users of ISO/IEC 15288 may decide to recursively apply ISO/IEC 15288 to software elements of the system, as indicated by the asterisk (*) in the list of lower-level processes.

7.1.1 Purpose

The purpose of the Software Implementation Process is to:

- Produce a specified system element implemented as a software product or service.
 - Transform specified behavior, interfaces, and implementation constraints into actions that create a software item.
 - Ensure the software item satisfies architectural design requirements through verification and stakeholder requirements through validation.
-

7.1.2 Outcomes

As a result of successful implementation of the Software Implementation Process:

- a) An implementation strategy is defined.
 - b) Implementation technology constraints on the design are identified.
 - c) A software item is realized.
 - d) The software item is packaged and stored in accordance with agreements for its supply.
-

7.1.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

NOTE: Activities and tasks may overlap, interact, and be performed iteratively or

recursively. Ideally, an organizationally defined life cycle model is used.

7.1.3.1 Software Implementation Strategy and Planning

This activity defines the approach, life cycle model, supporting processes, and plans used to implement the software product or service.

- **7.1.3.1.1** If not stipulated in the contract, define or select a life cycle model appropriate to the project's scope, magnitude, and complexity.
- **7.1.3.1.2** Ensure the life cycle model includes clearly defined stages, purposes, and outcomes, and map the activities and tasks of the Software Implementation Process onto the model.
- **7.1.3.1.3** Document outputs in accordance with the **Software Documentation Management Process** (7.2.1).
- **7.1.3.1.4** Place outputs under the **Software Configuration Management Process** (7.2.2) and perform change control accordingly.
- **7.1.3.1.5** Document and resolve problems and non-conformances found in software products and tasks in accordance with the **Software Problem Resolution Process** (7.2.8).
- **7.1.3.1.6** Perform supporting processes as specified in the contract. Establish baselines and incorporate configuration items at appropriate times, as determined by the acquirer and supplier.
- **7.1.3.1.7** Select, tailor, and use documented, appropriate, and established standards, methods, tools, and programming languages (if not stipulated in the contract) for performing the Software Implementation Process and supporting processes.
- **7.1.3.1.8** Develop plans for conducting the Software Implementation Process activities. Plans shall specify standards, methods, tools, actions, responsibilities, and approaches for developing and qualifying all requirements, including safety and security. Separate plans may be developed if necessary. These plans shall be documented and executed.
- **7.1.3.1.9** If non-deliverable items are employed in software development or maintenance, ensure the operation and maintenance of the deliverable software product after delivery is planned and supported.

7.1.3.2 Lower-Level Software Implementation Processes

The Software Implementation Process is supported by the following lower-level processes:

- **Software Requirements Analysis Process (7.1.2)**
- **Software Architectural Design Process (7.1.3)**

- **Software Detailed Design Process (7.1.4)**
- **Software Construction Process (7.1.5)**
- **Software Integration Process (7.1.6)**
- **Software Qualification Testing Process (7.1.7)**

These may be applied **recursively** to software elements of the system if following ISO/IEC 15288.

7.1.2 Software Requirements Analysis Process

The Software Requirements Analysis Process is a lower-level process of the Software Implementation Process. Users of ISO/IEC 15288 may choose to provide this process recursively using the Requirements Analysis Process of ISO/IEC 15288.

7.1.2.1 Purpose

The purpose of the Software Requirements Analysis Process is to establish the requirements of the software elements of the system.

7.1.2.2 Outcomes

As a result of successful implementation of the Software Requirements Analysis Process:

- a) Requirements for software elements and their interfaces are defined.
 - b) Software requirements are analyzed for correctness and testability.
 - c) The impact of software requirements on the operating environment is understood.
 - d) Consistency and traceability between software and system requirements are established.
 - e) Prioritization for implementing software requirements is determined.
 - f) Software requirements are approved and updated as needed.
 - g) Changes to software requirements are evaluated for cost, schedule, and technical impact.
 - h) Software requirements are baselined and communicated to all affected parties.
-

7.1.2.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

7.1.2.3.1 Software Requirements Analysis

This activity defines and documents software requirements for each software item or configuration item.

- **7.1.2.3.1.1** Establish and document software requirements, including quality characteristics specifications.
- **7.1.2.3.1.2** Specify functional and capability requirements, performance requirements, physical characteristics, and environmental conditions for software item performance.
- **7.1.2.3.1.3** Define interfaces external to the software item.
- **7.1.2.3.1.4** Specify qualification requirements.
- **7.1.2.3.1.5** Specify safety requirements, including methods of operation, maintenance, environmental influences, and personnel injury prevention.
- **7.1.2.3.1.6** Specify security requirements, including prevention of compromise of sensitive information.
- **7.1.2.3.1.7** Specify human-factors engineering requirements, including manual operations, human–equipment interactions, personnel constraints, and areas sensitive to human error and training.
- **7.1.2.3.1.8** Specify data definition and database requirements.
- **7.1.2.3.1.9** Specify installation and acceptance requirements at the operation and maintenance sites.
- **7.1.2.3.1.10** Specify user documentation requirements.
- **7.1.2.3.1.11** Specify user operation and execution requirements.
- **7.1.2.3.1.12** Specify user maintenance requirements.
- **7.1.2.3.1.13** Determine and document the implementation priority of software requirements.
- **7.1.2.3.1.14** Record requirements and related artifacts in accordance with the **Software Documentation Management Process** (7.2.1) and place them under configuration control in accordance with the **Software Configuration Management Process** (7.2.2).

NOTE 1: Guidance for specifying quality characteristics is found in ISO/IEC 9126-1.

NOTE 2: If usability is an important requirement, recommendations for obtaining a desired level of usability are found in ISO/TR 18529, *Ergonomics of human-system interaction — Human-centred lifecycle process descriptions*. Annex E contains a process view focused on usability.

7.1.2.3.2 Requirements Evaluation and Approval

This activity evaluates the defined requirements, conducts reviews, and establishes baselines.

- **7.1.2.3.2.1** Evaluate software requirements using the following criteria and document the results:

- a) Traceability to system requirements and system design.
 - b) External consistency with system requirements.
 - c) Internal consistency.
 - d) Testability.
 - e) Feasibility of software design.
 - f) Feasibility of operation and maintenance.
 - **7.1.2.3.2.2** Conduct formal reviews of software requirements in accordance with the **Software Review Process (7.2.6)**.
 - **7.1.2.3.2.3** Upon successful evaluation and review, approve and baseline software requirements.
 - **7.1.2.3.2.4** Communicate approved baselines to all affected parties.
 - **7.1.2.3.2.5** Evaluate subsequent changes to baselined software requirements for cost, schedule, and technical impact before approval.
-

Cross-References

This process interacts with the following:

- **Software Documentation Management Process (7.2.1)** — for documenting requirements.
- **Software Configuration Management Process (7.2.2)** — for controlling requirement baselines.
- **Software Review Process (7.2.6)** — for conducting requirement reviews.
- **Software Architectural Design Process (7.1.3)** — for allocating software requirements to architecture.

7.1.3 Software Architectural Design Process

7.1.3.1 Purpose

The purpose of the Software Architectural Design Process is to provide a software design that implements the software requirements and can be verified and validated against those requirements.

7.1.3.2 Outcomes

As a result of successful implementation of the Software Architectural Design Process:

- a) A software architectural design is developed that describes the software items implementing the software requirements.
 - b) Internal and external interfaces of each software item are defined.
 - c) Consistency and traceability are established between software requirements and software design.
-

7.1.3.3 Activities and Tasks

The project shall implement the following activities in accordance with applicable organizational policies and procedures.

7.1.3.3.1 Software Architectural Design Development

This activity develops and documents the architectural structure of the software, including interfaces and databases.

- **7.1.3.3.1.1** Develop and document the software architectural design for each software item or configuration item, if identified.
- **7.1.3.3.1.2** Develop and document a top-level design for interfaces external to the software item.
- **7.1.3.3.1.3** Develop and document a top-level design for the interfaces between software components of the software item.
- **7.1.3.3.1.4** Develop and document a top-level design for the database associated with the software item.

- **7.1.3.3.1.5** Develop and document preliminary user documentation.
 - **7.1.3.3.1.6** Define and document preliminary test requirements and schedules for software integration activities.
-

7.1.3.3.2 Evaluation and Review of Architectural Design

This activity evaluates the architectural design to ensure compliance, consistency, and feasibility.

- **7.1.3.3.2.1** Evaluate the software item's architecture, interface, and database designs against the following criteria:
 - a) Traceability to the software item's requirements.
 - b) External consistency with the software item's requirements.
 - c) Internal consistency between software components.
 - d) Appropriateness of design methods and standards.
 - e) Feasibility of detailed design.
 - f) Feasibility of operation and maintenance.
 - **7.1.3.3.2.2** Conduct formal review(s) of the software architectural design in accordance with the **Software Review Process** (7.2.6).
 - **7.1.3.3.2.3** Record and control outputs and baselines through the **Software Documentation Management Process** (7.2.1) and the **Software Configuration Management Process** (7.2.2).
-

Cross-References

This process interacts with the following:

- **Software Requirements Analysis Process (7.1.2)** — for traceability and input requirements.
- **Software Detailed Design Process (7.1.4)** — for elaboration of the architecture into unit-level design.
- **Software Documentation Management Process (7.2.1)** — for documenting architectural artifacts.
- **Software Review Process (7.2.6)** — for formal reviews.

7.1.4 Software Detailed Design Process

7.1.4.1 Purpose

The purpose of the Software Detailed Design Process is to elaborate the software architectural design into a detailed design of software components and units that can be implemented, verified, and integrated. This process refines architectural elements into software units suitable for coding, testing, and integration.

7.1.4.2 Outcomes

As a result of successful implementation of the Software Detailed Design Process:

- a) A detailed design of each software component is developed, describing the software units to be built.
 - b) External interfaces of each software unit are defined.
 - c) Consistency and traceability are established between the detailed design, software requirements, and architectural design.
-

7.1.4.3 Activities and Tasks

The project shall implement the following activities in accordance with applicable organizational policies and procedures.

7.1.4.3.1 Detailed Design Development

This activity develops the detailed design of software components, interfaces, databases, documentation, and testing plans.

- **7.1.4.3.1.1** Develop a detailed design for each software component of the software item. Refine components into lower levels containing software units that can be coded, compiled, and tested. Ensure all software requirements are allocated from components to units. Document the detailed design.
- **7.1.4.3.1.2** Develop and document detailed designs for interfaces external to the software item, between software components, and between software units. The designs shall be detailed enough to permit coding without further information.

- **7.1.4.3.1.3** Develop and document a detailed design for the database associated with the software item.
 - **7.1.4.3.1.4** Update user documentation as necessary to reflect detailed design decisions.
 - **7.1.4.3.1.5** Define and document test requirements and schedules for testing software units, including stressing the software unit at its operational limits.
 - **7.1.4.3.1.6** Update the test requirements and schedule for Software Integration to reflect detailed design decisions.
 - **7.1.4.3.1.7** Record detailed design artifacts in accordance with the **Software Documentation Management Process** (7.2.1) and place them under configuration control in accordance with the **Software Configuration Management Process** (7.2.2).
-

7.1.4.3.2 Evaluation and Review of Detailed Design

This activity evaluates the detailed design to ensure its quality, feasibility, and alignment with requirements and architecture.

- **7.1.4.3.2.1** Evaluate the software design and test requirements against the following criteria and document the results:
 - a) Traceability to the software item's requirements.
 - b) External consistency with the architectural design.
 - c) Internal consistency between software components and units.
 - d) Appropriateness of design methods and standards.
 - e) Feasibility of testing.
 - f) Feasibility of operation and maintenance.
 - **7.1.4.3.2.2** Conduct formal review(s) of the software detailed design in accordance with the **Software Review Process** (7.2.6).
-

Cross-References

This process interacts with the following:

- **Software Requirements Analysis Process (7.1.2)** — for allocation and traceability of requirements to units.
- **Software Architectural Design Process (7.1.3)** — for consistency with architectural structures.
- **Software Documentation Management Process (7.2.1)** — for recording design artifacts.
- **Software Configuration Management Process (7.2.2)** — for baselining detailed design.

- **Software Review Process (7.2.6)** — for conducting detailed design reviews.

7.1.5 Software Construction Process

NOTE — The Software Construction Process in this International Standard is a lower-level process of the Software Implementation Process.

7.1.5.1 Purpose

The purpose of the Software Construction Process is to produce **executable software units** that properly reflect the software design, ensuring correctness, consistency, and readiness for integration.

7.1.5.2 Outcomes

As a result of successful implementation of the Software Construction Process:

- a) verification criteria are defined for all software units against their requirements;
 - b) software units defined by the design are produced;
 - c) consistency and traceability are established between software units, requirements, and design; and
 - d) verification of the software units against the requirements and the design is accomplished.
-

7.1.5.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures with respect to the Software Construction Process.

7.1.5.3.1 Software Construction

For each software item (or configuration item, if identified), the software construction activity consists of the following tasks:

7.1.5.3.1.1

The implementer shall develop and document the following:

- a) each software unit and database;
- b) test procedures and data for testing each software unit and database.

7.1.5.3.1.2

The implementer shall test each software unit and database to ensure that it satisfies its requirements. The test results shall be documented.

NOTE — Unit test evidence (e.g., Unit Test Reports, logs) shall be stored in /docs/ audit/construction/ or an equivalent controlled repository to maintain traceability with 7.1.5.5 (Audit and Configuration Control).

7.1.5.3.1.3

The implementer shall update the user documentation as necessary to reflect the software units produced.

7.1.5.3.1.4

The implementer shall update the test requirements and the schedule for Software Integration to reflect the constructed units.

Integration Readiness Checklist should include:

- 100 % unit test pass rate;
- All Non-Conformance Reports (NCRs) related to construction closed;
- Verification results signed by QA;
- Updated configuration baselines.

7.1.5.3.1.5

The implementer shall evaluate software code and test results considering the criteria listed below. The results of the evaluations shall be documented:

- a) traceability to the requirements and design of the software item;
 - b) external consistency with the requirements and design of the software item;
 - c) internal consistency between unit requirements;
 - d) test coverage of units;
 - e) appropriateness of coding methods and standards used;
 - f) feasibility of software integration and testing;
 - g) feasibility of operation and maintenance.
-

7.1.5.3.1.6

The implementer shall conduct **peer reviews, code inspections, and static analysis** to identify defects, security vulnerabilities, and maintainability issues early. Code quality shall conform to organizational standards (e.g., maximum cyclomatic complexity thresholds). Findings and corrective actions shall be recorded and tracked.

7.1.5.4 Deliverables

The following deliverables shall be produced during this process:

- Source code and database artifacts for all software units;
- Unit test procedures and data;
- Unit Test Reports and logs;
- Code review and static analysis reports;
- Updated user documentation (if applicable);
- Integration readiness checklist and schedule updates;
- Construction Audit Log (CAL), documenting peer review findings and QA

approvals per build.

7.1.5.5 Audit and Configuration Control

All construction-related evidence (e.g., source code, test results, review logs, CAL entries) shall be placed under configuration management. Audit evidence shall be retained for at least one release cycle or as required by applicable regulatory or compliance policy. Digital signatures or equivalent controls shall be applied to ensure authenticity and integrity of records.

7.1.6 Software Integration Process

NOTE — The Software Integration Process in this International Standard is a lower-level process of the Software Implementation Process.

Users of ISO/IEC 15288 may decide that this process is to be provided by the Integration Process of ISO/IEC 15288 in a recursive application of that standard.

7.1.6.1 Purpose

The purpose of the Software Integration Process is to **combine software units and software components** into integrated software items, consistent with the software design, and to demonstrate that the **functional and non-functional software requirements** are satisfied on an equivalent or complete operational platform.

7.1.6.2 Outcomes

As a result of successful implementation of the Software Integration Process:

- a) an integration strategy is developed for software units consistent with the software design and prioritized software requirements;
 - b) verification criteria for software items are developed to ensure compliance with the software requirements allocated to the items;
 - c) software items are verified using the defined criteria;
 - d) software items defined by the integration strategy are produced;
 - e) results of integration testing are recorded;
 - f) consistency and traceability are established between the software design and the integrated software items; and
 - g) a regression strategy is developed and applied for re-verifying software items when changes to software units (including associated requirements, design, and code) occur.
-

7.1.6.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures with respect to the Software Integration Process.

7.1.6.3.1 Software Integration

For each software item (or configuration item, if identified), the software integration activity consists of the following tasks:

7.1.6.3.1.1 Integration Planning

The implementer shall develop and document an **integration plan** describing:

- the integration strategy and sequence;
- roles and responsibilities;
- integration schedule;
- required resources, tools, and environments;
- test requirements, procedures, and data;
- acceptance criteria for each integration increment; and
- regression testing strategy.

The integration plan shall be approved and placed under configuration management.

7.1.6.3.1.2 Integration Execution

The implementer shall integrate the software units and software components in accordance with the integration plan. Testing shall be performed incrementally as aggregates are built to ensure each aggregate satisfies its allocated requirements. Upon completion of the activity, the integrated software item shall be fully assembled. Integration and test results shall be documented.

NOTE — Integration test reports, logs, and evidence should be stored in /docs/audit/integration/ or an equivalent controlled repository.

7.1.6.3.1.3 Documentation Update

The implementer shall update the user documentation as necessary to reflect the integrated software item.

7.1.6.3.1.4 Regression Testing

The implementer shall **develop and apply a regression testing strategy** to re-verify the software items when changes are made to software units, requirements,

design, or code. Regression test results shall be recorded and archived.

7.1.6.3.1.5 Evaluation of Integration Artifacts

The implementer shall evaluate the integration plan, design, code, integration test procedures and reports, test results, and user documentation considering the criteria listed below. The results of the evaluations shall be documented:

- a) traceability to the system and software requirements;
 - b) external consistency with the system and software requirements;
 - c) internal consistency of the integrated software item;
 - d) test coverage of the software item's requirements;
 - e) appropriateness of test standards and methods used;
 - f) conformance of test results to expected outcomes;
 - g) feasibility of subsequent software qualification testing;
 - h) feasibility of operation and maintenance.
-

7.1.6.3.1.6 Reviews

The implementer shall conduct reviews in accordance with subclause 7.2.6.

Examples of relevant reviews include integration readiness reviews, design reviews, and regression strategy reviews.

7.1.6.4 Deliverables

The following deliverables shall be produced during the Software Integration Process:

- Approved Integration Plan
 - Integrated software item(s) and configuration baselines
 - Integration test procedures, data, and results
 - Regression test strategy and results
 - Updated user documentation
 - Integration evaluation reports
 - Integration review records (e.g., readiness review minutes)
 - Integration Audit Log (IAL)
-

7.1.6.5 Audit and Configuration Control

All integration artifacts (integration plan, procedures, results, review records, regression strategy, etc.) shall be placed under configuration management. Audit evidence shall be retained for at least one release cycle or longer if required by organizational or regulatory policy. Digital signatures or equivalent measures shall be used to ensure evidence authenticity and integrity. Updates to configuration baselines shall be documented after each integration increment.

7.1.7 Software Qualification Testing Process

NOTE — The Software Qualification Testing Process in this International Standard is a lower-level process of the Software Implementation Process.

Users of ISO/IEC 15288 may decide that this process is to be provided by the Verification Process of ISO/IEC 15288 in a recursive application of that standard.

7.1.7.1 Purpose

The purpose of the Software Qualification Testing Process is to **confirm that the integrated software product meets its defined requirements** through formal testing conducted in a controlled environment.

7.1.7.2 Outcomes

As a result of successful implementation of the Software Qualification Testing Process:

- a) criteria for qualification testing of the integrated software are developed that demonstrate compliance with the software requirements;
- b) integrated software is verified using the defined qualification criteria;
- c) test results are recorded and stored as part of the project's audit trail; and
- d) a regression strategy is developed and applied for re-testing the integrated software when changes to software items occur.

NOTE — The regression strategy ensures that modifications to software items, requirements, design, or code trigger appropriate re-verification to maintain qualification status.

7.1.7.3 Activities and Tasks

The project shall implement the following activities in accordance with applicable organizational policies and procedures with respect to the Software Qualification Testing Process.

7.1.7.3.1 Software Qualification Testing

For each software item (or configuration item, if identified), the software

qualification testing activity consists of the following tasks:

7.1.7.3.1.0 Qualification Test Planning

The implementer shall define and document the **qualification test plan**, including:

- scope and objectives of qualification testing;
- mapping of qualification test cases to software requirements;
- responsibilities and schedule;
- test environments, tools, and data;
- acceptance criteria; and
- regression testing strategy.

The qualification test plan shall be reviewed, approved, and placed under configuration management.

7.1.7.3.1.1 Qualification Test Execution

The implementer shall conduct qualification testing in accordance with the qualification requirements and plan for the software item.

It shall be ensured that **the implementation of each software requirement is tested for compliance**. All qualification test results shall be documented and stored in /docs/audit/qualification/ or an equivalent controlled repository.

7.1.7.3.1.2 Documentation Update

The implementer shall update the user documentation as necessary to reflect the qualified software product.

7.1.7.3.1.3 Evaluation of Qualification Artifacts

The implementer shall evaluate the design, code, qualification test procedures, test results, and user documentation considering the following criteria. The results of the evaluations shall be documented:

- a) test coverage of all allocated software requirements;
- b) traceability between requirements, test cases, and results;
- c) conformance to expected results;
- d) completeness and accuracy of test procedures and data;
- e) adherence to applicable test standards and methods;
- f) readiness of the software item for subsequent system integration or acceptance

testing;
g) feasibility of operation and maintenance.

7.1.7.3.1.4 Regression Testing

The implementer shall **apply the regression testing strategy** to re-verify the integrated software whenever changes occur to software items, requirements, design, or code. Regression test results shall be recorded and stored with qualification evidence.

7.1.7.3.1.5 Audit Support

The implementer shall support audit(s) in accordance with subclause 7.2.7 (Software Audit Process).

The results of the audits shall be documented.

If both hardware and software are under concurrent development or integration, software qualification audits may be postponed until System Qualification Testing.

7.1.7.3.1.6 Handover Preparation

Upon successful completion of qualification testing and any required audits, the implementer shall prepare and update deliverables to support downstream processes such as System Integration, System Qualification Testing, Software Installation, or Software Acceptance Support, as applicable.

NOTE — The Software Qualification Testing Process may also be used to support the Software Verification Process (subclause 7.2.4) or the Software Validation Process (subclause 7.2.5).

7.1.7.4 Deliverables

The following deliverables shall be produced during the Software Qualification Testing Process:

- Qualification Test Plan
- Qualification test procedures, data, and environment specifications
- Qualification Test Reports and logs
- Regression testing strategy and results
- Updated user documentation
- Evaluation reports and traceability matrices

- Audit records (if applicable)
- Qualification Audit Log (QAL)
- Handover package for downstream system testing or acceptance activities

7.1.7.5 Audit and Configuration Control

All qualification testing artifacts shall be placed under configuration management. Audit evidence (plans, procedures, results, logs, QAL entries) shall be retained for at least one release cycle or per applicable regulatory requirements. Digital signatures or equivalent mechanisms shall be applied to preserve the integrity and authenticity of qualification evidence.

7.2 Software Support Processes

NOTE — The support processes listed under this subclause are specific to software and are labelled **Software Support Processes**. Although they play an integral role in assisting the Software Implementation Process, they may also provide services to other processes, such as **Agreement Processes, System Qualification Testing, Software Acceptance Support, Software Operation, and Software Maintenance**.

7.2.1 Software Documentation Management Process

The Software Documentation Management Process is a specialization of the **Information Management Process** from the Project Process Group in this International Standard. It governs the **planning, development, production, distribution, maintenance, and configuration management of all software-related documentation** produced throughout the software life cycle.

NOTE — ISO/IEC 15289 provides more detailed content for life cycle process information items (documentation).

7.2.1.1 Purpose

The purpose of the Software Documentation Management Process is to **develop and maintain recorded software information** produced by a process, ensuring that it is accurate, consistent, accessible, and traceable across the software life cycle.

7.2.1.2 Outcomes

As a result of successful implementation of the Software Documentation Management Process:

- a) a **documentation strategy** identifying the documentation to be produced during the life cycle of the software product or service is developed;
- b) the **standards and formats** to be applied for the development of software documentation are identified;
- c) required documentation for each process or project is identified;
- d) the **content and purpose** of all documentation is specified, reviewed, and

approved;

e) documentation is developed and made available in accordance with identified standards; and

f) documentation is maintained in accordance with defined criteria for retention, versioning, and change control.

7.2.1.3 Activities and Tasks

The project shall implement the following activities in accordance with applicable organizational policies and procedures with respect to the Software Documentation Management Process.

7.2.1.3.1 Process Implementation

This activity consists of the following task:

7.2.1.3.1.1

A **documentation plan**, identifying the documents to be produced during the life cycle of the software product, shall be developed, documented, and implemented. For each identified document, the following shall be addressed:

a) title or name;

b) purpose and content;

c) intended audience;

d) procedures and responsibilities for inputs, development, review, modification, approval, production, storage, distribution, maintenance, and configuration management;

e) schedule for intermediate and final versions.

7.2.1.3.2 Design and Development

This activity consists of the following tasks:

7.2.1.3.2.1

Each identified document shall be designed in accordance with applicable documentation standards covering **medium, content description, page numbering, figure/table placement, proprietary or security markings, packaging, accessibility, and other presentation requirements**.

7.2.1.3.2.2

Automated documentation generation tools may be used where appropriate. Input data used for document generation shall be **validated for correctness and completeness** before processing.

7.2.1.3.2.3

Prepared documents shall undergo **format, technical, and linguistic reviews** by qualified personnel prior to approval and release. Review outcomes shall be documented.

7.2.1.3.3 Production

This activity consists of the following tasks:

7.2.1.3.3.1

Documents shall be produced and distributed in accordance with the documentation plan. Production and distribution may use paper, electronic, or other media.

All documents shall include appropriate **metadata (e.g., version number, date, author, classification)**.

7.2.1.3.3.2

Master copies shall be stored according to organizational requirements for **record retention, security, maintenance, and backup**.

7.2.1.3.3.3

Controls shall be established in accordance with the **Software Configuration Management Process (subclause 7.2.2)** to ensure controlled baselining, versioning, and retrieval of documentation.

7.2.1.3.4 Maintenance

This activity consists of the following task:

7.2.1.3.4.1

When documentation is to be modified, the tasks of the **Software Maintenance Process** (subclause 6.4.10) shall be performed.

For documents under configuration management, modifications shall be:

- initiated through approved change requests,

- reviewed and re-approved according to the documentation plan, and
 - managed in accordance with the **Software Configuration Management Process (subclause 7.2.2)**.
-

7.2.1.4 Deliverables

The following deliverables shall be produced during this process:

- Documentation Plan;
 - Document design templates and standards;
 - All identified software documentation (e.g., design docs, test plans, audit logs, release notes);
 - Review records and approval logs;
 - Metadata indexes or documentation registries;
 - Configuration-managed master copies;
 - Documentation Change Logs.
-

7.2.1.5 Audit and Configuration Control

All documentation shall be placed under appropriate **configuration management**. Audit evidence (e.g., document versions, approval records, metadata logs) shall be retained in accordance with organizational policy or regulatory requirements. Digital signatures or equivalent mechanisms shall be applied to ensure integrity and authenticity. Documentation baselines shall be re-established after each approved change.

7.2.2 Software Configuration Management Process

NOTE — The Software Configuration Management Process is a specialization of the **Configuration Management Process** from the Project Process Group in this International Standard.

7.2.2.1 Purpose

The purpose of the Software Configuration Management (SCM) Process is to **establish and maintain the integrity of software items** throughout the life cycle and to make controlled versions and releases available to authorized parties in a timely and traceable manner.

7.2.2.2 Outcomes

As a result of successful implementation of the Software Configuration Management Process:

- a) a software configuration management strategy is developed;
 - b) items generated by the process or project are identified, defined, and baselined;
 - c) modifications and releases of the items are controlled;
 - d) modifications and releases are made available to affected parties;
 - e) the status of the items and modifications are recorded and reported;
 - f) the completeness and consistency of the items are ensured; and
 - g) the storage, handling, and delivery of the items are controlled.
-

7.2.2.3 Activities and Tasks

The project shall implement the following activities in accordance with applicable organizational policies and procedures with respect to the Software Configuration Management Process.

7.2.2.3.1 Process Implementation

This activity consists of the following task:

7.2.2.3.1.1

A **Software Configuration Management Plan (SCMP)** shall be developed. The plan shall describe:

- SCM activities, responsibilities, procedures, and schedule;
- organizational roles and interfaces (e.g., system CM, suppliers, subcontractors);
- configuration item (CI) identification scheme;
- change control procedures and escalation paths;
- configuration status accounting and evaluation procedures;
- release and delivery policies and tools used (e.g., version control systems, CMDB).

The plan shall be documented, approved, and implemented.

NOTE — The SCMP may be a part of the system configuration management plan.

7.2.2.3.2 Configuration Identification

This activity consists of the following task:

7.2.2.3.2.1

A **scheme shall be established for identification of software items and their versions** to be controlled. For each item and version, the following shall be identified:

- unique item identifiers;
- documentation that establishes the baseline;
- version references and labeling conventions;
- classification (e.g., safety-critical, security-sensitive, standard);
- relationships to other configuration items (e.g., parent-child, dependencies);
- associated metadata (author, date, branch, environment).

Baselines shall be defined at appropriate project milestones (e.g., development, integration, qualification, release) and approved prior to use.

7.2.2.3.3 Configuration Control

This activity consists of the following task:

7.2.2.3.3.1

The project shall implement structured **change control** procedures, including:

- identification and recording of change requests (CRs);
- impact analysis (technical, cost, schedule, risk);
- review and decision by the Change Control Board (CCB) or equivalent authority;

- approval or disapproval of the change;
- implementation of approved changes;
- verification of the implementation and regression testing as applicable;
- formal closure of the change record.

The **Software Problem Resolution Process (7.2.8)** may provide support for this activity.

7.2.2.3.4 Configuration Status Accounting

This activity consists of the following task:

7.2.2.3.4.1

The **status of configuration items, change requests, baselines, and releases** shall be recorded and reported. Status accounting shall provide timely and accurate information about:

- CI versions and their relationships;
- current baseline configurations;
- outstanding and completed changes;
- release history;
- traceability between CIs and documentation.

Configuration status records shall be accessible to authorized parties and maintained in the project's configuration management repository.

7.2.2.3.5 Configuration Evaluation

This activity consists of the following task:

7.2.2.3.5.1

Configuration evaluations shall determine and ensure:

- the **functional completeness** of software items against their requirements;

- the **physical completeness** of software items (i.e., design and code accurately reflect current technical descriptions);
- conformance to the approved baselines;
- resolution of discrepancies identified in prior evaluations or audits.

Periodic configuration audits shall be performed to maintain integrity and traceability.

7.2.2.3.6 Release Management and Delivery

This activity consists of the following task:

7.2.2.3.6.1

The **release and delivery** of software products and documentation shall be formally controlled. Specifically:

- master copies of code and documentation shall be maintained for the lifetime of the software product;
 - release packages shall be uniquely identified, documented, and traceable to their baseline;
 - safety- or security-critical items shall be handled, stored, packaged, and delivered in accordance with organizational policies;
 - digital signatures or checksums shall be applied to ensure integrity;
 - delivery records and rollback information shall be maintained.
-

7.2.2.4 Deliverables

The following deliverables shall be produced during this process:

- Software Configuration Management Plan (SCMP);
 - Configuration Item Identification Scheme;
 - Baseline records and configuration item lists;
 - Change Requests (CR) and associated impact analyses;
 - Configuration Status Reports and Logs;
 - Configuration Evaluation Reports and Audit Records;
 - Release packages and delivery records;
 - Digital signatures or integrity check artifacts;
 - SCM tool configuration and access control records.
-

7.2.2.5 Audit and Configuration Control

All SCM artifacts (plans, baselines, change records, status logs, releases) shall be placed under configuration management themselves.

Audit evidence shall be retained for at least one release cycle or per regulatory requirements.

Regular **configuration audits** shall ensure traceability, baseline integrity, and change compliance.

Access to SCM repositories shall be controlled, logged, and periodically reviewed.

7.2.3 Software Quality Assurance Process

NOTE — The Software Quality Assurance (SQA) Process provides assurance that work products and processes comply with predefined provisions, standards, and plans. It is closely coordinated with supporting processes such as Software Verification, Software Validation, Software Review, Software Audit, and Software Problem Resolution.

7.2.3.1 Purpose

The purpose of the Software Quality Assurance Process is to **provide assurance that work products and processes comply with predefined provisions and plans**, ensuring conformance to contractual, regulatory, and organizational requirements.

7.2.3.2 Outcomes

As a result of successful implementation of the Software Quality Assurance Process:

- a) a strategy for conducting quality assurance is developed;
 - b) evidence of software quality assurance is produced and maintained;
 - c) problems and/or non-conformances with requirements are identified and recorded; and
 - d) adherence of products, processes, and activities to the applicable standards, procedures, and requirements is verified.
-

7.2.3.3 Activities and Tasks

The project shall implement the following activities in accordance with applicable organizational policies and procedures with respect to the Software Quality Assurance Process.

7.2.3.3.1 Process Implementation

This activity consists of the following tasks:

7.2.3.3.1.1

A quality assurance process suited to the project shall be established. The objectives shall be to ensure that software products and processes comply with their established requirements and adhere to their established plans.

7.2.3.3.1.2

The quality assurance process shall be coordinated with the related **Software Verification (7.2.4)**, **Software Validation (7.2.5)**, **Software Review (7.2.6)**, and **Software Audit (7.2.7)** processes to ensure comprehensive quality oversight.

7.2.3.3.1.3

A **Software Quality Assurance Plan** shall be developed, documented, implemented, and maintained for the life of the project or contract. The plan shall include:

- a) quality standards, methodologies, procedures, and tools for performing the quality assurance activities (or references to official organizational documentation);
- b) procedures for contract review and coordination thereof;
- c) procedures for identification, collection, filing, maintenance, and disposition of quality records;
- d) resources, schedule, and responsibilities for conducting quality assurance activities;
- e) selected activities and tasks from supporting processes such as Software Verification (7.2.4), Software Validation (7.2.5), Software Review (7.2.6), Software Audit (7.2.7), and Software Problem Resolution (7.2.8).

7.2.3.3.1.4

Scheduled and on-going quality assurance activities and tasks shall be executed. When problems or non-conformances with contract or project requirements are detected, they shall be documented and serve as input to the **Software Problem Resolution Process (7.2.8)**. Records of these activities, their execution, problems, and resolutions shall be prepared and maintained.

7.2.3.3.1.5

Records of quality assurance activities and tasks shall be made available to the acquirer or designated authority, as specified in the contract.

7.2.3.3.1.6

It shall be assured that persons responsible for assuring compliance with the contract requirements have the organizational freedom, resources, and authority to perform objective evaluations and to initiate, effect, resolve, and verify problem resolutions.

7.2.3.3.2 Product Assurance

This activity consists of the following tasks:

7.2.3.3.2.1

It shall be assured that all plans required by the contract are documented, comply with the contract, are mutually consistent, and are being executed as required.

7.2.3.3.2.2

It shall be assured that software products and related documentation comply with the contract and adhere to the approved plans.

7.2.3.3.2.3

In preparation for delivery of the software products, it shall be assured that they have fully satisfied their contractual requirements and are acceptable to the acquirer.

7.2.3.3.3 Process Assurance

This activity consists of the following tasks:

7.2.3.3.3.1

It shall be assured that internal software engineering practices, development environments, test environments, and libraries comply with contract requirements, applicable standards, and organizational procedures.

7.2.3.3.3.2

It shall be assured that applicable prime-contract requirements are flowed down to subcontractors, and that subcontractors' software products satisfy prime-contract requirements.

7.2.3.3.3.3

It shall be assured that the acquirer and other parties are provided with the required support and cooperation in accordance with the contract, negotiations, and project plans.

7.2.3.3.3.4

It shall be assured that software product and process measurements are performed in accordance with established standards and procedures.

7.2.3.3.5

It shall be assured that staff assigned to the project possess the necessary skills and knowledge to meet the project requirements and receive required training as needed.

7.2.3.3.4 Assurance of Quality Systems

This activity consists of the following task:

7.2.3.3.4.1

Additional quality management activities may be performed in accordance with the clauses of **ISO 9001** or equivalent organizational quality management standards to ensure alignment between the project's quality assurance activities and broader organizational quality objectives.

7.2.3.4 Deliverables

The following deliverables shall be produced during this process:

- Software Quality Assurance Plan;
 - Records of QA activities and audits;
 - Non-conformance reports and problem tracking logs;
 - Quality review reports and metrics;
 - Evidence of coordination with verification, validation, review, and audit processes;
 - Qualification and training records for QA personnel.
-

7.2.3.5 Audit and Configuration Control

All SQA artifacts (plans, records, reports, logs) shall be placed under appropriate configuration management. Audit evidence shall be retained for at least one release cycle or in accordance with organizational and regulatory requirements. QA records shall be protected against unauthorized alteration or deletion, and access shall be controlled and logged.

Cross-References

This process interacts with the following:

- **Software Requirements Analysis Process (7.1.2)** — for traceability and input requirements.
- **Software Construction Process (7.1.5)** — to verify adherence to coding standards and construction plans.
- **Software Integration Process (7.1.6)** — to assure integration procedures and outcomes comply with requirements.
- **Software Qualification Testing Process (7.1.7)** — to verify testing results and ensure quality evidence.
- **Software Verification Process (7.2.4)** — for coordination of test evidence and evaluation activities.
- **Software Validation Process (7.2.5)** — to ensure QA evidence aligns with validation outcomes.
- **Software Review (7.2.6)** and **Software Audit (7.2.7)** — for formal inspections and audits.
- **Software Problem Resolution Process (7.2.8)** — for handling and tracking non-conformances.

7.2.4 Software Verification Process

NOTE — The Software Verification Process confirms that software work products and services accurately and completely reflect their specified requirements. Verification may be performed internally or by an independent organization, depending on project criticality, risk, and contractual obligations.

7.2.4.1 Purpose

The purpose of the Software Verification Process is to **confirm that each software work product and/or service of a process or project properly reflects the specified requirements**, ensuring correctness, completeness, and conformance throughout the software life cycle.

7.2.4.2 Outcomes

As a result of successful implementation of the Software Verification Process:

- a) a verification strategy is developed and implemented;
 - b) criteria for verification of all required software work products are identified;
 - c) required verification activities are performed;
 - d) defects are identified and recorded; and
 - e) results of the verification activities are made available to the customer and other involved parties.
-

7.2.4.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures with respect to the Software Verification Process.

7.2.4.3.1 Process Implementation

This activity consists of the following tasks:

7.2.4.3.1.1 Verification Strategy Determination

The need and extent of verification shall be determined based on the **criticality** of the software. Criticality may be gauged in terms of:

- a) the potential of an undetected error in a system or software requirement to cause death or personal injury, mission failure, or financial or catastrophic equipment loss or damage;
 - b) the maturity of and risks associated with the software technology to be used; and
 - c) the availability of funds and resources.
-

7.2.4.3.1.2 Verification Process Establishment

If the project warrants a verification effort, a verification process shall be established to verify the software product. This process shall cover planning, execution, evaluation, and reporting activities.

7.2.4.3.1.3 Independent Verification

If the project warrants an **independent verification effort**, a qualified organization responsible for conducting verification shall be selected. This organization shall be assured of **organizational independence and sufficient authority** to perform verification activities objectively.

7.2.4.3.1.4 Verification Planning

A **Verification Plan** shall be developed and documented. The plan shall specify:

- activities, methods, techniques, and tools for performing verification tasks (as defined in 7.2.4.3.2);
 - the life cycle activities and software products to be verified;
 - responsibilities, resources, schedule, and verification levels;
 - acceptance and completion criteria for each verification activity;
 - reporting mechanisms and escalation paths.
-

7.2.4.3.1.5 Verification Implementation and Reporting

The verification plan shall be implemented as documented. Problems and non-conformances detected by the verification effort shall be entered into the **Software Problem Resolution Process (7.2.8)**. All identified issues shall be tracked to closure.

Results of the verification activities shall be documented and made available to the

acquirer and other involved organizations.

7.2.4.3.2 Verification

This activity consists of the following tasks:

7.2.4.3.2.1 Requirements Verification

The requirements shall be verified considering the following criteria:

- a) system requirements are consistent, feasible, and testable;
 - b) system requirements have been appropriately allocated to hardware items, software items, and manual operations according to design criteria;
 - c) software requirements are consistent, feasible, testable, and accurately reflect system requirements;
 - d) software requirements related to safety, security, and criticality are correct, as demonstrated by suitably rigorous methods.
-

7.2.4.3.2.2 Design Verification

The design shall be verified considering the following criteria:

- a) the design is correct, consistent with, and traceable to requirements;
 - b) the design implements the proper sequence of events, inputs, outputs, interfaces, logic flow, timing and sizing budgets, and error definition, isolation, and recovery;
 - c) the selected design can be derived from requirements;
 - d) the design implements safety, security, and other critical requirements correctly, as shown by suitably rigorous methods.
-

7.2.4.3.2.3 Code Verification

The code shall be verified considering the following criteria:

- a) the code is traceable to design and requirements, testable, correct, and compliant with requirements and coding standards;
- b) the code implements the proper event sequence, consistent interfaces, correct data and control flow, completeness, appropriate allocation, timing and sizing budgets, and error definition, isolation, and recovery;

- c) the selected code can be derived from design or requirements;
 - d) the code implements safety, security, and other critical requirements correctly, as shown by suitably rigorous methods.
-

7.2.4.3.2.4 Integration Verification

The integration shall be verified considering the following criteria:

- a) software components and units of each software item have been completely and correctly integrated into the software item;
 - b) hardware items, software items, and manual operations of the system have been completely and correctly integrated into the system;
 - c) integration tasks have been performed in accordance with the integration plan.
-

7.2.4.3.2.5 Documentation Verification

The documentation shall be verified considering the following criteria:

- a) documentation is adequate, complete, and consistent;
 - b) documentation preparation is timely;
 - c) configuration management of documents follows specified procedures.
-

7.2.4.4 Deliverables

The following deliverables shall be produced during this process:

- Software Verification Plan;
 - Verification procedures, checklists, and tools;
 - Requirements, design, code, integration, and documentation verification reports;
 - Verification logs and defect reports;
 - Independent verification reports (if applicable);
 - Traceability matrices linking requirements, design, code, and tests;
 - Verification summary report for acquirer or oversight bodies.
-

7.2.4.5 Audit and Configuration Control

All verification plans, reports, logs, and related artifacts shall be placed under appropriate configuration management.

Audit evidence shall be retained in accordance with organizational and regulatory policies.

Verification evidence shall be digitally signed or otherwise protected to ensure integrity.

Verification baselines shall be updated to reflect resolved defects and approved changes.

Cross-References

This process interacts with the following:

- **Software Requirements Analysis Process (7.1.2)** — provides input requirements to be verified.
- **Software Architectural Design (7.1.3)** and **Detailed Design (7.1.4)** — provide design artifacts for verification.
- **Software Construction (7.1.5)** — provides code and unit artifacts for code verification.
- **Software Integration (7.1.6)** — provides integration results for verification of assembled components.
- **Software Qualification Testing (7.1.7)** — consumes verification results to plan qualification tests.
- **Software Quality Assurance (7.2.3)** — coordinates verification activities for compliance and reporting.
- **Software Validation (7.2.5)** — uses verification evidence to support validation efforts.
- **Software Review (7.2.6)** and **Software Audit (7.2.7)** — review verification outputs.
- **Software Problem Resolution (7.2.8)** — receives defects and non-conformances found during verification for resolution tracking.

7.2.5 Software Validation Process

NOTE — The Software Validation Process confirms that software products fulfill their intended use in the target operational environment. Validation may involve testing, analysis, simulation, modeling, and operational evaluation. Validation is typically performed after verification is complete, but may be conducted concurrently for incremental or iterative life cycles.

7.2.5.1 Purpose

The purpose of the Software Validation Process is to **confirm that the requirements for a specific intended use of the software work product are fulfilled**, demonstrating that the software meets user needs and intended operational scenarios.

7.2.5.2 Outcomes

As a result of successful implementation of the Software Validation Process:

- a) a validation strategy is developed and implemented;
 - b) criteria for validation of all required work products are identified;
 - c) required validation activities are performed;
 - d) problems are identified and recorded;
 - e) evidence is provided that the software work products, as developed, are suitable for their intended use; and
 - f) results of the validation activities are made available to the customer and other involved parties.
-

7.2.5.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures with respect to the Software Validation Process.

7.2.5.3.1 Process Implementation

This activity consists of the following tasks:

7.2.5.3.1.1 Validation Need and Independence Determination

A determination shall be made whether the project warrants a validation effort and the degree of **organizational independence** required for that effort, based on criticality, risk, and contractual or regulatory requirements.

7.2.5.3.1.2 Validation Process Establishment

If the project warrants a validation effort, a validation process shall be established to validate the system or software product. Validation tasks defined in 7.2.5.3.2, including associated methods, techniques, and tools, shall be selected based on the target environment and intended use.

7.2.5.3.1.3 Independent Validation Organization

If the project warrants an independent validation effort, a qualified organization responsible for conducting the effort shall be selected. This organization shall be assured of the **independence and authority** to perform the validation tasks objectively.

7.2.5.3.1.4 Validation Planning

A **Validation Plan** shall be developed and documented. The plan shall include, but is not limited to, the following:

- a) items subject to validation;
 - b) validation tasks to be performed;
 - c) resources, responsibilities, and schedule for validation;
 - d) procedures for forwarding validation reports to the acquirer and other involved parties;
 - e) acceptance criteria and success metrics for each validation activity;
 - f) test environments, tools, and data required to conduct validation.
-

7.2.5.3.1.5 Validation Execution and Reporting

The validation plan shall be implemented. Problems and non-conformances detected by the validation effort shall be entered into the **Software Problem Resolution Process (7.2.8)**. All problems and non-conformances shall be

resolved. Results of the validation activities shall be documented and made available to the acquirer and other involved organizations.

7.2.5.3.2 Validation

NOTE — Other means besides testing (such as analysis, modeling, simulation, inspections, or operational demonstrations) may be employed for validation where appropriate.

This activity consists of the following tasks:

7.2.5.3.2.1 Test Preparation

Prepare selected test requirements, test cases, and test specifications for analyzing validation results. These shall be based on intended use cases, operational scenarios, and end-user needs.

7.2.5.3.2.2 Test Case Alignment with Intended Use

Ensure that validation test requirements, test cases, and test specifications **reflect the particular requirements for the specific intended use** of the software product, including realistic operational environments and workflows.

7.2.5.3.2.3 Validation Test Execution

Conduct the validation tests prepared under 7.2.5.3.2.1 and 7.2.5.3.2.2, including:

- a) testing with **stress, boundary, and singular inputs**;
 - b) testing the software product for its ability to **isolate and minimize the effect of errors**, including graceful degradation under stress, boundary, or singular conditions, and appropriate operator assistance requests;
 - c) testing that **representative users** can successfully achieve their intended tasks using the software product under realistic operational conditions.
-

7.2.5.3.2.4 Validation of Intended Use

Validate that the software product satisfies its intended use by evaluating test outcomes, operational scenarios, and user feedback against validation criteria and

acceptance metrics.

7.2.5.3.2.5 Target Environment Validation

Where appropriate, test the software product in **selected areas of the target environment** to confirm its suitability for deployment and operation in real or representative conditions.

7.2.5.4 Deliverables

The following deliverables shall be produced during this process:

- Software Validation Plan;
 - Validation test requirements, test cases, and specifications;
 - Validation test procedures and reports;
 - Records of problems, non-conformances, and their resolution;
 - Validation summary report and acceptance evidence;
 - Operational test results and user evaluation feedback;
 - Independent validation reports (if applicable).
-

7.2.5.5 Audit and Configuration Control

All validation plans, test procedures, results, and reports shall be placed under appropriate **configuration management**.

Audit evidence shall be retained in accordance with organizational and regulatory policies.

Validation evidence shall be protected to ensure integrity, and baselines shall be updated upon resolution of issues and completion of validation activities.

Cross-References

This process interacts with the following:

- **Software Requirements Analysis (7.1.2)** — provides requirements and intended use cases for validation.
- **Software Architectural Design (7.1.3), Detailed Design (7.1.4), and Construction (7.1.5)** — provide work products subject to validation.
- **Software Integration (7.1.6) and Software Qualification Testing (7.1.7)** — provide integrated products and test evidence for validation.

- **Software Quality Assurance (7.2.3)** — coordinates validation activities for compliance with plans.
- **Software Verification (7.2.4)** — provides verification evidence used to support validation planning.
- **Software Review (7.2.6)** and **Software Audit (7.2.7)** — provide additional oversight and independent evaluations.
- **Software Problem Resolution (7.2.8)** — receives problems and non-conformances found during validation for resolution tracking.

7.2.6 Software Review Process

NOTE — Software reviews are held at both **project management** and **technical levels** throughout the software life cycle. Their purpose is to maintain a common understanding among stakeholders of project status, risks, and objectives, and to determine actions necessary to ensure that the software product satisfies stakeholder expectations.

7.2.6.1 Purpose

The purpose of the Software Review Process is to **maintain a common understanding with the stakeholders of the progress against the objectives of the agreement** and to determine what should be done to help ensure development of a product that satisfies the stakeholders. Software reviews provide structured opportunities for early detection of issues and collaborative decision-making.

7.2.6.2 Outcomes

As a result of successful implementation of the Software Review Process:

- a) management and technical reviews are held based on the needs of the project;
 - b) the status and products of an activity or process are evaluated through review activities;
 - c) review results are made known to all affected parties;
 - d) action items resulting from reviews are tracked to closure; and
 - e) risks and problems are identified and recorded.
-

7.2.6.3 Activities and Tasks

The project shall implement the following activities in accordance with applicable organizational policies and procedures with respect to the Software Review Process.

7.2.6.3.1 Process Implementation

This activity consists of the following tasks:

7.2.6.3.1.1 Review Scheduling

Periodic reviews shall be held at **predetermined milestones** as specified in the project plan(s). Stakeholders may also convene **ad hoc reviews** as necessary.

7.2.6.3.1.2 Resource Provisioning

All resources required to conduct reviews shall be provided. These resources include personnel, locations, facilities, hardware, software, and tools needed to support effective review sessions.

7.2.6.3.1.3 Review Preparation

The participating parties should agree on the following for each review:

- meeting agenda;
 - software products (work products/results) and problems to be reviewed;
 - scope and procedures;
 - **entry and exit criteria** for the review.
-

7.2.6.3.1.4 Problem Recording

Problems detected during reviews shall be **recorded and entered into the Software Problem Resolution Process (7.2.8)** for tracking and resolution.

7.2.6.3.1.5 Review Results Documentation

The results of the review shall be documented and distributed to all relevant parties. This includes formal review outcomes such as **approval, disapproval, or contingent approval**.

7.2.6.3.1.6 Action Tracking and Closure

Participating parties shall agree on the **outcome of the review**, assign action item responsibilities, and define **closure criteria**. Action items shall be tracked to completion.

7.2.6.3.2 Project Management Reviews

This activity consists of the following task:

7.2.6.3.2.1 Project Status Evaluation

Project status shall be evaluated relative to the applicable project plans, schedules, standards, and guidelines. The outcome of the review should be considered by appropriate management and may include:

- a) ensuring activities progress according to plan, based on an evaluation of activity or product status;
 - b) maintaining overall control of the project through adequate allocation of resources;
 - c) changing project direction or determining the need for alternate planning;
 - d) evaluating and managing risk issues that may jeopardize the success of the project.
-

7.2.6.3.3 Technical Reviews

This activity consists of the following task:

7.2.6.3.3.1 Product and Process Evaluation

Technical reviews shall be held to evaluate the software products or services under consideration and provide evidence that:

- a) they are **complete**;
 - b) they comply with their **standards and specifications**;
 - c) changes are properly implemented and affect only those areas identified by the **Software Configuration Management Process (7.2.2)**;
 - d) they adhere to applicable schedules;
 - e) they are **ready for the next planned activity**; and
 - f) the development, operation, or maintenance is being conducted according to the plans, schedules, standards, and guidelines of the project.
-

7.2.6.4 Deliverables

The following deliverables shall be produced during this process:

- Review schedule and agenda;
- Review checklists and criteria;

- Meeting minutes and attendance records;
 - Problem and action item logs;
 - Review results documentation (approval, disapproval, contingent approval);
 - Risk and issue records identified during reviews.
-

7.2.6.5 Audit and Configuration Control

All review records, minutes, action item logs, and review reports shall be placed under **configuration management**.

Audit evidence shall be retained according to organizational and regulatory policies. Review artifacts shall be versioned and linked to corresponding baselines and work products to maintain traceability between reviews and project milestones.

Cross-References

This process interacts with the following:

- **Software Requirements Analysis (7.1.2)** — reviewed to ensure correctness and stakeholder alignment.
- **Software Architectural Design (7.1.3)** and **Detailed Design (7.1.4)** — reviewed at design milestones for completeness and compliance.
- **Software Construction (7.1.5), Integration (7.1.6), and Qualification Testing (7.1.7)** — reviewed to assess development and integration progress.
- **Software Configuration Management (7.2.2)** — provides baseline references and ensures controlled change review.
- **Software Quality Assurance (7.2.3)** — coordinates reviews to ensure quality oversight.
- **Software Verification (7.2.4)** and **Validation (7.2.5)** — provide evidence and work products subject to review.
- **Software Audit (7.2.7)** — may follow reviews to independently assess compliance.
- **Software Problem Resolution (7.2.8)** — receives issues and action items generated by reviews.

7.2.7 Software Audit Process

NOTE — The Software Audit Process provides **independent determination** of the degree to which software products and processes comply with specified requirements, plans, and agreements. It supports objective assessments, identifies non-conformances, and ensures follow-up actions are taken and verified.

7.2.7.1 Purpose

The purpose of the Software Audit Process is to **independently determine compliance of selected products and processes with the applicable requirements, plans, and agreements**, as appropriate.

7.2.7.2 Outcomes

As a result of successful implementation of the Software Audit Process:

- a) an audit strategy is developed and implemented;
 - b) compliance of selected software work products and/or services or processes with requirements, plans, and agreements is determined according to the audit strategy;
 - c) audits are conducted by an appropriate independent party; and
 - d) problems detected during an audit are identified and communicated to those responsible for corrective action and resolution.
-

7.2.7.3 Activities and Tasks

The project shall implement the following activities in accordance with applicable organizational policies and procedures with respect to the Software Audit Process.

7.2.7.3.1 Process Implementation

This activity consists of the following tasks:

7.2.7.3.1.1 Audit Scheduling

Audits shall be scheduled at **predetermined milestones**, as specified in project or quality assurance plans. Additional audits may be conducted when significant changes occur or at the request of stakeholders.

7.2.7.3.1.2 Auditor Independence

Auditing personnel shall not have any **direct responsibility** for the software products and activities they audit. They shall have sufficient authority to access necessary information and raise findings without conflicts of interest.

7.2.7.3.1.3 Audit Resources

All resources required to conduct the audits shall be agreed upon by the parties. These resources may include support personnel, locations, facilities, hardware, software, tools, and access to documentation and repositories.

7.2.7.3.1.4 Audit Preparation and Agreement

Prior to each audit, the parties shall agree on the following:

- agenda;
 - software products and/or process outputs to be reviewed;
 - audit scope and procedures;
 - applicable standards and checklists;
 - **entry and exit criteria** for the audit.
-

7.2.7.3.1.5 Problem Recording and Resolution

Problems detected during the audits shall be recorded and entered into the **Software Problem Resolution Process (7.2.8)** for tracking and resolution.

7.2.7.3.1.6 Audit Reporting

After completing an audit, the audit results shall be **documented and provided to the audited party**. The audited party shall acknowledge to the auditing party any problems found and describe planned resolutions.

7.2.7.3.1.7 Audit Closure

The parties shall agree on the **outcomes of the audit**, responsibilities for action items, and **closure criteria** for findings. Closure of audit findings shall be verified and documented.

7.2.7.3.2 Software Audit

This activity consists of the following task:

7.2.7.3.2.1

Software audits shall be conducted to ensure that:

- a) as coded, software products (such as software items) reflect the approved design documentation;
 - b) acceptance review and testing requirements prescribed by the documentation are adequate for software product acceptance;
 - c) test data comply with specifications;
 - d) software products were successfully tested and meet their specifications;
 - e) test reports are correct and discrepancies between actual and expected results have been resolved;
 - f) user documentation complies with applicable standards;
 - g) activities have been conducted according to applicable requirements, plans, and contracts;
 - h) costs and schedules adhere to the established plans.
-

7.2.7.4 Deliverables

The following deliverables shall be produced during this process:

- Software Audit Strategy or Plan (if separate);
 - Audit schedules and agendas;
 - Audit checklists and criteria;
 - Audit reports and findings logs;
 - Records of non-conformances entered into the Problem Resolution system;
 - Audit closure records and evidence of corrective actions;
 - Independent auditor statements or certifications (if applicable).
-

7.2.7.5 Audit and Configuration Control

All audit plans, reports, logs, and findings shall be placed under **configuration management**.

Audit evidence shall be retained in accordance with organizational and regulatory requirements.

Digital signatures or equivalent mechanisms shall be applied to preserve authenticity and integrity.

Audit records shall be accessible to authorized parties for traceability and compliance audits.

Cross-References

This process interacts with the following:

- **Software Requirements Analysis (7.1.2)** — provides baseline requirements for audit checks.
- **Software Construction (7.1.5), Integration (7.1.6), and Qualification Testing (7.1.7)** — provide work products and test evidence subject to audit.
- **Software Quality Assurance (7.2.3)** — coordinates audit scheduling and scope as part of QA planning.
- **Software Verification (7.2.4)** and **Validation (7.2.5)** — provide artifacts and results that audits examine for compliance.
- **Software Review (7.2.6)** — complements audits with peer evaluations during development.
- **Software Problem Resolution (7.2.8)** — receives problems and non-conformances found during audits for resolution tracking.

7.2.8 Software Problem Resolution Process

NOTE — The Software Problem Resolution Process may also be used or easily adapted to manage, track, and control **software change requests** and other issue categories (e.g., anomalies, defects, non-conformances) throughout the life cycle.

7.2.8.1 Purpose

The purpose of the Software Problem Resolution Process is to **ensure that all discovered problems are identified, analyzed, managed, and controlled to resolution** in a consistent, traceable, and auditable manner.

7.2.8.2 Outcomes

As a result of successful implementation of the Software Problem Resolution Process:

- a) a problem management strategy is developed;
 - b) problems are recorded, identified, and classified;
 - c) problems are analyzed and assessed to identify acceptable solutions;
 - d) problem resolutions are implemented;
 - e) problems are tracked to closure; and
 - f) the status of all problems reported is known.
-

7.2.8.3 Activities and Tasks

The project shall implement the following activities in accordance with applicable organizational policies and procedures with respect to the Software Problem Resolution Process.

7.2.8.3.1 Process Implementation

This activity consists of the following tasks:

7.2.8.3.1.1 Establish Problem Resolution Process

A **problem resolution process** shall be established for handling all problems

(including non-conformances) detected in software products, processes, and activities. The process shall:

- a) ensure that each problem is documented, analyzed, resolved, and dispositioned;
 - b) ensure that status is tracked and reported through closure;
 - c) maintain problem records as stipulated in contracts, organizational procedures, or regulatory requirements.
-

7.2.8.3.1.2 Categorization and Prioritization Scheme

A scheme shall be established for **categorizing and prioritizing problems**. Each problem shall be classified by category (e.g., defect, requirement issue, documentation error) and by priority (e.g., critical, major, minor) to support **trend analysis, triage, and resolution planning**.

7.2.8.3.1.3 Trend Analysis

Regular **trend analyses** shall be performed to detect recurring patterns or systemic issues in reported problems. Trends shall be documented and fed back into **process improvement, risk management, and QA activities**.

7.2.8.3.1.4 Resolution Evaluation

Problem resolutions and dispositions shall be evaluated to ensure that:

- problems have been fully resolved;
 - adverse trends have been reversed or mitigated;
 - changes have been correctly implemented in the appropriate software products and processes;
 - no new problems have been introduced as a result of resolution actions.
-

7.2.8.3.2 Problem Resolution

This activity consists of the following tasks:

7.2.8.3.2.1 Problem Reporting

When problems (including non-conformances) are detected in a software product or an activity, a **problem report** shall be prepared to describe each problem. The report shall include:

- unique identifier;
 - description of the problem;
 - context and environment in which it was detected;
 - severity and priority;
 - affected configuration items and processes;
 - references to related requirements, designs, or tests.
-

7.2.8.3.2.2 Investigation and Analysis

Each problem shall be **investigated and analyzed** to determine root cause, impact, and appropriate corrective actions. This may involve defect reproduction, traceability analysis, or impact assessments across system elements.

7.2.8.3.2.3 Resolution Implementation

Approved resolution actions shall be implemented, verified, and recorded. All related **configuration items, documents, and test artifacts** shall be updated as needed. Resolution may include corrective, preventive, or workaround actions.

7.2.8.3.2.4 Closure and Tracking

Each problem shall be tracked to closure through a **closed-loop process**, from detection through investigation, resolution, verification, and closure confirmation. Closure criteria shall be defined in the Problem Management Strategy or relevant organizational procedures.

7.2.8.3.2.5 Communication and Reporting

The status of all open and closed problems shall be **monitored and reported** to stakeholders (e.g., project management, QA, acquirer) at regular intervals. Reports shall include summaries of critical problems, trends, and resolution timelines.

7.2.8.4 Deliverables

The following deliverables shall be produced during this process:

- Problem Management Strategy or Plan;
- Categorization and prioritization scheme;
- Problem Reports (PRs) with unique identifiers;

- Root cause analyses and impact assessments;
 - Problem trend analysis reports;
 - Problem resolution records and closure logs;
 - Status reports and dashboards for internal and external stakeholders;
 - Updated configuration and test artifacts linked to resolved problems.
-

7.2.8.5 Audit and Configuration Control

All problem reports, analyses, resolutions, and related evidence shall be placed under **configuration management**.

Problem resolution records shall be retained for the duration required by organizational policy or regulatory obligations.

Audits shall periodically verify that problems are categorized correctly, resolved in compliance with procedures, and tracked through closure.

Digital signatures, access control, and traceability links shall be used to protect problem records against tampering or loss.

Cross-References

This process interacts with the following:

- **Software Requirements Analysis (7.1.2)** — for identifying and resolving requirement-related defects.
- **Software Construction (7.1.5), Integration (7.1.6), and Qualification Testing (7.1.7)** — for reporting, tracking, and resolving issues discovered during development and testing.
- **Software Quality Assurance (7.2.3)** — for tracking non-conformances and feeding results into QA records.
- **Software Verification (7.2.4) and Software Validation (7.2.5)** — for recording and resolving defects discovered during testing and operational evaluation.
- **Software Review (7.2.6) and Software Audit (7.2.7)** — for capturing and processing findings from reviews and audits.
- **Software Configuration Management (7.2.2)** — for controlling changes arising from problem resolutions and ensuring proper baselining.
- **Project Risk and Measurement Processes (6.3)** — for feeding trend data into project-level risk mitigation and continuous improvement.

7.3 Software Reuse Processes

NOTE — Users of this International Standard who wish to adopt organizational software reuse practices may supplement these provisions with those of **IEEE Std 1517™-1999**, *IEEE Standard for Information Technology — Software Life Cycle Processes — Reuse Processes*, which provides more detailed activities and tasks for institutionalizing reuse.

7.3.1 Domain Engineering Process

7.3.1.1 Purpose

The purpose of the Domain Engineering Process is to **develop and maintain domain models, domain architectures, and reusable assets for a domain**. This process supports systematic reuse by identifying commonalities and variabilities across a family of systems, subsystems, or applications.

7.3.1.2 Outcomes

As a result of successful implementation of the Domain Engineering Process:

- a) the representation forms for the domain models and domain architectures are selected;
- b) the boundaries of the domain and its relationships to other domains are established;
- c) a domain model that captures the essential common and differing features, capabilities, concepts, and functions in the domain is developed;
- d) a domain architecture describing the family of systems within the domain, including their commonalities and variabilities, is developed;
- e) assets belonging to the domain are specified;
- f) assets belonging to the domain are acquired or developed and maintained throughout their life cycles; and
- g) the domain models and architectures are maintained throughout their life cycles.

NOTE 1 — Domain models and architectures provide the foundation for creating reusable assets for a class of systems, subsystems, or applications.

NOTE 2 — The Domain Engineering Process may overlap with development and maintenance processes that use assets produced by the Domain Engineering Process.

7.3.1.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures with respect to the Domain Engineering Process.

Reference — IEEE Std 1517™ provides a more detailed set of activities and tasks that align with those defined below.

7.3.1.3.1 Process Implementation

This activity consists of the following tasks:

7.3.1.3.1.1

The domain engineer shall create and execute a **Domain Engineering Plan**, describing scope, objectives, methods, responsibilities, and schedules for domain analysis, design, and asset provisioning.

7.3.1.3.1.2

The domain engineer shall select the **representation forms** to be used for domain architectures and models (e.g., feature models, reference architectures, ontologies, UML profiles).

7.3.1.3.1.3

The domain engineer shall establish procedures for **receiving, resolving, and providing feedback** to the asset manager whenever problems or change requests occur for assets developed by the domain engineer.

7.3.1.3.2 Domain Analysis

This activity consists of the following tasks:

7.3.1.3.2.1

The domain engineer shall define the **boundaries of the domain** and the relationships between this domain and other related domains.

7.3.1.3.2.2

The domain engineer shall identify the **current and anticipated needs of stakeholders** of software products within this domain.

7.3.1.3.2.3

The domain engineer shall build the **domain models** using the selected representation forms. These models shall describe essential domain concepts, commonalities, variabilities, and interactions.

7.3.1.3.2.4

The domain engineer shall construct a **vocabulary or ontology** that provides standardized terminology to describe important domain concepts and the relationships among common assets.

7.3.1.3.2.5

The domain engineer shall classify and document the domain models in accordance with the organization's modeling standards and asset acceptance procedures.

7.3.1.3.2.6

The domain engineer shall conduct **domain analysis reviews**, involving software developers, asset managers, domain experts, and users.

7.3.1.3.2.7

The domain engineer shall submit the approved domain models to the **asset manager** for storage, cataloging, and reuse distribution.

7.3.1.3.3 Domain Design

This activity consists of the following tasks:

7.3.1.3.3.1

The domain engineer shall create and document the **domain architecture**, consistent with the domain model and following organizational standards.

7.3.1.3.3.2

For each entity selected for design as a reusable asset, the domain engineer shall develop and document an **asset specification**, including functionality, interfaces, constraints, and variability points.

7.3.1.3.3.3

Each asset specification shall be evaluated in accordance with the organization's **asset acceptance and certification procedures**.

7.3.1.3.3.4

The domain engineer shall conduct **domain design reviews**, involving software developers, domain experts, and asset managers.

7.3.1.3.3.5

The domain engineer shall submit the approved domain architecture to the asset manager for inclusion in the organizational asset base.

7.3.1.3.4 Asset Provision

For each asset developed or acquired, this activity consists of the following tasks:

7.3.1.3.4.1

The domain engineer shall obtain the asset by **acquisition or development**.

7.3.1.3.4.2

The domain engineer shall **document and classify** the asset according to the organization's asset taxonomy (e.g., reusable component, design pattern, domain model, service, library).

7.3.1.3.4.3

The domain engineer shall evaluate the asset in accordance with the organization's **asset acceptance and certification procedures**, including testing, documentation review, and interface conformance.

7.3.1.3.4.4

The domain engineer shall conduct **asset reviews**, involving software developers and asset managers.

7.3.1.3.4.5

The domain engineer shall submit the approved asset to the **asset manager** for cataloging, storage, and controlled distribution.

7.3.1.3.5 Asset Maintenance (Reuse-Related)

The following reuse-related task shall be added to the **Software Maintenance Process (6.4.10)**:

7.3.1.3.5.1

When analyzing requests for asset modification and choosing implementation options, the domain engineer shall consider:

- a) conformance with the domain models and domain architecture;
 - b) impact on the systems and software products that use the asset;
 - c) impact on future users of the asset; and
 - d) impact on the **reusability** of the asset.
-

7.3.1.4 Deliverables

The following deliverables shall be produced during this process:

- Domain Engineering Plan;
 - Domain models, vocabularies, and ontologies;
 - Domain architecture documentation;
 - Asset specifications, classification records, and acceptance reports;
 - Domain and asset review records;
 - Asset submission and cataloging records;
 - Maintenance impact assessments for modified assets.
-

7.3.1.5 Audit and Configuration Control

All domain engineering artifacts (models, architectures, specifications, and assets) shall be placed under **configuration management**.

Audit evidence, including review records and acceptance reports, shall be retained in accordance with organizational and regulatory requirements.

Changes to domain models or assets shall follow the organization's change control procedures and be fully traceable.

Cross-References

This process interacts with the following:

- **Software Requirements Analysis (7.1.2)** — provides requirements and domain insights for asset modeling.
- **Software Architectural Design (7.1.3)** — consumes domain architectures and reusable design assets.
- **Software Construction (7.1.5)** — reuses domain assets during implementation.
- **Software Configuration Management (7.2.2)** — manages versions and

baselines of reusable assets.

- **Software Quality Assurance (7.2.3)** — ensures assets and domain models meet quality and acceptance standards.
- **Software Verification (7.2.4)** and **Software Validation (7.2.5)** — verify and validate assets and domain architectures prior to reuse.
- **Software Maintenance (6.4.10)** — applies reuse-specific maintenance rules for asset evolution.

7.3.2 Reuse Asset Management Process

NOTE — The Reuse Asset Management Process governs the **full life cycle of reusable assets**, from inception to retirement. It ensures that assets are properly classified, stored, retrieved, maintained, versioned, and retired, supporting efficient and controlled reuse across projects and domains.

7.3.2.1 Purpose

The purpose of the Reuse Asset Management Process is to **manage the life cycle of reusable assets from conception to retirement**, ensuring that assets are properly classified, controlled, maintained, and made available for effective reuse.

7.3.2.2 Outcomes

As a result of successful implementation of the Reuse Asset Management Process:

- a) an asset management strategy is documented;
 - b) an asset classification scheme is established;
 - c) criteria for asset acceptance, certification, and retirement are defined;
 - d) an asset storage and retrieval mechanism is operated;
 - e) the use of assets is recorded;
 - f) changes to the assets are controlled; and
 - g) users of assets are notified of problems detected, modifications made, new versions created, and deletions of assets from the storage and retrieval mechanism.
-

7.3.2.3 Activities and Tasks

The organization shall implement the following activities in accordance with applicable organizational policies and procedures with respect to the Reuse Asset Management Process.

7.3.2.3.1 Process Implementation

This activity consists of the following tasks:

7.3.2.3.1.1 Asset Management Planning

The **Asset Manager** shall create an **Asset Management Plan** to define the resources, responsibilities, and procedures for managing assets throughout their life cycle. The plan shall include:

- roles and responsibilities (e.g., asset manager, domain engineer, reuse program administrator);
 - procedures for asset acceptance, classification, storage, retrieval, modification, notification, and retirement;
 - tools and repositories used to manage reusable assets;
 - review and audit mechanisms.
-

7.3.2.3.1.2 Asset Management Plan Execution

The asset manager shall execute the plan, ensuring consistent application of defined procedures across all managed assets.

7.3.2.3.1.3 Plan Review

The asset management plan shall be reviewed in accordance with the **Software Review Process (7.2.6)**. Reuse program administrators and domain engineers shall participate in these reviews to ensure alignment with organizational reuse objectives.

7.3.2.3.2 Asset Storage and Retrieval Definition

This activity consists of the following tasks:

7.3.2.3.2.1 Storage and Retrieval Mechanism Implementation

The asset manager shall **implement and maintain an asset storage and retrieval mechanism** (e.g., asset library, repository, registry) that supports secure, organized, and efficient access to reusable assets.

7.3.2.3.2.2 Asset Classification Scheme

The asset manager shall **develop, document, and maintain a classification scheme** for assets. The scheme may include:

- functional domain;
- maturity level (e.g., prototype, certified, retired);

- quality or certification status;
 - applicable standards;
 - intended usage context.
-

7.3.2.3.2.3 Storage and Retrieval Reviews

The asset manager shall conduct periodic reviews of the storage and retrieval mechanism in accordance with the **Software Review Process (7.2.6)**. Reuse program administrators and domain engineers shall be included in these reviews to ensure relevance, usability, and completeness.

7.3.2.3.3 Asset Management and Control

For each asset, this activity consists of the following tasks:

7.3.2.3.3.1 Asset Acceptance and Certification

Each submitted asset shall be **evaluated against asset acceptance and certification criteria**, which may include quality, documentation completeness, licensing, and compliance with reuse standards.

7.3.2.3.3.2 Asset Availability

For each accepted asset, it shall be **made available for reuse** through the asset storage and retrieval mechanism in accordance with established access and security policies.

7.3.2.3.3.3 Asset Classification

Each accepted asset shall be **classified** according to the established reuse classification scheme to support discoverability and appropriate reuse decisions.

7.3.2.3.3.4 Configuration Management of Assets

The asset manager shall perform **configuration management** of each asset using the **Software Configuration Management Process (7.2.2)**. This includes version control, change tracking, and baselining.

7.3.2.3.3.5 Reuse Tracking

The asset manager shall **track each reuse** of the asset and report relevant information (e.g., projects, versions used, usage contexts) to the **Domain Engineer** and reuse program administrators.

7.3.2.3.3.6 Notification of Asset Changes

The asset manager shall **monitor and record asset-related requests and problem reports**.

All **asset reusers and the domain engineer shall be notified** of:

- problems detected in the asset;
 - modifications made;
 - new versions created;
 - deletions of the asset from the storage and retrieval mechanism.
-

7.3.2.3.3.7 Asset Retirement

Assets shall be **retired** from the storage and retrieval mechanism according to defined retirement procedures and criteria. Retirement may be triggered by obsolescence, quality degradation, security vulnerabilities, or replacement by newer assets.

7.3.2.4 Deliverables

The following deliverables shall be produced during this process:

- Asset Management Plan;
 - Asset classification scheme and taxonomy;
 - Storage and retrieval mechanism implementation and configuration documentation;
 - Asset acceptance and certification records;
 - Asset version histories and configuration management records;
 - Reuse usage logs and reports;
 - Notifications of asset changes and retirements;
 - Asset retirement records.
-

7.3.2.5 Audit and Configuration Control

All reusable assets, their metadata, and associated management artifacts shall be placed under appropriate **configuration management**.

Audit records of asset acceptance, modifications, reuse, and retirement shall be maintained according to organizational and regulatory requirements.

Access to asset repositories shall be logged and controlled. Periodic audits shall ensure integrity, consistency, and completeness of stored assets.

Cross-References

This process interacts with the following:

- **Domain Engineering Process (7.3.1)** — provides domains, models, and reusable asset definitions.
- **Reuse Program Management Process (7.3.3)** — defines organizational strategy and policies for reuse.
- **Software Configuration Management (7.2.2)** — provides version control and change management for assets.
- **Software Review Process (7.2.6)** — used for plan and mechanism reviews.
- **Software Problem Resolution (7.2.8)** — may be invoked to handle problems with reusable assets.
- **Software Verification (7.2.4)** and **Validation (7.2.5)** — may provide evidence for asset certification.

7.3.3 Reuse Program Management Process

7.3.3.1 Purpose

The purpose of the Reuse Program Management Process is to plan, establish, manage, control, and monitor an organization's reuse program and to systematically exploit reuse opportunities.

7.3.3.2 Outcomes

As a result of successful implementation of the Reuse Program Management Process:

- a) the organization's reuse strategy, including its purpose, scope, goals, and objectives, is defined;
- b) the domains for potential reuse opportunities are identified;
- c) the organization's systematic reuse capability is assessed;
- d) the reuse potential of each domain is assessed;
- e) reuse proposals are evaluated to ensure the reuse product is suitable for the proposed application;
- f) the reuse strategy is implemented in the organization;
- g) feedback, communication, and notification mechanisms that operate between affected parties are established; and
- h) the reuse program is monitored and evaluated.

NOTE — The affected parties may include reuse program administrators, asset managers, domain engineers, developers, operators, and maintainers.

7.3.3.3 Activities and Tasks

The project shall implement the following activities in accordance with applicable organizational policies and procedures with respect to the Reuse Program Management Process.

7.3.3.3.1 Initiation

This activity consists of the following tasks:

7.3.3.3.1.1 The reuse program for an organization shall be initiated by establishing the organization's reuse strategy that includes its reuse goals, purposes, objectives, and scope.

7.3.3.3.1.2 A reuse sponsor should be named.

7.3.3.3.1.3 Reuse program participants shall be identified and their roles shall be assigned.

7.3.3.3.1.4 A reuse steering function shall be established to assume the authority and responsibility for the organization's reuse program.

7.3.3.3.1.5 A reuse program support function shall be established.

7.3.3.3.2 Domain Identification

This activity consists of the following tasks:

7.3.3.3.2.1 The reuse program administrator, aided by the appropriate managers, domain engineers, users, and software developers, shall identify and document the domains in which to investigate reuse opportunities or in which the organization intends to practice reuse.

7.3.3.3.2.2 The reuse program administrator, aided by the appropriate managers, domain engineers, users, and software developers, shall evaluate the domains to assure that they accurately reflect the organization's reuse strategy.

7.3.3.3.2.3 The reuse program administrator shall conduct reviews in accordance with the Software Review Process. Software developers, domain engineers, and users shall be included in the reviews.

7.3.3.3.2.4 As more information about the organization's domains and plans for future software products becomes available or when the domains are analyzed, the domains may be refined and rescope by the reuse program administrator.

7.3.3.3.3 Reuse Assessment

This activity consists of the following tasks:

7.3.3.3.3.1 The reuse program administrator shall assess the organization's systematic reuse capability.

7.3.3.3.3.2 The reuse program administrator shall assess each domain being considered for reuse to determine the potential for reuse success in the domain.

7.3.3.3.3.3 The reuse program administrator shall make recommendations for refining the organization's reuse strategy and reuse program implementation plan based on the results of the reuse assessments.

7.3.3.3.3.4 The reuse program administrator, in conjunction with the appropriate acquirers, suppliers, developers, operators, maintainers, asset managers, and domain engineers, shall incrementally improve the skills, technology, reuse processes, organizational structure, and metrics that together comprise the reuse infrastructure.

7.3.3.3.4 Planning

This activity consists of the following tasks:

7.3.3.3.4.1 A reuse program implementation plan shall be created, documented, and maintained to define the resources and procedures for implementing a reuse

program.

7.3.3.3.4.2 The plan shall be reviewed and evaluated for completeness, feasibility of implementation, and ability to realize the organization's reuse strategy. Those evaluating the plan should include members of the reuse steering function and the appropriate managers.

7.3.3.3.4.3 Approval and support for the reuse program implementation plan shall be obtained from the reuse steering function and the appropriate managers.

7.3.3.3.4.4 The reuse program administrator shall conduct review(s) in accordance with the Software Review Process. Members of the reuse steering function and the appropriate managers shall be included in the reviews.

7.3.3.3.5 Execution and Control

This activity consists of the following tasks:

7.3.3.3.5.1 Activities in the reuse program implementation plan shall be executed in accordance with the plan.

7.3.3.3.5.2 The reuse program administrator shall monitor the progress of the reuse program against the organization's reuse strategy and make any necessary adjustments to the plan to realize the strategy.

7.3.3.3.5.3 The reuse program administrator shall record and report significant variances, risks, and issues encountered during the execution of the reuse program.

7.3.3.3.5.4 The reuse program administrator shall periodically reaffirm management sponsorship, support, and commitment to the reuse program.

7.3.3.3.6 Review and Evaluation

This activity consists of the following tasks:

7.3.3.3.6.1 The reuse program administrator shall conduct periodic reviews of the organization's reuse program in accordance with the Software Review Process to determine program effectiveness and alignment with organizational goals.

7.3.3.3.6.2 The reuse program administrator shall provide assessment results and lessons learned to the reuse steering function and appropriate managers.

7.3.3.3.6.3 The reuse program administrator shall recommend and make changes to the reuse program strategy, plans, or supporting infrastructure to address identified issues, improve performance, and exploit emerging reuse opportunities.

 **This completes section 7.3.3** in full ISO/IEC 12207 style, with restored structure, numbering, and language consistency.

Would you like me to generate a **"Compliance Checklist Table"** (Activity ↔ Evidence ↔ Responsible Role) for this section, Master?

8 Annex A

(normative)

Tailoring Process

A.1 Introduction

This Annex provides requirements for the tailoring of this International Standard. No conformance claim shall be made if a tailoring of this International Standard results in the failure to meet the intent of its conformance or tailoring provisions. Tailoring is to be determined and recorded as part of the agreement or organizational application of this standard.

A.2 Tailoring Process

A.2.1 Purpose of the Tailoring Process

The purpose of the Tailoring Process is to adapt the processes of this International Standard to satisfy particular circumstances or factors that:

- a) surround an organization that is employing this International Standard in an agreement;
 - b) influence a project that is required to meet an agreement in which this International Standard is referenced; or
 - c) reflect the needs of an organization in order to supply products or services.
-

A.2.2 Tailoring Process Outcomes

As a result of the successful implementation of the Tailoring Process:

- a) modified life cycle processes are defined to achieve the purposes and outcomes of a life cycle model, while maintaining conformance with the intent of the International Standard.
-

A.2.3 Tailoring Process Activities

If this International Standard is tailored, then the organization or project shall

implement the following tasks in accordance with applicable policies and procedures with respect to the Tailoring Process, as required:

A.2.3.1 Identify Tailoring Influences

Identify and document the circumstances that influence tailoring. These influences include, but are not limited to:

- a) stability of, and variety in, operational environments;
 - b) risks, commercial or performance, to the concern of interested parties;
 - c) novelty, size, and complexity;
 - d) starting date and duration of utilization;
 - e) integrity issues such as safety, security, privacy, usability, and availability;
 - f) emerging technology opportunities;
 - g) profile of budget and organizational resources available;
 - h) availability of the services of enabling systems;
 - i) roles and responsibilities in the overall life cycle of the system; and
 - j) the need to conform to other standards.
-

A.2.3.2 Consider Critical Properties

In the case of properties critical to the system, due account shall be taken of the life cycle structures recommended or mandated by standards relevant to the dimension of the criticality (e.g., safety-critical, security-critical systems).

A.2.3.3 Obtain Stakeholder Input

Obtain input from all parties affected by the tailoring decisions. This includes, but is not limited to:

- a) the system stakeholders;
 - b) the interested parties to an agreement made by the organization; and
 - c) the contributing organizational functions.
-

A.2.3.4 Make Tailoring Decisions

Make tailoring decisions in accordance with the Decision Management Process to achieve the purposes and outcomes of the selected life cycle model.

NOTE 1 — Organizations typically establish standard life cycle models as part of

the Life Cycle Model Management Process. It may be appropriate for an organization to tailor processes of this International Standard in order to achieve the purposes and outcomes of the stages of a selected life cycle model.

A.2.3.5 Select Processes and Tailor

Select the life cycle processes that require tailoring and delete selected outcomes, activities, or tasks as appropriate. Irrespective of tailoring, organizations and projects are always permitted to implement processes that achieve additional outcomes or implement additional activities and tasks beyond those required for conformance to this standard.

NOTE 2 — An organization or project may encounter a situation where there is a desire to modify a provision of this International Standard. Modification should be avoided because it may have unanticipated consequences on other processes, outcomes, activities, or tasks. If necessary, modification is performed by deleting the provision (making the appropriate claim of tailored conformance) and, with careful consideration of consequences, implementing a process that achieves additional outcomes or performs additional activities and tasks beyond those of the tailored standard.

 This **completed Annex A** is now clean, normative, and ready for inclusion in the tailorable section of a conformance claim or SDLC policy manual.

Would you like me to follow this with a **Tailoring Decision Record Template** (per A.2.3.4) that can be inserted into your project's CMDB or SDLC baseline documentation, Master?

8 Annex B

(normative)

Process Reference Model (PRM) for Assessment Purposes

B.1 Introduction

It is understood that some users of this International Standard may desire to assess the implemented processes in accordance with ISO/IEC 15504-2, *Information Technology — Process Assessment — Part 2: Performing an Assessment*.

This Annex provides a Process Reference Model (PRM) suitable for use in conjunction with ISO/IEC 15504-2.

The source for the processes in this model is the set of processes defined in the body of this International Standard. In each case, the name, statement of purpose, and statement of outcomes for each process in the body of this standard have been referenced for use in this Annex. In some cases, the processes in the body of the standard have been explicitly identified as lower-level processes or replacement processes for equivalent clauses in ISO/IEC 15288.

B.2 Conformance with ISO/IEC 15504-2

B.2.1 General

The Process Reference Model included in this Annex is suitable for use in process assessments performed in accordance with ISO/IEC 15504-2, *Information Technology — Process Assessment — Part 2: Performing an Assessment*.

Each of the following clauses includes italicized text quoting the requirements from ISO/IEC 15504-2, followed by upright text describing how these requirements are satisfied by this International Standard.

B.2.2 Requirements for Process Reference Models

A Process Reference Model shall contain:

a) A declaration of the domain of the Process Reference Model.

This is provided in Clause 1 of this International Standard.

b) A description of the processes within the scope of the Process Reference Model.

This is provided in Annex B.3, meeting the requirements of subclause 6.2.4 of this International Standard.

c) A description of the relationship between the Process Reference Model and its intended context of use.

This is provided in Clause 5 of this International Standard.

d) A description of the relationship between the processes defined within the Process Reference Model.

This is provided in Annex B.3 in the description of each process. For example, some process descriptions in this Annex include statements that the process is a lower-level process and that it replaces equivalent clauses in ISO/IEC 15288.

The Process Reference Model shall document the community of interest of the model and the actions taken to achieve consensus within that community of interest:

a) The relevant community of interest shall be characterized or specified.

The relevant community of interest comprises the users of ISO/IEC 15288 and ISO/IEC 12207.

b) The extent of achievement of consensus shall be documented.

Both ISO/IEC 15288 and ISO/IEC 12207 are International Standards developed and approved through the ISO/IEC JTC 1 process, which satisfies the formal consensus requirements.

c) If no actions are taken to achieve consensus, a statement to this effect shall be documented.

This is not applicable, as the required consensus actions have been undertaken through the international standardization process.

The processes defined within a Process Reference Model shall have **unique**

process descriptions and identification.

The process descriptions provided in this Annex are unique. Identification is ensured by unique process names and by the clause numbering of this Annex.

B.2.3 Process Descriptions

The fundamental elements of a Process Reference Model are the descriptions of the processes within the scope of the model.

The process descriptions in this PRM incorporate:

- a **statement of purpose** for each process, which describes at a high level the overall objectives of performing the process; and
- a **set of outcomes**, which demonstrate successful achievement of the process purpose.

These process descriptions shall meet the following requirements:

- a) A process shall be described in terms of its purpose and outcomes.
- b) In any process description, the set of process outcomes shall be **necessary and sufficient** to achieve the purpose of the process.
- c) Process descriptions shall be such that **no aspects of the Measurement Framework** as described in Clause 5 of ISO/IEC 15504-2 beyond **Capability Level 1** are contained or implied.

An **outcome statement** describes one of the following:

- the production of an artefact;
- a significant change of state; or
- the meeting of specified constraints (e.g., requirements, goals, conditions).

These requirements are met by the process descriptions in this Annex. Some outcomes might be interpreted as contributing to levels of capability above Level 1; however, a conforming implementation of the relevant processes does not require achievement of these higher levels of capability.

B.2.4 Common Process Attributes for Capability Determination

The attributes described in 5.1.9 of this International Standard characterize the specificity of each process. When an implemented process conforms to these attributes, the process' defined purpose and outcomes are achieved through the

implementation of its defined activities.

In addition to these basic attributes, processes may be characterized by **other attributes common to all processes**. These common attributes contribute to the achievement of higher levels of process capability as defined in ISO/IEC 15504-2.

There are six levels of process capability in the Measurement Framework of ISO/IEC 15504-2, as described in the following table:

Table B.1 — Six Levels of Process Capability	
Capability Level	Process Capability
0	Incomplete Process
1	Performed Process
2	Managed Process
3	Established Process
4	Predictable Process
5	Optimizing Process

B.2.5 Common Process Attributes for Higher Capability Levels

The achievement of higher-level attributes and capabilities is enabled by the interaction of the process with support and organizational processes such as **Documentation, Configuration Management, Quality Assurance, Measurement, and Improvement**.

ISO/IEC 15504-2 identifies the following **common process attributes (PA)** affiliated with the achievement of higher levels of process capability:

- **Performance Management (PA 2.1)** — Determines the extent to which the performance of the process is managed.
The achievement of this attribute involves planning, monitoring, and adjusting the process performance.
- **Work Product Management (PA 2.2)** — Determines the extent to which the work products produced by the process are appropriately managed.
The achievement of this attribute ensures that work products are properly established, controlled, and maintained.
- **Process Definition (PA 3.1)** — Determines the extent to which the

process is established as a **standard process** within the organization.

The achievement of this attribute involves the definition of the process in terms of:

- required competencies and roles for performing the process;
- required infrastructure and work environment;
- methods for monitoring its effectiveness and suitability; and
- tailoring guidelines.
- **Process Deployment (PA 3.2)** — Determines the extent to which the

process is effectively deployed as a tailored instance of the standard process.

The achievement of this attribute is reflected in fidelity to the standard process, effective deployment of resources to the implementation of the process, and the collection and analysis of data for understanding and refining process behaviour.

- **Process Measurement (PA 4.1)** — Determines the extent to which process measurements are used to ensure that process performance supports the achievement of defined business goals.

The achievement of this attribute is concerned with the existence of an effective system for the **collection of measures** relevant to the performance of the process and the quality of the work products. The measures are applied to determine the extent of achievement of the organization's business goals.

- **Process Control (PA 4.2)** — Determines the extent to which the process is quantitatively managed to produce a process that is stable, capable, and predictable within defined limits.

The achievement of this attribute implies the application of analysis and control techniques to ensure that the process performs within defined limits and that corrective actions are taken to address deviations.

- **Process Innovation (PA 5.1)** — Determines the extent to which changes to the process are identified from analysis of variation in performance and from investigations of innovative approaches to process definition and implementation. The achievement of this attribute is concerned with the existence of a **proactive focus on continuous improvement** to fulfil both current and projected business goals.

- **Process Optimization (PA 5.2)** — Determines the extent to which changes to the definition, management, and performance of the process result in effective impact that achieves relevant process improvement objectives.

The achievement of this attribute is concerned with an **orderly and proactive approach** to identifying and introducing appropriate changes to the process, minimizing undesired disruption, and evaluating their effects to ensure improvements are realized.

B.3 Process Reference Model

The **Process Reference Model** is composed of the **statement of purpose** and **set of outcomes** for each of the processes included in **Clause 6** and **Clause 7** of this International Standard.

These processes are organized by their life cycle categories (e.g., Agreement, Organizational, Project, Technical, and Support), and each process description includes:

- the **process name**;
- the **process purpose statement**; and
- the **process outcomes**, which collectively provide a **necessary and sufficient** set to achieve the process purpose.

The set of processes is listed in **Table B.2**.

Table B.2 — ISO/IEC 12207:2008 Processes

ISO/IEC 12207 Clause Number	ISO/IEC 12207:2008 Process Name
6	<i>System Life Cycle Processes</i>
6.1	<i>Agreement Processes</i>
6.1.1	<i>Acquisition Process</i>
6.1.2	<i>Supply Process</i>
6.2	<i>Organizational Project-Enabling Processes</i>
6.2.1	<i>Life Cycle Model Management Process</i>
6.2.2	<i>Infrastructure Management Process</i>
6.2.3	<i>Project Portfolio Management Process</i>
6.2.4	<i>Human Resource Management Process</i>
6.2.5	<i>Quality Management Process</i>
6.3	<i>Project Processes</i>
6.3.1	<i>Project Planning Process</i>
6.3.2	<i>Project Assessment and Control Process</i>
6.3.3	<i>Decision Management Process</i>
6.3.4	<i>Risk Management Process</i>
6.3.5	<i>Configuration Management Process</i>
6.3.6	<i>Information Management Process</i>
6.3.7	<i>Measurement Process</i>
6.4	<i>Technical Processes</i>
6.4.1	<i>Stakeholder Requirements Definition Process</i>
6.4.2	<i>System Requirements Analysis</i>
6.4.3	<i>System Architectural Design</i>
6.4.4	<i>Implementation Process</i>
6.4.5	<i>System Integration Process</i>
6.4.6	<i>System Qualification Testing Process</i>
6.4.7	<i>Software Installation</i>
6.4.8	<i>Software Acceptance Support</i>
6.4.9	<i>Software Operation Process</i>
6.4.10	<i>Software Maintenance Process</i>
6.4.11	<i>Software Disposal Process</i>
7	<i>Software Life Cycle Processes</i>
7.1	<i>Software Implementation Processes</i>
7.1.1	<i>Software Implementation Process</i>
7.1.2	<i>Software Requirements Analysis Process</i>
7.1.3	<i>Software Architectural Design Process</i>
7.1.4	<i>Software Detailed Design Process</i>
7.1.5	<i>Software Construction Process</i>
7.1.6	<i>Software Integration Process</i>

ISO/IEC 12207 Clause Number	ISO/IEC 12207:2008 Process Name
7.1.7	<i>Software Qualification Testing Process</i>
7.2	<i>Software Support Processes</i>
7.2.1	<i>Software Documentation Management Process</i>
7.2.2	<i>Software Configuration Management Process</i>
7.2.3	<i>Software Quality Assurance Process</i>
7.2.4	<i>Software Verification Process</i>
7.2.5	<i>Software Validation Process</i>
7.2.6	<i>Software Review Process</i>
7.2.7	<i>Software Audit Process</i>
7.2.8	<i>Software Problem Resolution Process</i>
7.3	<i>Software Reuse Processes</i>
7.3.1	<i>Domain Engineering Process</i>
7.3.2	<i>Reuse Asset Management Process</i>
7.3.3	<i>Reuse Program Management Process</i>

NOTE — These process descriptions provide the basis for process assessment at **Capability Level 1** in accordance with ISO/IEC 15504-2. Higher levels of capability are evaluated by applying the common process attributes defined in Clause B.2.5 in conjunction with this PRM.

Some activities of the processes in Clauses 6 and 7 are replaced with corresponding **lower-level processes**. The descriptions of these lower-level processes are provided below.

B.3.1 Acquisition Process Lower-Level Processes

B.3.1.1 Acquisition Preparation Process

This process is a lower-level process of the Acquisition Process. It replaces the *Acquisition Preparation* activity (6.1.1.3.1).

B.3.1.1.1 Purpose

The purpose of the Acquisition Preparation Process is to establish the needs and goals of the acquisition and to communicate these with potential suppliers.

B.3.1.1.2 Outcomes

As a result of successful implementation of the Acquisition Preparation Process:

- a) the concept or the need for the acquisition, development, or enhancement is established;
- b) stakeholder requirements are defined;
- c) an acquisition strategy is developed; and
- d) supplier selection criteria are defined.

B.3.1.2 Supplier Selection Process

This process is a lower-level process of the Acquisition Process. It replaces the *Supplier Selection* activity (6.1.1.3.3).

B.3.1.2.1 Purpose

The purpose of the Supplier Selection Process is to choose the organization that is to be responsible for the delivery of the project requirements.

B.3.1.2.2 Outcomes

As a result of successful implementation of the Supplier Selection Process:

- a) the supplier selection criteria are established and used to evaluate potential suppliers;
- b) the supplier is selected based on evaluation of proposals, process capabilities, and other relevant factors; and
- c) an agreement is established and negotiated between the acquirer and the supplier.

B.3.1.3 Agreement Monitoring Process

This process is a lower-level process of the Acquisition Process. It replaces the *Agreement Monitoring* activity (6.1.1.3.5).

B.3.1.3.1 Purpose

The purpose of the Agreement Monitoring Process is to track and assess the performance of the supplier against agreed requirements.

B.3.1.3.2 Outcomes

As a result of successful implementation of the Agreement Monitoring Process:

- a) joint activities between the acquirer and the supplier are performed as needed;
- b) information on technical progress is exchanged regularly with the supplier;
- c) supplier performance is monitored against agreed requirements; and
- d) agreement changes, if needed, are negotiated between the acquirer and supplier and documented.

B.3.1.4 Acquirer Acceptance Process

This process is a lower-level process of the Acquisition Process. It replaces the *Acquirer Acceptance* activity (6.1.1.3.6).

B.3.1.4.1 Purpose

The purpose of the Acquirer Acceptance Process is to approve the supplier's deliverable when all acceptance criteria are satisfied.

B.3.1.4.2 Outcomes

As a result of successful implementation of the Acquirer Acceptance Process:

- a) the delivered software product and/or service is evaluated with respect to the agreement;
- b) the acquirer's acceptance is based on the agreed acceptance criteria; and
- c) the software product and/or service is accepted by the acquirer.

B.3.2 Supply Process Lower-Level Processes

B.3.2.1 Supplier Tendering Process

This process is a lower-level process of the Supply Process. It replaces the *Supplier Tendering* activity (6.1.2.3.2).

B.3.2.1.1 Purpose

The purpose of the Supplier Tendering Process is to establish an interface to respond to acquirer inquiries and requests for proposal, and to prepare and submit proposals.

B.3.2.1.2 Outcomes

As a result of successful implementation of the Supplier Tendering Process:

- a) a communication interface is established and maintained to respond to acquirer inquiries and RFPs;
 - b) requests for proposal are evaluated to determine whether to submit a proposal;
 - c) the need for preliminary surveys or feasibility studies is determined;
 - d) suitable resources are identified to perform the proposed work; and
 - e) a supplier proposal is prepared and submitted.
-

B.3.2.2 Contract Agreement Process

This process is a lower-level process of the Supply Process. It replaces the *Contract Agreement* activity (6.1.2.3.4).

B.3.2.2.1 Purpose

The purpose of the Contract Agreement Process is to negotiate and approve a contract/agreement that clearly specifies expectations, responsibilities, deliverables, and liabilities of both the supplier and the acquirer.

B.3.2.2.2 Outcomes

As a result of successful implementation of the Contract Agreement Process:

- a) a contract/agreement is negotiated, reviewed, approved, and awarded;
- b) mechanisms for monitoring supplier performance and mitigating risks are reviewed and considered for inclusion;
- c) proposers/tenderers are notified of selection results; and
- d) formal confirmation of agreement is obtained.

The Contract Agreement Process is used to obtain formal confirmation of assignments offered during the Supplier Tendering Process.

B.3.2.3 Product/Service Delivery and Support Process

This process is a lower-level process of the Supply Process. It replaces the *Product/Service Delivery and Support* activity (6.1.2.3.6).

B.3.2.3.1 Purpose

The purpose of the Product/Service Delivery and Support Process is to provide the specified product or service to the acquirer with appropriate support to ensure that the requirements have been met.

B.3.2.3.2 Outcomes

As a result of successful implementation of the Product/Service Delivery and Support Process:

- a) the contents of the product release are determined;
- b) the release is assembled from configured items;
- c) the release documentation is produced;
- d) the delivery mechanism and media are determined;
- e) release approval is effected against defined criteria;
- f) the product release is made available to the acquirer;
- g) confirmation of release is obtained;

- h) the product is delivered to the acquirer;
- i) acquirer acceptance tests and reviews are supported;
- j) the product is put into operation in the customer's environment; and
- k) problems detected during acceptance are identified and communicated for resolution.

NOTE — Incremental delivery may be performed in completed increments.

B.3.3 Life Cycle Model Management Process Lower-Level Processes

B.3.3.1 Process Establishment Process

This process is a lower-level process of the Life Cycle Model Management Process. It replaces the *Process Establishment* activity (6.2.1.3.1).

B.3.3.1.1 Purpose

The purpose of the Process Establishment Process is to establish a suite of organizational processes for all life cycle processes as they apply to business activities.

B.3.3.1.2 Outcomes

As a result of successful implementation of the Process Establishment Process:

- a) a defined and maintained standard set of processes is established, with indications of applicability;
 - b) the detailed tasks, activities, and work products of the standard process are identified, with expected performance characteristics;
 - c) a strategy for tailoring and applying the standard process to products or services is developed; and
 - d) information and data related to use of the standard process for specific projects are maintained.
-

B.3.3.2 Process Assessment Process

This process is a lower-level process of the Life Cycle Model Management Process. It replaces the *Process Assessment* activity (6.2.1.3.2).

B.3.3.2.1 Purpose

The purpose of the Process Assessment Process is to determine the extent of process implementation and identify opportunities for continuous improvement.

B.3.3.2.2 Outcomes

As a result of successful implementation of the Process Assessment Process:

- a) information and data related to the use of the standard process for specific projects exist and are maintained;
 - b) the strengths and weaknesses of the organization's standard processes are understood; and
 - c) accurate and accessible assessment records are maintained.
-

B.3.3.3 Process Improvement Process

This process is a lower-level process of the Life Cycle Model Management Process. It replaces the *Process Improvement* activity (6.2.1.3.3).

B.3.3.3.1 Purpose

The purpose of the Process Improvement Process is to continually improve organizational effectiveness and efficiency through improvements to processes aligned with business needs.

B.3.3.3.2 Outcomes

As a result of successful implementation of the Process Improvement Process:

- a) commitment is established to provide resources to sustain improvement actions;
- b) internal and external issues are identified as improvement opportunities;
- c) the current status of processes is analyzed to identify improvement stimuli;
- d) improvement goals are identified and prioritized, and changes are implemented;
- e) the effects of improvements are monitored and confirmed against goals;
- f) knowledge gained from improvements is communicated throughout the organization; and
- g) improvements are evaluated for reuse elsewhere in the organization.

NOTE 1 — Information sources include assessments, audits, customer feedback, cost of quality, and organizational performance reports.

NOTE 2 — Process assessment may be used to determine current process status.

B.3.4 Human Resource Management Process Lower-Level Processes

B.3.4.1 Skill Development Process

This process is a lower-level process of the Human Resource Management

Process. It replaces the *Skill Development* activity (6.2.4.3.2).

B.3.4.1.1 Purpose

The purpose of the Skill Development Process is to provide individuals with the skills and knowledge needed to perform their roles effectively.

B.3.4.1.2 Outcomes

As a result of successful implementation of the Skill Development Process:

- a) training is developed or acquired to meet organizational and project needs; and
 - b) training is conducted to ensure individuals can perform their assignments effectively.
-

B.3.4.2 Skill Acquisition and Provision Process

This process is a lower-level process of the Human Resource Management Process. It replaces the *Skill Acquisition and Provision* activity (6.2.4.3.3).

B.3.4.2.1 Purpose

The purpose of the Skill Acquisition and Provision Process is to provide the organization and projects with individuals who possess the necessary skills and knowledge to work effectively as a cohesive group.

B.3.4.2.2 Outcomes

As a result of successful implementation of the Skill Acquisition and Provision Process:

- a) individuals with the required competencies are identified and recruited;
 - b) effective interaction between individuals and groups is supported;
 - c) the workforce has the skills to share information and coordinate efficiently; and
 - d) objective performance criteria are defined and used for feedback and enhancement.
-

B.3.4.3 Knowledge Management Process

This process is a lower-level process of the Human Resource Management Process. It replaces the *Knowledge Management* activity (6.2.4.3.4).

B.3.4.3.1 Purpose

The purpose of the Knowledge Management Process is to ensure that knowledge, information, and skills are collected, shared, reused, and improved throughout the organization.

B.3.4.3.2 Outcomes

As a result of successful implementation of the Knowledge Management Process:

- a) infrastructure is established and maintained for sharing information across the organization;
 - b) knowledge is readily available and shared; and
 - c) an appropriate knowledge management strategy is selected.
-

B.3.5 Software Operation Process Lower-Level Processes

B.3.5.1 Operational Use Process

This process is a lower-level process of the Software Operation Process. It replaces the *Operational Use* activity (6.4.9.3.3).

B.3.5.1.1 Purpose

The purpose of the Operational Use Process is to ensure the correct and efficient operation of the product during its intended usage.

B.3.5.1.2 Outcomes

As a result of successful implementation of the Operational Use Process:

- a) operational risks for product introduction and operation are identified and monitored;
 - b) the product is operated in its intended environment according to requirements; and
 - c) criteria for operational use are developed to demonstrate compliance with agreed requirements.
-

B.3.5.2 Customer Support Process

This process is a lower-level process of the Software Operation Process. It replaces the *Customer Support* activity (6.4.9.3.4).

B.3.5.2.1 Purpose

The purpose of the Customer Support Process is to ensure that responsive and effective support is provided to customers during operational use.

B.3.5.2.2 Outcomes

As a result of successful implementation of the Customer Support Process:

- a) service needs for customer support are identified and monitored;

- b) customer satisfaction with support services and the product is evaluated on an ongoing basis;
- c) operational support is provided by handling customer inquiries, requests, and resolving problems; and
- d) customer support needs are met through appropriate services.